Epic-FHIR — API category: Clinical Specialty (sections 13 – 13.3.2)

Source: Epic on FHIR (fhir.epic.com) documentation, mirrored 2026-11. Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

13 API category: Clinical Specialty

File: site/apis/13-Clinical-Specialty/index.html

R4 (46)

API | Method | URL template | Type
Bodystructure.Create (Radiotherapy Volume) (R4) | POST | /api/FHIR/R4/BodyStructure | Industry-Standard
BodyStructure.Read (Organ) (R4) | GET | /api/FHIR/R4/BodyStructure/{ID} | Industry-Standard
Bodystructure.Read (Radiotherapy Volume) (R4) | GET | /api/FHIR/R4/BodyStructure/{ID} | Industry-Standard
BodyStructure.Read (Tooth) (R4) | GET | /api/FHIR/R4/BodyStructure/{ID} | Industry-Standard
BodyStructure.Search (Organ) (R4) | GET | /api/FHIR/R4/BodyStructure?location={location}&morphology={morphology}&patient={patient}&subject={subject} | Industry-Standard
Bodystructure.Search (Radiotherapy Volume) (R4) | GET | /api/FHIR/R4/BodyStructure?identifier={identifier}&location={location}&morphology={morphology}&patient={patient}&subject={subject} | Industry-Standard
BodyStructure.Search (Tooth) (R4) | GET | /api/FHIR/R4/BodyStructure?location={location}&morphology={morphology}&patient={patient}&subject={subject}&_id={_id} | Industry-Standard
Bodystructure.Update (Radiotherapy Volume) (R4) | PUT | /api/FHIR/R4/BodyStructure/{ID} | Industry-Standard
CarePlan.Read (Dental) (R4) (described under Clinical Care Provision) | GET | /api/FHIR/R4/CarePlan/{ID} | Industry-Standard
CarePlan.Read (Oncology) (R4) (described under Clinical Care Provision) | GET | /api/FHIR/R4/CarePlan/{ID} | Industry-Standard
CarePlan.Search (Dental) (R4) (described under Clinical Care Provision) | GET | /api/FHIR/R4/CarePlan?category={category}&patient={patient}&subject={subject} | Industry-Standard
CarePlan.Search (Oncology) (R4) (described under Clinical Care Provision) | GET | /api/FHIR/R4/CarePlan?category={category}&patient={patient}&status={status} | Industry-Standard
Condition.Read (Dental Finding) (R4) (described under Clinical General) | GET | /api/FHIR/R4/Condition/{ID} | Industry-Standard
Condition.Search (Dental Finding) (R4) (described under Clinical General) | GET | /api/FHIR/R4/Condition?category={category}&clinical-status={clinical-status}&patient={patient} | Industry-Standard
Flag.Read (Isolation) (R4) | GET | /api/FHIR/R4/Flag/{ID} | Industry-Standard
Flag.Search (Isolation) (R4) | GET | /api/FHIR/R4/Flag?category={category}&encounter={encounter}&patient={patient}&status={status}&subject={subject} | Industry-Standard
NutritionOrder.Read (R4) | GET | /api/FHIR/R4/NutritionOrder/{ID} | Industry-Standard
NutritionOrder.Search (R4) | GET | /api/FHIR/R4/NutritionOrder?patient={patient} | Industry-Standard
Observation.Read (Assessments) (R4) | GET | /api/FHIR/R4/Observation/{ID} | USCDI
Observation.Read (Family Situation) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (Labor and Delivery) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (Newborn Delivery) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (Obstetrics and Gynecology) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (Periodontal) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Search (Assessments) (R4) | GET | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count} | USCDI
Observation.Search (Family Situation) (R4) | GET | /api/FHIR/R4/Observation?category={category}&code={code}&issued={issued}&patient={patient}&subject={subject} | Industry-Standard
Observation.Search (Labor and Delivery) (R4) | GET | /api/FHIR/R4/Observation?category={category}&code={code}&date={date}&patient={patient} | Industry-Standard
Observation.Search (Newborn Delivery) (R4) | GET | /api/FHIR/R4/Observation?category={category}&code={code}&patient={patient} | Industry-Standard
Observation.Search (Obstetrics and Gynecology) (R4) | GET | /api/FHIR/R4/Observation?&category={category}&code={code}&date={date}&focus={focus}&patient={patient}&subject={subject} | Industry-Standard
Observation.Search (Periodontal) (R4) | GET | /api/FHIR/R4/Observation?category={category}&date={date}&patient={patient}&subject={subject} | Industry-Standard
Procedure.Create (External Radiotherapy Summary) (R4) | POST | /api/FHIR/R4/Procedure | Industry-Standard
Procedure.Read (External Radiotherapy Summary) (R4) | GET | /api/FHIR/R4/Procedure/{ID} | Industry-Standard
Procedure.Search (External Radiotherapy Summary) (R4) | GET | /api/FHIR/R4/Procedure?category={category}&code={code}&date={date}&identifier={identifier}&patient={patient}&subject={subject} | Industry-Standard
Procedure.Update (External Radiotherapy Summary) (R4) | PUT | /api/FHIR/R4/Procedure/{ID} | Industry-Standard
RequestGroup.Read (Dental Visit) (R4) (described under Clinical Care Provision) | GET | /api/FHIR/R4/RequestGroup/{ID} | Industry-Standard
RequestGroup.Read (Oncology Plan Day) (R4) | GET | /api/FHIR/R4/RequestGroup/{ID} | Industry-Standard
RequestGroup.Search (Dental Visit) (R4) | GET | /api/FHIR/R4/RequestGroup?_id={_id} | Industry-Standard
RequestGroup.Search (Oncology Plan Day) (R4) | GET | /api/FHIR/R4/RequestGroup?_id={_id} | Industry-Standard
ServiceRequest.Create (External Radiotherapy Summary) (R4) | POST | /api/FHIR/R4/ServiceRequest | Industry-Standard
ServiceRequest.Read (Dental Procedure) (R4) (described under Clinical Care Provision) | | api/FHIR/R4/ServiceRequest/{ID} | Industry-Standard
ServiceRequest.Read (External Radiotherapy Summary) (R4) | GET | /api/FHIR/R4/ServiceRequest/{ID} | Industry-Standard
ServiceRequest.Read (Pregnancy Plans) (R4) | GET | /api/FHIR/R4/ServiceRequest/{ID} | Industry-Standard

ServiceRequest.Search (Dental Procedure) (R4) (described under Clinical Care Provision) | GET | /api/FHIR/R4/ServiceRequest?category={category}&onlyscannable={onlyscannable}&patient={patient}&status={status} | Industry-Standard
ServiceRequest.Search (External Radiotherapy Summary) (R4) | GET | /api/FHIR/R4/ServiceRequest?authored={authored}&category={category}&code={code}&encounter={encounter}&identifier={identifier}&onlyscannable={onlyscannable}&patient={patient}&requester={requester}&status={status}&subject={subject} | Industry-Standard
ServiceRequest.Search (Pregnancy Plans) (R4) | GET | api/FHIR/R4/ServiceRequest?patient={patient}&subject={subject}&category={category}&status={status} | Industry-Standard
ServiceRequest.Update (External Radiotherapy Summary) (R4) | PUT | /api/FHIR/R4/ServiceRequest/{ID} | Industry-Standard

STU3 (4)

API | Method | URL template | Type
NutritionOrder.Read (STU3) | GET | /api/FHIR/STU3/NutritionOrder/{ID} | Industry-Standard
NutritionOrder.Search (STU3) | GET | /api/FHIR/STU3/NutritionOrder?patient={patient} | Industry-Standard
Observation.Read (Family Situation) (STU3) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Search (Family Situation) (STU3) | GET | /api/FHIR/STU3/Observation?category={category}&code={code}&issued={issued}&patient={patient}&subject={subject} | Industry-Standard

Non-FHIR (2)

API | Method | URL template | Type
GetDepartmentWaitTimes | POST | api/epic/2014/Scheduling/Utility/GETDEPARTMENTWAITTIMES/GetDepartmentWaitTimes | Epic Public
GetWaitTimes | GET | api/epic/2010/Specialty/EmergencyDepartment/GETWAITTIMES/EmergencyDepartment/WaitTimes?DepartmentID={DepartmentID}&DepartmentIDType={DepartmentIDType} | Epic Public

13.1 Clinical Specialty: R4

File: site/apis/13-Clinical-Specialty/1-R4/index.html

* Bodystructure.Create (Radiotherapy Volume) (R4)
* BodyStructure.Read (Organ) (R4)
* Bodystructure.Read (Radiotherapy Volume) (R4)
* BodyStructure.Read (Tooth) (R4)
* BodyStructure.Search (Organ) (R4)
* Bodystructure.Search (Radiotherapy Volume) (R4)
* BodyStructure.Search (Tooth) (R4)
* Bodystructure.Update (Radiotherapy Volume) (R4)
* Flag.Read (Isolation) (R4)
* Flag.Search (Isolation) (R4)
* NutritionOrder.Read (R4)
* NutritionOrder.Search (R4)
* Observation.Read (Assessments) (R4)
* Observation.Read (Family Situation) (R4)
* Observation.Read (Labor and Delivery) (R4)
* Observation.Read (Newborn Delivery) (R4)
* Observation.Read (Obstetrics and Gynecology) (R4)
* Observation.Read (Periodontal) (R4)
* Observation.Search (Assessments) (R4)
* Observation.Search (Family Situation) (R4)
* Observation.Search (Labor and Delivery) (R4)
* Observation.Search (Newborn Delivery) (R4)
* Observation.Search (Obstetrics and Gynecology) (R4)
* Observation.Search (Periodontal) (R4)
* Procedure.Create (External Radiotherapy Summary) (R4)
* Procedure.Read (External Radiotherapy Summary) (R4)
* Procedure.Search (External Radiotherapy Summary) (R4)
* Procedure.Update (External Radiotherapy Summary) (R4)
* RequestGroup.Read (Oncology Plan Day) (R4)
* RequestGroup.Search (Dental Visit) (R4)
* RequestGroup.Search (Oncology Plan Day) (R4)
* ServiceRequest.Create (External Radiotherapy Summary) (R4)

* ServiceRequest.Read (External Radiotherapy Summary) (R4)
* ServiceRequest.Read (Pregnancy Plans) (R4)
* ServiceRequest.Search (External Radiotherapy Summary) (R4)
* ServiceRequest.Search (Pregnancy Plans) (R4)
* ServiceRequest.Update (External Radiotherapy Summary) (R4)

13.1.1 Bodystructure.Create (Radiotherapy Volume) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11040-Bodystructure-Create-Radiotherapy-Volume-R4.html
Epic API ID | 11040
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | POST
URL template | /api/FHIR/R4/BodyStructure
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11040

Description

A volume of the body that represents the target of radiotherapy planning or treatment delivery. BodyStructure resources are patient-specific and carry details about the type of structure in question and various optional qualifiers such as laterality.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
BodyStructure (R4) | BodyStructure | | | | An instance of the R4 BodyStructure resource.
BodyStructure (R4).description | string | | | | A description of the volume.
BodyStructure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy volume.
BodyStructure (R4).identifier.system | String | | | 0..1 | The identification system or namespace from which the identifier is assigned. For a DICOM UID, the system value must be: urn:dicom:uid
BodyStructure (R4).identifier.use | String | | | 1..1 | The use or function of the identifier. The following as supported values for use: DICOM UID should use official. Human-readable display names should use usual.
BodyStructure (R4).identifier.value | String | | | 1..1 | The value of the identifier. Either a DICOM UID or a human-readable display name depending on the specified identifier.use.
BodyStructure (R4).location | CodeableConcept | | | | SNOMED code for body structure being targeted.
BodyStructure (R4).locationQualifier | CodeableConcept | | Yes | | Additional qualifier SNOMED codes to describe more specifically how the body structure is targeted.
BodyStructure (R4).meta | Meta
BodyStructure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).
BodyStructure (R4).meta.profile | String | | Yes | 1..* | Must declare conformance to http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-volume
BodyStructure (R4).morphology | CodeableConcept | | | 0..1 | The type of anatomical volume that this body structure represents. The SNOMED code set is found in http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-volume-type-vs.
BodyStructure (R4).patient | Reference | | | 1..1 | A FHIR reference to the patient whose anatomy is delineated by this volume.

Response elements

Name | Type | Optional | Array | Cardinality | Description
BodyStructure (R4) | BodyStructure | | | | An instance of the R4 BodyStructure resource.
BodyStructure (R4).description | string | | | 0..1 | A description of the volume.
BodyStructure (R4).id | id | | | 0..1 | The FHIR id for the resource in Epic.
BodyStructure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy volume.
BodyStructure (R4).identifier.system | String | | | 0..1 | The identification system or namespace from which the identifier is assigned. For a DICOM UID, the system value must be: urn:dicom:uid
BodyStructure (R4).identifier.use | String | | | 1..1 | The use or function of the identifier. The following as supported values for use: DICOM UID should use official. Human-readable display names should use usual.
BodyStructure (R4).identifier.value | String | | | 1..1 | The value of the identifier. Either a DICOM UID or a human-readable display name depending on the specified identifier.use.
BodyStructure (R4).location | CodeableConcept | | | 0..1 | SNOMED code for body structure being targeted.
BodyStructure (R4).locationQualifier | CodeableConcept | | Yes | 0..1 | Additional qualifier SNOMED codes to describe more specifically how the body structure is targeted.
BodyStructure (R4).meta | Meta
BodyStructure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).

BodyStructure (R4).meta.profile | String | | Yes | 1..* | Must declare conformance to http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-volume

BodyStructure (R4).morphology | CodeableConcept | | | 0..1 | The type of anatomical volume that this body structure represents. The SNOMED code set is found in http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-volume-type-vs.

BodyStructure (R4).patient | Reference | | | 1..1 | A FHIR reference to the patient whose anatomy is delineated by this volume.

Sample request

```
POST https://hostname/instance/api/FHIR/R4/BodyStructure
{
  "resourceType": "BodyStructure",
  "meta": {
    "versionId": "1",
    "lastUpdated": "2020-07-03T10:07:41.050+02:00",
    "profile": [
      "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-volume",
      "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-volume"
    ]
  },
  "identifier": [
    {
      "use": "official",
      "system": "urn:dicom:uid",
      "value": "urn:oid:1.2.246.352.71.842418.2121.20150602151825"
    },
    {
      "use": "usual",
      "system": "http://example.com/fhir/identifier/radiotherapyDoseReferenceId",
      "value": "Prostate"
    }
  ],
  "morphology": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "228793007",
        "display": "Planning target volume"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "228792002",
        "display": "Clinical target volume"
      }
    ]
  },
  "location": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "41216001",
        "display": "Prostate"
      }
    ]
  },
  "locationQualifier": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "261183002",
          "display": "Upper"
        }
      ]
    },
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "7771000",
          "display": "Left"
        }
      ]
    }
  ],
  "description": "This is for testing purpose",
  "patient": {
    "reference": "Patient/eA7-kIKUJpr4kKMBTt633Lg3"
  }
}
```

Sample response

```
Status code: Created (201)
Content-Type: application/json; charset=utf-8
Location: BodyStructure/frcgE1EPRC0yCfY6JK0tiJQ4
```

13.1.2 BodyStructure.Read (Organ) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10280-BodyStructure-Read-Organ-R4.html
Epic API ID | 10280
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/BodyStructure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10280

Response elements

Name | Type | Optional | Array | Cardinality | Description
BodyStructure | BodyStructure | False | | | An instance of the R4 BodyStructure resource.
BodyStructure.active | Boolean | False | | | Whether the organ is attached to an episode.
BodyStructure.description | string | False | | | The display name of the organ.
BodyStructure.identifier | Identifier | False | Yes | | BodyStructure FHIR ID.
BodyStructure.location | CodeableConcept | True | | | SNOMED code and display name for the organ.
BodyStructure.morphology | CodeableConcept | True | | | SNOMED code 280115004 (Acquired body structure (body structure)) if the patient is the recipient of the organ or 91723000 (Anatomical structure (body structure)) if the patient is the organ donor.
BodyStructure.extension (also: organ-status) | Extension | True | Yes | | The status of the organ.
BodyStructure.patient | Reference | False | | | The organ donor or recipient.

Sample request

```
https://hostname/instance/api/FHIR/R4/BodyStructure/eB962RufIdue04nYAGf1HNCIl7CI54FFqvZpawIRgp.S1YsihI9sXlVyxh.iAXl0D3
```

Sample response


```
{
  "resourceType": "BodyStructure",
  "id": "eB962RufIdue04nYAGf1HNCIl7CI54FFqvZpawIRgp.S1YsihI9sXlVyxh.iAXl0D3",
  "extension": [
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.10.798271.415",
            "code": "4",
            "display": "Transplanted"
          }
        ],
        "text": "Transplanted"
      },
      "url": "http://open.epic.com/FHIR/StructureDefinition/extension/organ-status"
    }
  ],
  "active": true,
  "morphology": {
    "coding": [
      {
        "system": "urn:oid:2.16.840.1.113883.6.96",
        "code": "280115004",
        "display": "Acquired body structure (body structure)"
      }
    ],
    "text": "Acquired body structure (body structure)"
  },
  "location": {
    "coding": [
      {
        "system": "urn:oid:2.16.840.1.113883.6.96",
        "code": "18639004",
        "display": "Left kidney structure (body structure)"
      }
    ],
    "text": "Left kidney structure (body structure)"
  },
  "description": "Left Kidney",
  "patient": {
    "reference": "Patient/eRKAMfSfGUZ1HFGl2QGrHlQ3",
    "display": "Test, Patient"
  }
}
```


13.1.3 Bodystructure.Read (Radiotherapy Volume) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11041-Bodystructure-Read-Radiotherapy-Volume-R4.html

Epic API ID | 11041
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/BodyStructure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11041

Description

A volume of the body that represents the target of radiotherapy planning or treatment delivery. BodyStructure resources are patient-specific and carry details about the type of structure in question and various optional qualifiers such as laterality.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The BodyStructure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
BodyStructure (R4) | BodyStructure | | | | An instance of the R4 BodyStructure resource.
BodyStructure (R4).description | string | | | 0..1 | A description of the volume.
BodyStructure (R4).id | id | | | 0..1 | The FHIR id for the resource in Epic.
BodyStructure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy volume.
BodyStructure (R4).identifier.system | String | | | 0..1 | The identification system or namespace from which the identifier is assigned. For a DICOM UID, the system value must be: urn:dicom:uid
BodyStructure (R4).identifier.use | String | | | 1..1 | The use or function of the identifier. The following as supported values for use: DICOM UID should use official. Human-readable display names should use usual.
BodyStructure (R4).identifier.value | String | | | 1..1 | The value of the identifier. Either a DICOM UID or a human-readable display name depending on the specified identifier.use.
BodyStructure (R4).location | CodeableConcept | | | 0..1 | SNOMED code for body structure being targeted.
BodyStructure (R4).locationQualifier | CodeableConcept | | Yes | 0..1 | Additional qualifier SNOMED codes to describe more specifically how the body structure is targeted.
BodyStructure (R4).meta | Meta
BodyStructure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).
BodyStructure (R4).meta.profile | String | | Yes | 1..* | Must declare conformance to http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-volume
BodyStructure (R4).morphology | CodeableConcept | | | 0..1 | The type of anatomical volume that this body structure represents. The SNOMED code set is found in http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-volume-type-vs.
BodyStructure (R4).patient | Reference | | | 1..1 | A FHIR reference to the patient whose anatomy is delineated by this volume.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/BodyStructure/frcgE1EPRC0yCfY6JK0tiJQ4
```

Sample response


```
{
  "resourceType": "BodyStructure",
  "id": "frcgE1EPRC0yCfY6JK0tiJQ4",
  "meta": {
    "versionId": "1",
    "lastUpdated": "2026-02-03T20:08:39Z",
    "profile": [
      "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-volume"
    ]
  },
  "identifier": [
    {
      "use": "official",
      "system": "urn:dicom:uid",
      "value": "3afc4b8e-da0d-4c5b-9883-dbc616c18c74"
    },
    {
      "use": "usual",
      "system": "http://example.com/fhir/identifier/radiotherapyDoseReferenceId",
      "value": "Left Breast"
    }
  ],
  "active": true,
  "morphology": {
```

```
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "228792002",
        "display": "Clinical target volume"
      }
    ]
  },
  "location": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "BREAST STRUCTURE"
      }
    ]
  },
  "locationQualifier": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "7771000",
          "display": "Left"
        }
      ]
    }
  ],
  "description": "Left Breast",
  "patient": {
    "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
    "display": "Test, Patient"
  }
}
```


13.1.4 BodyStructure.Read (Tooth) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10068-BodyStructure-Read-Tooth-R4.html
Epic API ID | 10068
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET
URL template | /api/FHIR/R4/BodyStructure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10068

Description

The BodyStructure (Tooth) resource returns a patient's tooth record. Tooth records are individual anatomic records including primary and supernumerary teeth.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | BodyStructure identifier.

Response elements

Name | Type | Optional | Array | Cardinality | Description
BodyStructure | BodyStructure | False | | | An instance of the R4 BodyStructure resource.
BodyStructure.active | Boolean | True | | | True if the tooth is actively in a patient's dentition. False if it has been removed.
BodyStructure.extension (also: dental-tooth-eruption-status) | Extension | True | Yes | | Tooth eruption status, including erupted, unerupted, partially erupted, and missing.
BodyStructure.location | CodeableConcept | True | | | Returns the SNOMED mapping and Epic ID for the tooth.
BodyStructure.morphology | CodeableConcept | True | | | Returns SNOMED code 302214001 or "Tooth" if SNOMED codes are not available.
BodyStructure.patient | Reference | False | | | The patient whose tooth this is.

Sample request

```
https://hostname/instance/api/FHIR/R4/BodyStructure/eU1gdcwFZyKjNGyBZhKh989YxW3fhmjvviCKYH.RTCBw3
```

Sample response

```
<BodyStructure xmlns="http://hl7.org/fhir">
<id value="eU1gdcwFZyKjNGyBZhKh989YxW3fhmjvviCKYH.RTCBw3" />
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-tooth-eruption-status">
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.72.1.7.7.10.826983.17460" />
<code value="1" />
</coding>
<text value="Erupted" />
</valueCodeableConcept>
</extension>
<active value="true" />
<morphology>
<coding>
<system value="urn:oid:2.16.840.1.113883.6.96" />
<code value="302214001" />
</coding>
<text value="Entire tooth (body structure)" />
</morphology>
<location>
<coding>
<system value="urn:oid:2.16.840.1.113883.6.96" />
<code value="245566003" />
<display value="Entire permanent maxillary right third molar tooth (body structure)" />
</coding>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.3.866976.94010" />
<code value="1" />
</coding>
<text value="Tooth (Universal): 1" />
</location>
<description value="Tooth (Universal): 1" />
<patient>
<reference value="Patient/erOOucv8EeF9xNKxjL..kSQ3" />
<type value="Patient" />
<display value="Johnson, Gail”/>
</patient>
</BodyStructure>
```


13.1.5 BodyStructure.Search (Organ) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10281-BodyStructure-Search-Organ-R4.html
Epic API ID | 10281
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/BodyStructure?location={location}&morphology={morphology}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10281

Request parameters

Name | Type | Optional | Array | Cardinality | Description
location | token | False | | | Limits the search to a particular organ location. Format: “http://snomed.info/sct|SNOMEDCODE” where SNOMEDCODE is the numeric portion of the code associated with a particular body structure. For example, enter “http://snomed.info/sct|18639004” to search for left kidney structures. May send multiple entries separated by commas.
morphology | token | False | | | Limits the search to all organs, or to specifically donor or recipient organs. Enter “organs” for all organs, “http://snomed.info/sct|91723000” for donor organs, or “http://snomed.info/sct|280115004” for recipient organs.
patient | reference | Either patient or subject is required | | | The patient for which to perform the search.
subject | reference | Either patient or subject is required | | | The patient for which to perform the search.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.BodyStructure | BodyStructure | False | | | An instance of the R4 BodyStructure resource.
Bundle.entry.resource.BodyStructure.active | Boolean | True | | | Whether the organ is attached to any active episode.
Bundle.entry.resource.BodyStructure.description | string | False | | | The display name of the organ.
Bundle.entry.resource.BodyStructure.id | id | False | | | BodyStructure FHIR ID.
Bundle.entry.resource.BodyStructure.location | CodeableConcept | False | | | SNOMED code and display name for the organ.
Bundle.entry.resource.BodyStructure.morphology | CodeableConcept | False | | | SNOMED code 280115004 (Acquired body structure (body structure)) if the patient is the recipient of the organ or 91723000 (Anatomical structure (body structure)) if the patient is the organ donor.
Bundle.entry.resource.BodyStructure.extension (also: organ-status) | Extension | False | Yes | | The status of the organ.

Bundle.entry.resource.BodyStructure.patient | Reference | False | | | The organ donor or recipient.

Sample request

```
https://hostname/instance/api/FHIR/R4/BodyStructure?morphology=organs&patient=eRKAMfSfGUZ1HFGl2QGrHlQ3&location=http://s
    nomed.info/sct|18639004
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/BodyStructure?morphology=organs&patient=eRKAMfSfGUZ1HFGl2QGrHlQ3&loc
    ation=http://snomed.info/sct%7C18639004"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url":
    "https://hostname/instance/api/FHIR/R4/BodyStructure/eB962RufIdue04nYAGf1HNCIl7CI54FFqvZpawIRgp.S1YsihI9sXlVyxh.iAXl
    0D3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/BodyStructure/eB962RufIdue04nYAGf1HNCIl7CI54FFqvZpawIRgp.S1YsihI
    9sXlVyxh.iAXl0D3",
      "resource": {
        "resourceType": "BodyStructure",
        "id": "eB962RufIdue04nYAGf1HNCIl7CI54FFqvZpawIRgp.S1YsihI9sXlVyxh.iAXl0D3",
        "extension": [
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.10.798271.415",
                  "code": "4",
                  "display": "Transplanted"
                }
              ],
              "text": "Transplanted"
            },
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/organ-status"
          }
        ],
        "active": true,
        "morphology": {
          "coding": [
            {
              "system": "urn:oid:2.16.840.1.113883.6.96",
              "code": "280115004",
              "display": "Acquired body structure (body structure)"
            }
          ],
          "text": "Acquired body structure (body structure)"
        },
        "location": {
          "coding": [
            {
              "system": "urn:oid:2.16.840.1.113883.6.96",
              "code": "18639004",
              "display": "Left kidney structure (body structure)"
            }
          ],
          "text": "Left kidney structure (body structure)"
        },
        "description": "Left Kidney",
        "patient": {
          "reference": "Patient/eRKAMfSfGUZ1HFGl2QGrHlQ3",
          "display": "Test, Patient"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


13.1.6 Bodystructure.Search (Radiotherapy Volume) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11042-Bodystructure-Search-Radiotherapy-Volume-R4.html

Epic API ID | 11042
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template |
/api/FHIR/R4/BodyStructure?identifier={identifier}&location={location}&morphology={morphology}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11042

Description

A volume of the body that represents the target of radiotherapy planning or treatment delivery. BodyStructure resources are patient-specific and carry details about the type of structure in question and various optional qualifiers such as laterality.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revIncludе. All others are ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
identifier | token | Optional | Yes | | Identifier for the volume. If provided, a patient is no longer required.
location | token | Optional | Yes | | If provided, returns all Radiotherapy Volumes from the given location
morphology | token | Optional | Yes | | If provided, returns all Radiotherapy Volumes with the given morphology.
patient | reference | Conditional | Yes | | A patient’s FHIR ID. Returns all Radiotherapy Volumes associated with the patient. Either patient or subject can be used to search on the patient. A patient or subject is required if searching without the identifier parameter.
subject | reference | Conditional | Yes | | A patient’s FHIR ID. Returns all Radiotherapy Volumes associated with the patient. Either patient or subject can be used to search on the patient. A patient or subject is required if searching without the identifier parameter.

Response elements

Name | Type | Optional | Array | Cardinality | Description
BodyStructure (R4) | BodyStructure | | | | An instance of the R4 BodyStructure resource.
BodyStructure (R4).description | string | | | 0..1 | A description of the volume.
BodyStructure (R4).id | id | | | 0..1 | The FHIR id for the resource in Epic.
BodyStructure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy volume.
BodyStructure (R4).identifier.system | String | | | 0..1 | The identification system or namespace from which the identifier is assigned. For a DICOM UID, the system value must be: urn:dicom:uid
BodyStructure (R4).identifier.use | String | | | 1..1 | The use or function of the identifier. The following as supported values for use: DICOM UID should use official. Human-readable display names should use usual.
BodyStructure (R4).identifier.value | String | | | 1..1 | The value of the identifier. Either a DICOM UID or a human-readable display name depending on the specified identifier.use.
BodyStructure (R4).location | CodeableConcept | | | 0..1 | SNOMED code for body structure being targeted.
BodyStructure (R4).locationQualifier | CodeableConcept | | Yes | 0..1 | Additional qualifier SNOMED codes to describe more specifically how the body structure is targeted.
BodyStructure (R4).meta | Meta
BodyStructure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).
BodyStructure (R4).meta.profile | String | | Yes | 1..* | Must declare conformance to http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-volume
BodyStructure (R4).morphology | CodeableConcept | | | 0..1 | The type of anatomical volume that this body structure represents. The SNOMED code set is found in http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-volume-type-vs.
BodyStructure (R4).patient | Reference | | | 1..1 | A FHIR reference to the patient whose anatomy is delineated by this volume.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/BodyStructure?patient=eP-BmFG0190obAH1wt4wjUw3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/BodyStructure?_id=fqYNbuWQOsf.f9Duiycny2g4"
    }
  ],
```

```
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/BodyStructure/fqYNbuWQOsf.f9Duiycny2g4",
      "resource": {
        "resourceType": "BodyStructure",
        "id": "fqYNbuWQOsf.f9Duiycny2g4",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2026-02-03T20:08:39Z",
          "profile": [
            "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-volume"
          ]
        },
        "identifier": [
          {
            "use": "official",
            "system": "urn:dicom:uid",
            "value": "3afc4b8e-da0d-4c5b-9883-dbc616c18c74"
          },
          {
            "use": "usual",
            "system": "http://example.com/fhir/identifier/radiotherapyDoseReferenceId",
            "value": "Left Breast"
          }
        ],
        "active": true,
        "morphology": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "228792002",
              "display": "Clinical target volume"
            }
          ]
        },
        "location": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "76752008",
              "display": "BREAST STRUCTURE"
            }
          ]
        },
        "locationQualifier": [
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "7771000",
                "display": "Left"
              }
            ]
          }
        ],
        "description": "Left Breast",
        "patient": {
          "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
          "display": "Test, Patient"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


13.1.7 BodyStructure.Search (Tooth) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10069-BodyStructure-Search-Tooth-R4.html
Epic API ID | 10069
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET

URL template | /api/FHIR/R4/BodyStructure?location={location}&morphology={morphology}&patient={patient}&subject={subject}&_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10069

Description

The BodyStructure (Tooth) resource returns a patient's tooth records. Tooth records are individual anatomic records including primary and supernumerary teeth.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
location | token | False | | | Accepts a string for a specific tooth in the patient's mouth. Format: "[Notation OID]|[tooth ID]", where "Notation OID" is the OID matching the tooth notation system used to communicate the tooth ID. May send multiple entries separated by commas. Supported notations are Universal (94010), FDI (94011) and Palmer (94013). Ex. "urn:oid:1.2.840.114350.1.13.861.1.7.3.866976.94010|32"
morphology | token | False | | | Points a search to the correct subresource of tooth. Either "teeth" or "302214001" are accepted. These values are functionally equivalent.
patient | reference | Either the "patient" or the "subject" parameter is required. | | | The patient to search for.
subject | reference | Either the "patient" or the "subject" parameter is required. | | | The patient to search for.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.BodyStructure | BodyStructure | False | | | An instance of the R4 BodyStructure resource.
Bundle.entry.resource.BodyStructure.active | Boolean | True | | | True if the tooth is actively in a patient's dentition. False if it has been removed.
Bundle.entry.resource.BodyStructure.extension (also: dental-tooth-eruption-status) | Extension | False | Yes | | Tooth eruption status, including erupted, unerupted, partially erupted, and missing.
Bundle.entry.resource.BodyStructure.location | CodeableConcept | False | | | Returns the SNOMED mapping and Epic ID for the tooth.
Bundle.entry.resource.BodyStructure.morphology | CodeableConcept | False | | | Returns SNOMED code 302214001 or "Tooth" if SNOMED codes are not available.
Bundle.entry.resource.BodyStructure.patient | Reference | False | | | The patient whose tooth this is.

Sample request

```
https://hostname/instance/api/FHIR/R4/BodyStructure?morphology=teeth&patient=erOOucv8EeF9xNKxjL..kSQ3
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
<type value="searchset" />
<total value="34" />
<link>
<relation value="self" />
<url value="https://hostname/instance/api/FHIR/R4/BodyStructure?morphology=teeth&amp;patient=erOOucv8EeF9xNKxjL..kSQ3"
    />
</link>
<entry>
<link>
<relation value="self" />
<url value="https://hostname/instance/api/FHIR/R4/BodyStructure/eU1gdcwFZyKjNGyBZhKh989YxW3fhmjvviCKYH.RTCBw3" />
</link>
<fullUrl value="https://hostname/instance/api/FHIR/R4/BodyStructure/eU1gdcwFZyKjNGyBZhKh989YxW3fhmjvviCKYH.RTCBw3" />
<resource>
<id value="eU1gdcwFZyKjNGyBZhKh989YxW3fhmjvviCKYH.RTCBw3" />
<extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-tooth-eruption-status">
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.72.1.7.7.10.826983.17460" />
<code value="1" />
</coding>
<text value="Erupted" />
</valueCodeableConcept>
</extension>
<active value="true" />
<morphology>
<coding>
<system value="urn:oid:2.16.840.1.113883.6.96" />
<code value="302214001" />
</coding>
<text value="Entire tooth (body structure)" />
</morphology>
<location>
<coding>
<system value="urn:oid:2.16.840.1.113883.6.96" />
<code value="245566003" />
<display value="Entire permanent maxillary right third molar tooth (body structure)" />
</coding>
```

```
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.3.866976.94010" />
<code value="1" />
</coding>
<text value="Tooth (Universal): 1" />
</location>
<description value="Tooth (Universal): 1" />
<patient>
<reference value="Patient/erOOucv8EeF9xNKxjL..kSQ3" />
<type value="Patient" />
<display value="Johnson, Gail" />
</patient>
</BodyStructure>
```

13.1.8 Bodystructure.Update (Radiotherapy Volume) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11043-Bodystructure-Update-Radiotherapy-Volume-R4.html
Epic API ID | 11043
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | PUT
URL template | /api/FHIR/R4/BodyStructure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11043

Description

A volume of the body that represents the target of radiotherapy planning or treatment delivery. BodyStructure resources are patient-specific and carry details about the type of structure in question and various optional qualifiers such as laterality.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
BodyStructure (R4) | BodyStructure | | | | An instance of the R4 BodyStructure resource.
BodyStructure (R4).description | string | | | | A description of the volume.
BodyStructure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy volume.
BodyStructure (R4).identifier.system | String | | | 0..1 | The identification system or namespace from which the identifier is assigned. For a DICOM UID, the system value must be: urn:dicom:uid
BodyStructure (R4).identifier.use | String | | | 1..1 | The use or function of the identifier. The following as supported values for use: DICOM UID should use official. Human-readable display names should use usual.
BodyStructure (R4).identifier.value | String | | | 1..1 | The value of the identifier. Either a DICOM UID or a human-readable display name depending on the specified identifier.use.
BodyStructure (R4).location | CodeableConcept | | | | SNOMED code for body structure being targeted.
BodyStructure (R4).locationQualifier | CodeableConcept | | Yes | | Additional qualifier SNOMED codes to describe more specifically how the body structure is targeted.
BodyStructure (R4).meta | Meta
BodyStructure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).
BodyStructure (R4).meta.profile | String | | Yes | 1..* | Must declare conformance to http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-volume
BodyStructure (R4).morphology | CodeableConcept | | | 0..1 | The type of anatomical volume that this body structure represents. The SNOMED code set is found in http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-volume-type-vs.
BodyStructure (R4).patient | Reference | | | 1..1 | A FHIR reference to the patient whose anatomy is delineated by this volume.

Sample request

```
PUT https://hostname/instance/api/FHIR/R4/BodyStructure/fqYNbuWQOsf.f9Duiycny2g4
{
    "resourceType": "BodyStructure",
    "id": fqYNbuWQOsf.f9Duiycny2g4,
    "meta": {
        "versionId": "1",
        "lastUpdated": "2026-02-03T20:08:39Z",
        "profile": [
            "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-volume"
        ]
    },
    "identifier": [
        {
            "use": "official",
            "system": "urn:dicom:uid",
            "value": "3afc4b8e-da0d-4c5b-9883-dbc616c18c74"
        },
```

```
        {
            "use": "usual",
            "system": "http://example.com/fhir/identifier/radiotherapyDoseReferenceId",
            "value": "Left Breast"
        }
    ],
    "active": true,
    "morphology": {
        "coding": [
            {
                "system": "http://snomed.info/sct",
                "code": "228792002",
                "display": "Clinical target volume"
            }
        ]
    },
    "location": {
        "coding": [
            {
                "system": "http://snomed.info/sct",
                "code": "76752008",
                "display": "BREAST STRUCTURE"
            }
        ]
    },
    "locationQualifier": [
        {
            "coding": [
                {
                    "system": "http://snomed.info/sct",
                    "code": "7771000",
                    "display": "Left"
                }
            ]
        }
    ],
    "description": "Left Breast",
    "patient": {
        "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
        "display": "Test, Patient"
    }
}
```


Sample response

```
Status code: OK (200)
Content-Type: application/json; charset=utf-8
```

13.1.9 Flag.Read (Isolation) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10196-Flag-Read-Isolation-R4.html
Epic API ID | 10196
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Flag/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10196

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for the target Flag resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Flag | Flag | False | | | A single instance of a Flag resource.
Flag.author | Reference | True | | | The practitioner who added the isolation to the patient’s chart.
Flag.category | CodeableConcept | True | Yes | | Always “isolation”.
Flag.code | CodeableConcept | False | | | The display name for the isolation and the SNOMED code if one exists.
Flag.encounter | Reference | True | | | The encounter associated with the isolation.
Flag.id | String | False | | | The isolation FHIR ID.
Flag.period | Period | False | | | The period during which the isolation was active. If the isolation is currently active, only period.start is populated.

Flag.period.end | String | True | | | Date when the isolation was removed from the patient’s chart.
Flag.period.start | String | False | | | Date when the isolation was added to the patient’s chart.
Flag.status | String | False | | | The status of the isolation.
Flag.subject | Reference | False | | | The patient associated with the isolation.

Sample request

```
https://hostname/instance/api/FHIR/R4/Flag/eBPewPslq6bBnIM765MKxr0GE06cHHVsds9fR0TLrQsI3
```

Sample response

```
<flag xmlns="http://hl7.org/fhir">
  <id value="eBPewPslq6bBnIM765MKxr0GE06cHHVsds9fR0TLrQsI3"/>
  <status value="inactive"/>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/flag-category"/>
      <code value="isolation"/>
      <display value="Isolation Flag"/>
    </coding>
    <text value="Isolation Flag"/>
  </category>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.10370"/>
      <code value="1"/>
      <display value="Contact"/>
    </coding>
    <text value="Contact"/>
  </code>
  <subject>
    <reference value="Patient/e41Miyc6Hkq7ZhziVk0GWDw3"/>
    <display value="Penrose, Alana"/>
  </subject>
  <period>
    <start value="2021-08-26T15:42:15Z"/>
    <end value="2021-08-31T20:04:48Z"/>
  </period>
  <encounter>
    <reference value="Encounter/embPAh4r2VvC3EaaypMrilon4jk2fNvBMa6iekfdeXFA3"/>
  </encounter>
  <author>
    <reference value="Practitioner/e4X2WQBx33yiYJ5QTGnHqog3"/>
    <display value="Ramon Escobar"/>
  </author>
</flag>
```

Change log

Epic Version | Description | Type
May 2023 (Special Updates to February 2023) | Code systems in the Flag.code element now return a URL in place of an OID. Typical code systems include: System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change
System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

13.1.10 Flag.Search (Isolation) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10195-Flag-Search-Isolation-R4.html
Epic API ID | 10195
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Flag?category={category}&encounter={encounter}&patient={patient}&status={status}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10195

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Always "isolation" for this sub-resource.
encounter | String | True | | | Reference to an Encounter resource associated with an isolation. If no encounter is specified, all isolations are returned.

patient | String | Either patient or subject must be provided. If both are provided, they must match. | | | Reference to a patient resource the isolation is for.
status | String | True | | | Status of the isolation. If no status is specified, all isolations are returned.
subject | String | Either patient or subject must be provided. If both are provided, they must match. | | | Reference to a patient resource the isolation is for.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
author | String | True | | | FHIR ID of practitioner who added the isolation to the patient's chart.
date | String | True | | | The period during which the isolation was active. If the isolation is currently active, only period.start is populated.
identifier | String | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Flag | Flag | False | | | A single instance of a Flag resource.
Bundle.entry.resource.Flag.author | Reference | False | | | The practitioner who added the isolation to the patient's chart.
Bundle.entry.resource.Flag.category | CodeableConcept | True | Yes | | Always "isolation".
Bundle.entry.resource.Flag.code | CodeableConcept | False | | | The display name for the isolation and the SNOMED code if one exists.
Bundle.entry.resource.Flag.encounter | Reference | True | | | The encounter associated with the isolation.
Bundle.entry.resource.Flag.id | String | False | | | The isolation FHIR ID.
Bundle.entry.resource.Flag.period | Period | True | | | The period during which the isolation was active. If the isolation is currently active, only period.start is populated.
Bundle.entry.resource.Flag.period.end | String | True | | | Date when the isolation was removed from the patient's chart.
Bundle.entry.resource.Flag.period.start | String | False | | | Date when the isolation was added to the patient's chart.
Bundle.entry.resource.Flag.status | String | False | | | The status of the isolation.
Bundle.entry.resource.Flag.subject | Reference | False | | | The patient associated with the isolation.

Sample request

```
https://hostname/instance/api/FHIR/R4/Flag?patient=eL7ZfbCw3Ablzwst7tis6iJSwobiwmyxawGIyK-odAFE3&category=isolation
```

Sample response

```
<Flag xmlns="http://hl7.org/fhir">
  <id value="exo1bsspIVf4CpSkWV8ixsEzfhbP3Oz8j5fAkHpv5Oec3"/>
  <status value="active"/>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/flag-category"/>
      <code value="isolation"/>
      <display value="Isolation Flag"/>
    </coding>
    <text value="Isolation Flag"/>
  </category>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.10.698084.10370"/>
      <code value="1"/>
      <display value="Contact"/>
    </coding>
    <text value="Contact"/>
  </code>
  <subject>
    <reference value="Patient/eL7ZfbCw3Ablzwst7tis6iJSwobiwmyxawGIyK-odAFE3"/>
    <display value="Norwalk, Winston"/>
  </subject>
  <period>
    <start value="2021-08-31T22:04:52Z"/>
  </period>
  <encounter>
    <reference value="Encounter/eeso2rgPmVowbzVAbGPHjgEBek-b6xi3-E59Drlpwghw3"/>
  </encounter>
  <author>
    <reference value="Practitioner/e4X2WQBx33yiYJ5QTGnHqog3"/>
    <display value="Ramon Escobar"/>
  </author>
</Flag>
```

Change log

Epic Version | Description | Type
May 2023 (Special Updates to February 2023) | Code systems in the Flag.code element now return a URL in place of an OID. Typical code systems include: System URL OID SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 LOINC http://loinc.org 2.16.840.1.113883.6.1 | Breaking change
System | URL | OID
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
LOINC | http://loinc.org | 2.16.840.1.113883.6.1

13.1.11 NutritionOrder.Read (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10224-NutritionOrder-Read-R4.html
Epic API ID | 10224
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/NutritionOrder/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10224

Description

This resource can read various types of nutrition orders, including oral diets, oral nutrition supplements, enteral nutrition (tube feedings), and infant formula. This resource can also include information from diet orders such as a patient's food allergies, intolerances, and personal/cultural requirements or preferences (such as being kosher or vegan), and includes data on texture and quantity when applicable.

Read interactions typically begin with a client having previously established a relationship, often through querying for NutritionOrder through the search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for a NutritionOrder resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
NutritionOrder | NutritionOrder | False | | | An instance of the R4 NutritionOrder resource.
NutritionOrder.dateTime | String | False | | | The date and time of the order.
NutritionOrder.id | String | False | | | Identifier.
NutritionOrder.intent | String | False | | | The current intention for an order. Supported values include: order plan proposal
NutritionOrder.note | Annotation | False | Yes | | An explanation of dietary recommendations.
NutritionOrder.oralDiet | OralDiet | True | | | The details of a diet order.
NutritionOrder.oralDiet.instruction | String | True | | | The name of a diet.
NutritionOrder.oralDiet.texture | Texture | True | Yes | | Texture information
NutritionOrder.oralDiet.texture.modifier | CodeableConcept | True | | | Describes the texture of food as part of a diet.
NutritionOrder.oralDiet.type | CodeableConcept | True | Yes | | The diet type.
NutritionOrder.patient | Reference | False | | | The patient whom the nutrition orders are related to.
NutritionOrder.status | String | False | | | The status of a nutrition order. Supported value include: draft active completed revoked unknown entered-in-error

Sample request

```
https://hostname/instance/api/FHIR/R4/NutritionOrder/emV2M.3TKuBneTxDOPf2VYUgryLaXH6Qus6BUOZ8Tg9k3
```

Sample response

```
<NutritionOrder xmlns="http://hl7.org/fhir">
  <id value="emV2M.3TKuBneTxDOPf2VYb5x3vlG-nAgKYCt9fijpOk3"/>
  <status value="active"/>
  <intent value="order"/>
  <patient>
    <reference value="eFLQ4DCYc6n424WRext9mTQ3"/>
    <display value="Patient, Epic"/>
  </patient>
  <dateTime value="2021-07-01T19:42:38Z"/>
  <oralDiet>
    <type>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.34365"/>
        <code value="3"/>
        <display value="Vegetarian"/>
      </coding>
      <text value="Vegetarian"/>
    </type>
    <texture>
      <modifier>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.34442"/>
          <code value="15"/>
          <display value="Solid"/>
        </coding>
```

```
        <text value="Solid"/>
      </modifier>
    </texture>
    <instruction value="GENERAL DIET PLAN"/>
  </oralDiet>
</NutritionOrder>
```

13.1.12 NutritionOrder.Search (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10227-NutritionOrder-Search-R4.html
Epic API ID | 10227
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/NutritionOrder?patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10227

Description

This resource can read various types of nutrition orders, including oral diets, oral nutrition supplements, enteral nutrition (tube feedings), and infant formula. This resource can also include information from diet orders such as a patient's food allergies, intolerances, and personal/cultural requirements or preferences (such as being kosher or vegan), and includes data on texture and quantity when applicable.

You can use the search interaction to query for nutrition orders based on a patient.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
patient | String | False | | | FHIR identifier for the patient whose nutrition order details are being requested.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
additive | String | True | | | Unsupported
datetime | String | True | | | Return nutrition orders requested on this date
encounter | String | True | | | Unsupported
formula | String | True | | | Unsupported
identifier | String | True | | | Unsupported
instantiates-canonical | String | True | | | Unsupported
oraldiet | String | True | | | Type of diet that can be consumed orally (i.e., take via the mouth).
provider | String | True | | | Unsupported
status | String | True | | | The status of a nutrition order. Supported value include: draft active completed revoked unknown entered-in-error
supplement | String | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.NutritionOrder | NutritionOrder | False | | | An instance of the R4 NutritionOrder resource.
Bundle.entry.resource.NutritionOrder.dateTime | String | False | | | The date and time of the order.
Bundle.entry.resource.NutritionOrder.id | String | False | | | Identifier.
Bundle.entry.resource.NutritionOrder.intent | String | False | | | The current intention for an order. Supported values include: order plan proposal
Bundle.entry.resource.NutritionOrder.note | Annotation | False | Yes | | An explanation of dietary recommendations.
Bundle.entry.resource.NutritionOrder.oralDiet | OralDiet | True | | | Type of diet that can be consumed orally (i.e., take via the mouth).
Bundle.entry.resource.NutritionOrder.oralDiet.instruction | String | True | | | The name of a diet.
Bundle.entry.resource.NutritionOrder.oralDiet.texture | Texture | True | Yes | | Texture information
Bundle.entry.resource.NutritionOrder.oralDiet.texture.modifier | CodeableConcept | True | | | Describes the texture of food as part of a diet.
Bundle.entry.resource.NutritionOrder.oralDiet.type | CodeableConcept | True | Yes | | The diet type.
Bundle.entry.resource.NutritionOrder.patient | Reference | False | | | The patient whom the nutrition orders are related to.
Bundle.entry.resource.NutritionOrder.status | String | False | | | The status of a nutrition order. Supported value include: draft active completed revoked unknown entered-in-error

Sample request

```
https://hostname/instance/api/FHIR/R4/NutritionOrder?patient=eFLQ4DCYc6n424WRext9mTQ3
```

Sample response


```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/NutritionOrder?patient=eFLQ4DCYc6n424WRext9mTQ3"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/NutritionOrder/emV2M.3TKuBneTxDOPf2VYUgryLaXH6Qus6BUOZ8Tg9k3"/>
    </link>
    <fullUrl
    value="https://hostname/instance/api/FHIR/R4/NutritionOrder/emV2M.3TKuBneTxDOPf2VYUgryLaXH6Qus6BUOZ8Tg9k3"/>
    <resource>
      <NutritionOrder>
        <id value="emV2M.3TKuBneTxDOPf2VYUgryLaXH6Qus6BUOZ8Tg9k3"/>
        <status value="active"/>
        <intent value="order"/>
        <patient>
          <reference value="eFLQ4DCYc6n424WRext9mTQ3"/>
          <display value="Patient, Epic"/>
        </patient>
        <dateTime value="2021-07-01T19:42:38Z"/>
        <oralDiet>
          <type>
            <coding>
              <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.34365"/>
              <code value="3"/>
              <display value="Vegetarian"/>
            </coding>
            <text value="Vegetarian"/>
          </type>
          <texture>
            <modifier>
              <coding>
                <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.34442"/>
                <code value="8"/>
                <display value="Pureed"/>
              </coding>
              <text value="Pureed"/>
            </modifier>
          </texture>
          <instruction value="GENERAL DIET PLAN"/>
        </oralDiet>
      </NutritionOrder>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```


13.1.13 Observation.Read (Assessments) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11053-Observation-Read-Assessments-R4.html
Epic API ID | 11053
FHIR version | R4
Categories | Clinical Specialty
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11053

Description

This web service retrieves assessment data from a patient's chart. Assessment data includes structured clinical evaluation results, such as screening instruments (PHQ-9, GAD-7). This web service can only be queried using an Observation Assessment FHIR ID.

Assessments are returned as either a panel observation referencing member items via hasMember, or as item observations that reference the source panel via derivedFrom.

Panel example:

the full PHQ-9 (all questions)

Item example:

the PHQ-9 total score

Observation.Read (Assessments) can be used with organizations on August 2024 or later. The assessments available through this API depend on organizational configuration and assessment usage.

If your application is only interested in reading from SDOH assessments, use

Observation.Read (SDOH Assessments) (R4)

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation
Observation (R4).category | CodeableConcept | | Yes | 1..* | Always returns "survey". Also returns additional assessment-specific categories from the following list of categories: * functional-status * disability-status * cognitive-status * sdoh * activity * social-history
Observation (R4).code | CodeableConcept | | | 1..* | The LOINC code or encoded flowsheet ID that corresponds to the assessment item or panel.
Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an observation value is missing.
Observation (R4).derivedFrom | Reference | | Yes | 0..* | Displays any observations that are related to the returned observation. This is the observation panel for the observation item. Example: PHQ 9 Score -> PHQ 9 Panel
Observation (R4).effectiveDateTime | dateTime | | | 1..1 | The date and time the observation was taken.
Observation (R4).encounter | Reference | | | 1..1 | Reference to an encounter resource that this observation belongs to.
Observation (R4).extension (supportingInfo) (also: extension (supportingInfo)) | Extension | | Yes | 0..* | If the observation is linked to a note written by a clinician, this returns a reference to the DocumentReference resource. URL: http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo
Observation (R4).hasMember | Reference | | Yes | 0..* | The FHIR IDs for any observations related to the returned observation. This includes all documented observation items for an observation panel. Example: PHQ 9 Panel -> PHQ 9 (Question 1), PHQ 9 (Question 2) ...
Observation (R4).issued | instant | | | 1..1 | The date and time the observation was documented.
Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.
Observation (R4).performer | Reference | | Yes | 1..* | Reference to a FHIR resource for who filed the Observation. If the assessment was filed by a patient, this references a Patient resource. Otherwise, it references a Practitioner resource.
Observation (R4).status | code | | | 1..1 | The following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted
Observation (R4).subject | Reference | | | 1..1 | The FHIR ID of the patient associated with the observation.
Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | The value if the data type is a category or custom list. Empty otherwise. Only one of the value* elements is returned.
Observation (R4).valueDateTime | dateTime | | | 0..1 | The value if the data type is date. Only one of the value* elements is returned.
Observation (R4).valueQuantity | Quantity | | | 0..1 | The value if the data type has units, is numeric, or percentile. Only one of the value* elements is returned.
Observation (R4).valueString | string | | | 0..1 | The value if the data type is string or the value doesn't have units. Only one of the value* elements is returned.
Observation (R4).valueTime | time | | | 0..1 | The value if the data type is time. Only one of the value* elements is returned.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation/fUHql2pexBgZr2A7PF0HpdzYa1q5u0GAMp5ZG0-XsbS44
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "fUHql2pexBgZr2A7PF0HpdzYa1q5u0GAMp5ZG0-XsbS44",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "survey",
          "display": "Survey"
        }
      ]
    },
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
          "code": "disability-status",
          "display": "Disability Status"
        }
      ]
    },
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
          "code": "sdoh",
```

```
          "display": "SDOH"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
        "code": "tEu1HmVtV48oFQF5t6QINYA0",
        "display": "Feeling tired or having little energy"
      },
      {
        "system": "http://loinc.org",
        "code": "44254-1",
        "display": "Feeling tired or having little energy"
      }
    ]
  },
  "subject": {
    "reference": "Patient/ehA7q-qQThKVJOpfgyJyEEA3",
    "display": "Patient, Test"
  },
  "encounter": {
    "reference": "Encounter/eqth4R7xj1WnKOQyjsAdXGI2qcA2aVC1vTa9qmky4b2M3",
    "identifier": {
      "use": "usual",
      "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
      "value": "10011718547"
    },
    "display": "Office Visit"
  },
  "effectiveDateTime": "2025-06-25T21:00:00Z",
  "issued": "2025-06-25T21:55:37Z",
  "performer": [
    {
      "reference": "Practitioner/eoGCVeKi.9z.jLKV2dSWufQ3",
      "display": "FHIR, Test Provider"
    }
  ],
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "LA6568-5",
        "display": "Not at all"
      }
    ]
  },
  "derivedFrom": [
    {
      "reference": "Observation/fMPcvF0Xbp5L4vtFDW2jNpVwiTm2PMhkPe7XebMdr34g4"
    }
  ]
}
```


13.1.14 Observation.Read (Family Situation) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10455-Observation-Read-Family-Situation-R4.html
Epic API ID | 10455
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10455

Description

This resource can only be used with healthcare organizations in the Netherlands. This resource cannot be used in any other locales at this time.

The FHIR Observation Family Situation resource searches for patient observations which include all forms of cohabitation that form a recognizable social unit, of people who are or who are not related and who have long-lasting, affective bonds and provide each other with support and care.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | False | Yes | | This is always "social-history".
Observation.code | CodeableConcept | False | | | Hard coded to SNOMED code 36547003
Observation.component | BackboneElement | False | Yes | | Can return the following when reported data is available: Care responsibility: The activities the patient carries out to care for a dependent family member, composed of a valueString (SNOMED 406192005). Child: Record of a patient's child, including their birth date expressed as a valueDateTime. Only appears if a child's birth was documented with Stork (SNOMED 67822003). Family composition: Make up of a patient's household, which can include non-family members (SNOMED 224130005). Mapped to the values in GezinssamenstellingCodelijst. Number of children: The number of children a patient has, expressed as a valueQuantity (SNOMED 224118004). Number of children living at home: The number of children a patient has that live with them, expressed as a valueQuantity (SNOMED 55811000146107).
Observation.dataAbsentReason | CodeableConcept | False | | | Always "unknown," appears only if not data is retrieved.
Observation.status | code | False | | | This value is always "final".
Observation.subject | Reference | False | | | The patient FHIR ID.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/fH6C48xiz47Lxa9k1rwsFPA4
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "fH6C48xiz47Lxa9k1rwsFPA4",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "social-history",
          "display": "Social History"
        }
      ],
      "text": "Social History"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "365470003",
        "display": "Finding of family details and household composition (finding)"
      }
    ],
    "text": "Finding of family details and household composition (finding)"
  },
  "subject": {
    "reference": "Patient/eEvB0tF1AgkK4A2cbxByNpQ3",
    "display": "Mayer, M."
  },
  "note": [
    {
      "text": "Supportive family environment, no concerns\n"
    }
  ],
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "224130005",
            "display": "Household composition (observable entity)"
          }
        ],
        "text": "Household composition (observable entity)"
      },
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "408821002",
            "display": "Lives with partner"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "8871000146101",
            "display": "Lives with underage children"
          },
```

```
          {
            "system": "http://snomed.info/sct",
            "code": "224137008",
            "display": "Lives with parents"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "160756002",
            "display": "Lives with relatives"
          }
        ],
        "text": "spouse, children, mother, other family member"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "224118004",
            "display": "Number of offspring (observable entity)"
          }
        ],
        "text": "Number of offspring (observable entity)"
      },
      "valueQuantity": {
        "value": 2
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "55811000146107",
            "display": "Number of children living at home (observable entity)"
          }
        ],
        "text": "Number of children living at home (observable entity)"
      },
      "valueQuantity": {
        "value": 2
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "406192005",
            "display": "Caregiver behavior (observable entity)"
          }
        ],
        "text": "Caregiver behavior (observable entity)"
      },
      "valueString": "Ensures mother has transportation to appointments"
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "67822003",
            "display": "Child (person)"
          }
        ],
        "text": "Child (person)"
      },
      "valueDateTime": "2016-06-07"
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "67822003",
            "display": "Child (person)"
          }
        ],
        "text": "Child (person)"
      },
      "valueDateTime": "2019-06"
    }
  ]
}
```


13.1.15 Observation.Read (Labor and Delivery) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10300-Observation-Read-Labor-and-Delivery-R4.html
Epic API ID | 10300
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10300

Description

This web service returns delivery documentation for the delivering patient. It includes general information about the delivery encounter as well as specific documentation captured during the delivery. See

code

for details on what delivery documentation can be included.

This web service requires system-level configuration to determine which details within the delivery documentation should be available as documented delivery points.

Without a

code

specified, all documentation that is configured to share will be included in the response, with references to the Observations for each documented delivery point included within the

hasMember

element of the response.

Observation.Search (Newborn Delivery) (R4)

is a complimentary web service that returns delivery documentation for the newborn child.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Observation FHIR ID for the Labor and Delivery observation.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation
Observation (R4).category | CodeableConcept | | Yes | 1..* | This is always "labor-delivery".
Observation (R4).code | CodeableConcept | | | 1..1 | Supported SNOMED codes include: 249160009 (Onset of Second Stage of Labor) 169821004 (Duration of first stage of labor) 169822006 (Duration of second stage of labor) 169823001 (Duration of third stage of labor 249120008 (Onset of labor first stage) 397955000 (Analgesia in labor) 200158002 (Other complications of labor and delivery) 199745000 (Complication occurring during labor and delivery) 364297003 (Lacerations) 443848000 (15 minute Apgar score) 443849008 (20 minute Apgar score) Supported LOINC codes include: 48767-8 (Birth comments) 21112-8 (Birth date) 8293-3 (Chest circumference --inspiration) 89269-5 (Body height @ birth measured 8339-4 (Birth weight measured) 8290-9 (Head occipital-frontal circumference --at birth- by Tape measure) 73771-8 (Birth order) 55281-0 (Number of fetuses) 57075-4 (Newborn delivery information) 56829-5 (Time membranes ruptured) 57069-7 (Preterm labor symptoms) 8722-1 (Anesthesia method) 12116-0 (Cord vessels) 73762-7 (Final route and method of delivery) 75186-7 (Vital status) 73761-9 (Fetal presentation--at birth) 38386-9 (Color of amniotic fluid) 73760-1 (If cesarean, a trial of labor was attempted) 9272-6 (1 minute Apgar score) 9274-2 (5 minute Apgar score) 9271-8 (10 minute Apgar score) Other supported values include: Delivery comments Cord clamping instant Forceps attempted Vacuum attempted Start pushing instant Established respiration instant Placenta instant Breastfeeding initiated instant Decision instant Placental weight Rupture of membranes to delivery Repair packets Abdominal girth Estimated blood loss Augmentation start Cervical ripening instant Contraction start instant Dilation start instant Induction instant Skin to skin start instant Skin to skin end instant Augmentation type Breech type Ripening type Cord blood disposition Cord complication Gases sent Episiotomy Feeding intention Forceps location Induction type Labor identifier Labor Type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Primary reason for induction Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Antenatal steroid Vacuum location Labor antibiotics Adoption type Codings specified in External Concept Mappings are also included.
Observation (R4).component | BackboneElement | | Yes | 1..* | Populated only for patient-entered flowsheets.
Observation (R4).component.code | CodeableConcept | | | 1..1 | Returns the values documented for each component below: Augmentation type Ripening type Cord blood disposition Cord complication Episiotomy Forceps location Induction type Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Vacuum location Adoption type Hide birth mother's info Supported LOINC Codes include: 8722-1 (Anesthesia method) 38386-9 (Color of amniotic fluid) 32406-1 (1 minute Apgar color) 32407-9 (1 minute Apgar heart rate) 32409-5 (1 minute Apgar reflex irritability) 32408-7 (1 minute Apgar muscle tone) 32410-3 (1 minute Apgar respiratory effort) 32411-1 (5 minute Apgar color) 32412-9 (5 minute Apgar heart rate) 32414-5 (5 minute Apgar reflex irritability) 32413-7 (5 minute Apgar muscle tone) 32415-2 (5 minute Apgar respiratory effort) 32401-2 (10 minute Apgar color) 32402-0 (10 minute Apgar heart rate) 32404-6 (10 minute Apgar reflex irritability) 32403-8 (10 minute Apgar muscle tone) 32405-3 (10 minute Apgar respiratory effort) Supported SNOMED codes include: 397955000 (Analgesia in labor) 200158002 (Other complications of labor and delivery) 199745000 (Complication occurring during labor and delivery) 364297003 (Lacerations) 249227004 (15 minute Apgar color) 249223000 (15 minute Apgar heart rate) 249226008 (15 minute Apgar reflex irritability) 249225007 (15 minute Apgar muscle tone) 249224006 (15 minute Apgar respiratory effort) 249227004 (20 minute Apgar color) 249223000 (20 minute Apgar heart rate) 249226008 (20 minute Apgar reflex irritability) 249225007 (20 minute Apgar muscle tone) 249224006 (20 minute Apgar respiratory effort)

Observation (R4).component.valueBoolean | boolean | | | 0..1 | If set to true, the newborn name is not displayed. This is driven off of the adoption status for the newborn.

Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 1..1 | Returns the values documented for each component below: Augmentation type Ripening type Cord blood disposition Cord complication Episiotomy Forceps location Induction type Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Vacuum location Anesthesia method Other complications of labor and delivery Lacerations Rupture of amniotic fluid 1 minute Apgar color 1 minute Apgar heart rate 1 minute Apgar reflex irritability 1 minute Apgar muscle tone 1 minute Apgar respiratory effort 5 minute Apgar color 5 minute Apgar heart rate 5 minute Apgar reflex irritability 5 minute Apgar muscle tone 5 minute Apgar respiratory effort 10 minute Apgar color 10 minute Apgar heart rate 10 minute Apgar reflex irritability 10 minute Apgar muscle tone 10 minute Apgar respiratory effort 15 minute Apgar color 15 minute Apgar heart rate 15 minute Apgar reflex irritability 15 minute Apgar muscle tone 15 minute Apgar respiratory effort 20 minute Apgar color 20 minute Apgar heart rate 20 minute Apgar reflex irritability 20 minute Apgar muscle tone 20 minute Apgar respiratory effort Adoption type

Observation (R4).component.valueString | string | | | 1..1 | Returns analgesia comments or complication comments documented in the delivery documentation.

Observation (R4).dataAbsentReason | CodeableConcept | | | 1..1 | Indicates when documentation has been deleted or was never entered for a specific point within the delivery documentation. If the data has been removed for the delivery point, dataAbsentReason will indicate "Data for specified Observation no longer exists." If the data was never completed for the delivery point, dataAbsentReason will indicate "Specified Observation has never had data."

Observation (R4).effectiveDateTime | dateTime | | | 1..1 | Newborn birth date.

Observation (R4).encounter | Reference | | | 1..1 | The Encounter FHIR ID for the labor and delivery encounter.

Observation (R4).extension (delivery-personnel) (also: extension (delivery-personnel)) | Extension | | Yes | 1..* | Reference to the CareTeam resource, which will export this delivery's personnel. URL: http://open.epic.com/FHIR/StructureDefinition/extension/delivery-personnel

Observation (R4).extension (observation-focusCode) (also: extension (observation-focusCode)) | Extension | | Yes | 1..* | Identifies the relationship between the subject and the focus. This extension is included within the focus element. The relationship will always be "Child" when the relationship is allowed to be printed or shared. URL: http://hl7.org/fhir/StructureDefinition/observation-focusCode

Observation (R4).extension (workflow-reasonCode) (also: extension (workflow-reasonCode)) | Extension | | Yes | 0..* | Supports reasoning for other documentation: Reasons for augmentation Reasons for episiotomy Reasons for induction Reasons for use of forceps Reasons for use of vacuum Reasons for c-section URL: http://hl7.org/fhir/StructureDefinition/workflow-reasonCode

Observation (R4).focus | Reference | | Yes | 1..* | The Patient FHIR ID for the child.

Observation (R4).hasMember | Reference | | Yes | 1..* | Includes the Observation FHIR IDs for the delivery points or documentation associated with the delivery episode.

Observation (R4).id | id | | | | The Observation FHIR ID for the Labor and Delivery observation.

Observation (R4).identifier | Identifier | | Yes | 1..* | The delivery record ID (unique identifier for the birth record).

Observation (R4).issued | instant | | | 1..1 | The instant at which the documentation for the delivery was updated.

Observation (R4).note | Annotation | | Yes | 1..* | Returns strings for the following: Comment about analgesics used during labor Comment about complications occurring during labor

Observation (R4).performer | Reference | | Yes | 0..1 | The user who entered the documentation for the delivery, as a Practitioner FHIR ID. Only included in the response when the code request element is used to filter to a specific delivery point.

Observation (R4).status | code | | | 1..1 | The status of the delivery documentation.

Observation (R4).subject | Reference | | | 1..1 | The Patient FHIR ID.

Observation (R4).valueBoolean | boolean | | | 0..1 | Returns yes/no for the following: Labor antibiotics Forceps attempted Vacuum attempted Preterm labor symptoms

Observation (R4).valueCodeableConcept | CodeableConcept | | | 1..1 | Returns a value for the following elements in the delivery documentation: Breech type Gases sent Feeding intention Labor identifier Labor type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Cord vessels Final route and method of delivery Vital status Fetal presentation--at birth If cesarean, a trial of labor was attempted 1 minute Apgar score 5 minute Apgar score 10 minute Apgar Score 15 minute Apgar score 20 minute Apgar score Primary reason for induction

Observation (R4).valueDateTime | dateTime | | | 1..1 | A date or time associated with the component. Applies to the following component types: Augmentation start Cervical ripening instant Contraction start instant Dilation start instant Induction instant Onset of labor Time membranes ruptured Skin to skin start instant Skin to skin end instant Birth date Breastfeeding initiated instant Cord clamping instant Decision instant Established respiration instant Placenta instant Start pushing instant Dilation complete instant

Observation (R4).valueInteger | integer | | | 0..1 | Returns the count for the following: Number of fetuses - before delivery, this represents the number of fetuses in the pregnancy. After delivery has occurred, however, this may be updated to the number of viable deliveries in the pregnancy. Repair packets Birth position

Observation (R4).valueQuantity | Quantity | | | 1..1 | Returns measurements of the following: Abdominal girth Estimated blood loss Chest circumference --inspiration Body height @ birth measured Birth weight measured Head occipital-frontal circumference at birth - by tape measure Rupture of membranes to delivery Placental weight Duration of first stage of labor Duration of second stage of labor Duration of third stage of labor

Observation (R4).valueString | string | | | 1..1 | Returns the comments below: Delivery comments Birth comments

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation/edJ2RZlQDJIpO9k5GQH2Qij9k1I6C4a2nU9FDpMt1k5WuTVFfb-
    pRfBUACi0t4PLhyF.yrT3G6ocX8Ehyj4LcwA3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "ejm2jerbbtcmhcbmnm-peqwkaxxxwar4iopnrah3qycminmb84gupepkjp0l4tsusnk-1sijmj7opvm7hipxheq3",
  "extension": [
    {
      "valueReference": {
        "reference": "CareTeam/eobjqvjve-hwf.qiewcvjfk366llnygr4axqwgxhs0t83"
      },
      "url": "http://open.epic.com/fhir/StructureDefinition/extension/delivery-personnel"
    }
  ],
  "identifier": [
    {
      "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.728366",
```

```
      "value": "1046176"
    }
  ],
  "status": "preliminary",
  "category": [
    {
      "coding": [
        {
          "system": "http://open.epic.com/fhir/StructureDefinition/observation-category",
          "code": "labor-delivery",
          "display": "labor and delivery"
        }
      ],
      "text": "labor and delivery"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "57075-4",
        "display": "newborn delivery information"
      }
    ],
    "text": "newborn delivery information"
  },
  "subject": {
    "reference": "Patient/et-em40kxoqnnxriczj-gqyhszyyqaqfabrw3-xyq0dy3",
    "display": "zzfhir, fhiry"
  },
  "focus": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
                "code": "CHILD",
                "display": "child"
              }
            ],
            "text": "child"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/observation-focusCode"
        }
      ],
      "reference": "Patient/evrhk.d80nusyo7tggvep02w3dkds3pvx82vbierpy.q3",
      "display": "zzfhir, fhirybaby1"
    }
  ],
  "encounter": {
    "reference": "Encounter/eimo5bsmk.sb4u0hbo9gweort2-t.ij55dl6kdpoeqt03",
    "display": "hospital encounter"
  },
  "effectiveDateTime": "2021-12-29",
  "hasMember": [
    {
      "reference": "Observation/ejm2jerbbtcmhcbmnm-
    peqwkaxxxwar4iopnrah3qycminmb84gupepkjp0l4tsus5mrgqeolfgzem4gogcppya3",
      "display": "birth date"
    },
    {
      "reference": "Observation/ejm2jerbbtcmhcbmnm-
    peqwkaxxxwar4iopnrah3qycminmb84gupepkjp0l4tsusp69g9dkw6z4lpxj0jcm3cg3",
      "display": "vital status"
    }
  ]
}
```


13.1.16 Observation.Read (Newborn Delivery) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10306-Observation-Read-Newborn-Delivery-R4.html
Epic API ID | 10306
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10306

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | An instance of the R4 Observation resource.
Observation.category | CodeableConcept | False | Yes | | Always "newborn-delivery"
Observation.code | CodeableConcept | False | | | Supported SNOMED codes include: 249160009-Onset of Second Stage of Labor 169821004-Duration of first stage of labor 169822006-Duration of second stage of labor 169823001-Duration of third stage of labor 249120008-Onset of labor first stage 397955000-Analgesia in labor 200158002-Other complications of labor and delivery 199745000-Complication occurring during labor and delivery 364297003-Lacerations 443848000-15 minute Apgar score 443849008-20 minute Apgar score Supported LOINC codes include: 48767-8 Birth comments 21112-8 Birth date 8293-3 Chest circumference --inspiration 89269-5 Body height @ birth measured 8339-4 Birth weight measured 8290-9 Head occipital-frontal circumference --at birth- by Tape measure 73771-8 Birth order 55281-0 Number of fetuses 57075-4 Delivery personnel 56829-5 Time membranes ruptured 57069-7 Preterm labor symptoms 8722-1 Anesthesia method 12116-0 Cord vessels 73762-7 Final route and method of delivery 75186-7 Vital status 73761-9 Fetal presentation--at birth 38386-9 Color of amniotic fluid 73760-1 If cesarean, a trial of labor was attempted 9272-6 1 minute Apgar score 9274-2 5 minute Apgar score 9271-8 10 minute Apgar score Other supported values include: Delivery comments Cord clamping instant Forceps attempted Vacuum attempted Start pushing instant Established respiration instant Placenta instant Breastfeeding initiated instant Decision instant Placental weight Rupture of membranes to delivery Repair packets Abdominal girth Estimated blood loss Augmentation start Cervical ripening instant Contraction start instant Dilation start instant Induction instant Skin to skin start instant Skin to skin end instant Augmentation type Breech type Ripening type Cord blood disposition Cord complication Gases sent Episiotomy Feeding intention Forceps location Induction type Labor identifier Labor Type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Antenatal steroid Vacuum location Labor antibiotics Adoption type Codings specified in External Concept Mappings are also included.
Observation.component | BackboneElement | True | Yes | | Populated only for patient-entered flowsheets.
Observation.component.code | CodeableConcept | True | | | Returns the values documented for each component below: Augmentation type Ripening type Cord blood disposition Cord complication Episiotomy Forceps location Induction type Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Vacuum location Adoption type Hide birth mother's info Supported LOINC Codes include: 8722-1 (Anesthesia method) 38386-9 (Color of amniotic fluid) 32406-1 (1 minute Apgar color) 32407-9 (1 minute Apgar heart rate) 32409-5 (1 minute Apgar reflex irritability) 32408-7 (1 minute Apgar muscle tone) 32410-3 (1 minute Apgar respiratory effort) 32411-1 (5 minute Apgar color) 32412-9 (5 minute Apgar heart rate) 32414-5 (5 minute Apgar reflex irritability) 32413-7 (5 minute Apgar muscle tone) 32415-2 (5 minute Apgar respiratory effort) 32401-2 (10 minute Apgar color) 32402-0 (10 minute Apgar heart rate) 32404-6 (10 minute Apgar reflex irritability) 32403-8 (10 minute Apgar muscle tone) 32405-3 (10 minute Apgar respiratory effort) Supported SNOMED codes include: 397955000 (Analgesia in labor) 200158002 (Other complications of labor and delivery) 199745000 (Complication occurring during labor and delivery) 364297003 (Lacerations) 249227004 (15 minute Apgar color) 249223000 (15 minute Apgar heart rate) 249226008 (15 minute Apgar reflex irritability) 249225007 (15 minute Apgar muscle tone) 249224006 (15 minute Apgar respiratory effort) 249227004 (20 minute Apgar color) 249223000 (20 minute Apgar heart rate) 249226008 (20 minute Apgar reflex irritability) 249225007 (20 minute Apgar muscle tone) 249224006 (20 minute Apgar respiratory effort)
Observation.component.valueBoolean | boolean | True | | | Returns true/false for the following in the event of an adoption. If set to true, the newborn name is not displayed. Hide birth mother's info.
Observation.component.valueCodeableConcept | CodeableConcept | True | | | Returns the values documented for each component below: occurring during labor and delivery Augmentation type Ripening type Cord blood disposition Cord complication Episiotomy Forceps location Induction type Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Vacuum location Anesthesia method Other complications of labor and delivery Lacerations Rupture of amniotic fluid 1 minute Apgar color 1 minute Apgar heart rate 1 minute Apgar reflex irritability 1 minute Apgar muscle tone 1 minute Apgar respiratory effort 5 minute Apgar color 5 minute Apgar heart rate 5 minute Apgar reflex irritability 5 minute Apgar muscle tone 5 minute Apgar respiratory effort 10 minute Apgar color 10 minute Apgar heart rate 10 minute Apgar reflex irritability 10 minute Apgar muscle tone 10 minute Apgar respiratory effort 15 minute Apgar color 15 minute Apgar heart rate 15 minute Apgar reflex irritability 15 minute Apgar muscle tone 15 minute Apgar respiratory effort 20 minute Apgar color 20 minute Apgar heart rate 20 minute Apgar reflex irritability 20 minute Apgar muscle tone 20 minute Apgar respiratory effort Adoption type
Observation.component.valueString | string | True | | | Returns analgesia values documented in the Delivery Summary: Analgesia in labor
Observation.extension (also: delivery-personnel) | Extension | True | Yes | | Reference to the CareTeam resource, which will export this delivery's personnel.
Observation.effectiveDateTime | dateTime | True | | | Newborn birth date.
Observation.encounter | Reference | False | | | The newborn FHIR encounter ID.
Observation.focus | Reference | False | Yes | | The patient who delivered the newborn.
Observation.hasMember | Reference | True | Yes | | FHIR observation ID. Appears in place of the value*, component, issued, and performer properties when searching on all data points. SmartData elements values are exported with this element.
Observation.identifier | Identifier | False | Yes | | Delivery record HSB ID (unique identifier for the birth record).
Observation.issued | instant | True | | | Documentation instant.
Observation.note | Annotation | True | Yes | | Returns strings for the following: Comment about analgesics used during labor Comment about complications occurring during labor
Observation.extension (also: observation-focusCode) | Extension | False | Yes | | Identifies the relationship between the subject and the focus. This extension is included within the focus element.
Observation.performer | Reference | True | Yes | | User who entered the documentation.
Observation.status | code | False | | | Delivery Summary documentation is "final" if the Delivery Summary is signed. If the Delivery Summary is not currently signed or is open for addendum, it has a status of "preliminary." If documentation has been blanked out: "cancelled."
Observation.subject | Reference | False | | | The newborn patient
Observation.valueBoolean | boolean | True | | | Returns yes/no for the following: Labor antibiotics Forceps attempted Vacuum attempted Preterm labor symptoms
Observation.valueCodeableConcept | CodeableConcept | True | | | Returns a value for the following elements in Delivery Summary: Breech type Gases sent Feeding intention Labor identifier Labor type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Cord vessels Final route and method of delivery Vital status Fetal presentation--at birth If cesarean, a trial of labor was attempted 1 minute Apgar score 5 minute Apgar score 10 minute Apgar Score 15 minute Apgar score 20 minute Apgar score Primary reason for induction

Observation.valueDateTime | dateTime | True | | | A date or time associated with the component. Applies to the following component types: Augmentation start Cervical ripening instant Contraction start instant Dilation start instant Induction instant Onset of labor first stage Time membranes ruptured Skin to skin start instant Skin to skin end instant Birth date Breastfeeding initiated instant Cord clamping instant Decision instant Established respiration instant Placenta instant Start pushing instant

Observation.valueInteger | integer | True | | | Returns the count for the following: Number of fetuses - before delivery, this represents the number of fetuses in the pregnancy. After delivery has occurred, however, this may be updated to the number of viable deliveries in the pregnancy. Repair packets

Observation.valueQuantity | Quantity | True | | | Returns measurements of the following: Abdominal girth Estimated blood loss Chest circumference --inspiration Body height @ birth measured Birth weight measured Head occipital-frontal circumference --at birth- by Tape measure Rupture of membranes to delivery Placental weight Duration of first stage of labor Duration of second stage of labor Duration of third stage of labor

Observation.valueString | string | True | | | Returns the comments below: Delivery comments Birth comments

Observation.extension (also: workflow-reasonCode) | Extension | True | Yes | | Supports reasoning for other documentation: Reasons for augmentation Reasons for episiotomy Reasons for induction Reasons for use of forceps Reasons for use of vacuum Reasons for c-section

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/en7R12U6kUiuVuOmTEksSaozXV6tELtZGP4Na2n7XjxJ06PsmbXwGOe6KijbZc
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eDKo3D-Zpnp0V1-01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJrz6QEJAM05okUqGRbhBxHEw3",
  "extension": [
    {
      "valueReference": {
        "reference": "CareTeam/e-gJ-GXuSbEa1c2nI.NvFo6fX6aKeDXWg88FeVTN1vi43"
      },
      "url": "http://open.epic.com/FHIR/StructureDefinition/extension/delivery-personnel"
    }
  ],
  "status": "preliminary",
  "category": [
    {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
          "code": "newborn-delivery",
          "display": "Newborn Delivery"
        }
      ],
      "text": "Newborn Delivery"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "57075-4",
        "display": "Newborn delivery information"
      }
    ],
    "text": "Newborn delivery information"
  },
  "subject": {
    "reference": "Patient/evRHk.d80nuSYo7tggVEP02w3DkDs3pVx82VbiERpY.Q3",
    "display": "Cheny, LilaBaby1"
  },
  "focus": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
                "code": "MTH",
                "display": "Mother"
              }
            ],
            "text": "Mother"
          },
          "url": "http://hl7.org/fhir/StructureDefinition/observation-focusCode"
        }
      ],
      "reference": "Patient/eT-EM40KXOqnnxricZJ-gQyHsZYYqAQfaBrw3-xYq0dY3",
      "display": "Cheny, Lila"
    }
  ],
  "encounter": {
    "reference": "Encounter/e60GsidVkWiQC9uoYcT7w0sqcDD1cmSGm-FxakXOiHAQ3",
    "display": "Hospital Encounter"
  },
  "effectiveDateTime": "2021-12-29",
  "hasMember": [
    {
      "reference":
```

```
    "Observation/eDKo3D-Zpnp0V1-01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJrZCKg488O5goVMwYTgGqY1Q3",
      "display": "Birth date"
    },
    {
      "reference":
    "Observation/eDKo3D-Zpnp0V1-01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJreMnminM5UyDhRmLkbdl.6w3",
      "display": "Vital status"
    }
  ]
}
```


13.1.17 Observation.Read (Obstetrics and Gynecology) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10293-Observation-Read-Obstetrics-and-Gynecology-R4.html
Epic API ID | 10293
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10293

Description

This FHIR Observation resource returns obstetric history, gynecology history, measurements, visits, pregnancy episode data, and dating information.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | False | Yes | | This is always "obstetrics-gynecology."
Observation.code | CodeableConcept | False | | | The code or string of one of the following Observation Types: Age at Menarche (63888-2) Age at First Pregnancy (42797-1) Months Breastfeeding ("# of months breastfeeding") Age at Menopause (42802-9) Age at First Live Birth ("Age at first live birth") Gynecology History Comment (89471-7) Number of Fetuses (55281-0) Delivery Date Estimated (11778-8, 11779-6, 53692-0, 11781-2, 57063-0, 57064-8, 53694-6) Fetal Sex US (11882-8) Suspected Fetal Demise ("Suspected Fetal Demise") Prenatal Visits Count (68493-6) Obstetrics Summary Note (89470-9) Gravida (11996-6) Para (11977-6) Term (no code) Preterm (no code) Abortion (11612-9) SAB (11614-5) IAB (11613-7) Ectopic (33065-4) Molar (no code) Living (11638-4) Live Births (11636-8) Multiple (no code) General Perinatal Comment (89470-9) Delivery Record (118215003) Outcome (63893-2) Outcome Date/Time (93857-1) GA (18185-9) Length of Labor Stage 1 (169821004) Length of Labor Stage 2 (169822006) Birth Weight (8339-4) Baby Sex (76689-9) Delivery Method (73762-7) Anesthesia (8722-1) Preterm Labor (57069-7) Living Status – Current (75186-7) Apgar 1 (9272-6) Apgar 5 (9274-2) Baby Name (no code) Labor Complications (199745000) Additional Complications (200158002) Delivery Location (169812000) Pregravid Weight (8348-5) Pregravid Height (8302-2)
Observation.component | BackboneElement | False | Yes | | Returns the value based on the corresponding components: Components of of Fetuses Fetal chorion Fetal amnion Number of fetuses code Components of Estimated Delivery Date Last menstrual period date LMP precision On birth control. Estimated date of conception Cycle lngth Luteal length EDD basis event date GA at event. Either this or days to EDD from event is included, but not both. Days to EDD from event. Either this or GA at event is included, but not both. Components of Sex of Fetus A component is added for each fetal sex that has been documented.
Components of of Fetuses | Fetal chorion
Fetal amnion
Number of fetuses code
Components of Estimated Delivery Date | Last menstrual period date
LMP precision
On birth control.
Estimated date of conception
Cycle lngth
Luteal length
EDD basis event date
GA at event. Either this or days to EDD from event is included, but not both.
Days to EDD from event. Either this or GA at event is included, but not both.
Components of Sex of Fetus | A component is added for each fetal sex that has been documented.
Observation.component.code | CodeableConcept | False | | | Fetal Chorion - 92568-5 Fetal Amnion Last menstrual period start date - 8665-2 LMP Precision On Birth Control Conception date - 33067-0 Cycle Length Luteal Length EDD Basis Event Date GA at Event Days to EDD from Event Fetal Sex US - 11882-8 (Fetal Sex US)
Observation.component.valueBoolean | boolean | True | | | On Birth Control - Whether the patient was on birth control at the time of this LMP.
Observation.component.valueCodeableConcept | CodeableConcept | False | | | Fetal Chorion [1] Monochorionic: LA30047-7 [2] Dichorionic: LA30048-5 [3] Trichorionic: (no LOINC code) [99] Other: (no LOINC code) Value derived from I HSB 35571 – first part Fetal Amnion [1]

Monoamniotic [2] Diamniotic [3] Triamniotic [99] Other Value derived from I HSB 35571 – second part. Number of Fetuses Code The number of fetuses and chorionicity/amnionicity expressed as SNOMED codes. LMP Precision How precise the associated LMP date is, according to the patient. Sex of Individual Fetus The fetal sex. Surprise (I HSB 35574 code) Female (LA21135-1) Male (LA21134-4) See overview (I HSB 35574 code) The mapped SNOMED code for each surgical operation note anesthesia. Unmapped values are an Epic-based category code with the category value and display name. The living status at birth for the delivery. The current living status for the delivery. The mapped SNOMED code for each complication occurring during labor and delivery. Unmapped values show the Epic-based category code with the category value and display name. Uses EDG to SNOMED mappings for other complications of labor and delivery.

Fetal Chorion | [1] Monochorionic: LA30047-7 [2] Dichorionic: LA30048-5 [3] Trichorionic: (no LOINC code) [99] Other: (no LOINC code) Value derived from I HSB 35571 – first part

Fetal Amnion | [1] Monoamniotic [2] Diamniotic [3] Triamniotic [99] Other Value derived from I HSB 35571 – second part.

Number of Fetuses Code | The number of fetuses and chorionicity/amnionicity expressed as SNOMED codes.

LMP Precision | How precise the associated LMP date is, according to the patient.

Sex of Individual Fetus | The fetal sex. Surprise (I HSB 35574 code) Female (LA21135-1) Male (LA21134-4) See overview (I HSB 35574 code)

Observation.component.valueDateTime | String | False | | | Estimated Date of Conception - The estimated date of conception. EDD Basis Event Date - The date of the event. This, combined with either the “GA at Event” component or the “Days to EDD from Event” component, is used to calculate the EDD. For example, this could be the date than an ultrasound was performed. Last Menstrual Period Date – The first day of the patient’s last menstrual period.

Observation.component.valueInteger | Int32 | True | | | This is the order of the delivery within OB History, which is not necessarily the same as calculated “birth order” based on outcome date/time.

Observation.component.valueQuantity | Quantity | False | | | Cycle Length - The cycle length in number of days. Luteal Length - The luteal length in number of days. Gestational Age at Event - The Gestational age the patient was at the time of the event, represented in number of days. This, combined with the event date, is used to calculate the EDD.For example, this could be the gestational age determined during an ultrasound. Gestational Age is represented here in number of days Days to EDD from Event - The defined number of days to the EDD from the time of the event. This, combined with the event date, is used to calculate the EDD.

Observation.derivedFrom | Reference | False | Yes | | If this EDD observation is the main selected EDD for the pregnancy, then derivedFrom contains a reference to every EDD event that has been documented on that episode, regardless of whether it was selected as the main EDD to use for the pregnancy.

Observation.effectiveDateTime | dateTime | False | | | This is the outcome date/time of the delivery.

Observation.effectivePeriod | Period | False | | | The date range over which there have been prenatal visits.

Observation.focus | Reference | False | Yes | | The reference ID of one of the following: The pregnancy episode. The Delivery Resource Observation.

Observation.hasMember | Reference | False | Yes | | A hasMember reference is included for each supported data point that contains data.

Observation.interpretation | CodeableConcept | False | Yes | | Populated using SNOMED code 900000000000548007 (Preferred) when the entry represents the working expected delivery date. Otherwise, this property is absent.

Observation.issued | instant | False | | | Returns the instant of one of the following: When this observation was last edited. The Contact date of the eptDAT (which is a history contact). The instant of entry for this EDD.

Observation.method | CodeableConcept | False | | | The basis for this EDD. For example, the EDD could be based on an ultrasound.

Observation.note | Annotation | False | Yes | | The comment for one of the following: The comment associated with the EDD. Free-text birth comment for the delivery. Living status comment. Delivery location comment.

Observation.performer | Reference | False | Yes | | Returns the clinician's name for one of the following: The clinician who documented the observation. The delivery clinician.

Observation.status | code | False | | | Status is set to “final” except for special logic for the EDD: If the EDD is attached to the current pregnancy and the patient is marked as pregnant, this transmits as “preliminary.” Otherwise (the EDD is attached to a pregnancy and that pregnancy has been resolved or the patient is not pregnant), this transmits as “final.” If the pregnancy is deleted, then the Status is entered-in-error.

Observation.subject | Reference | False | | | The patient

Observation.extension (also: supportingInfo) | Extension | False | Yes | | The L&D Observation for this delivery. This is included only for real delivery records; historical delivery records do not have this focus.

Observation.valueBoolean | boolean | True | | | Yes/No answer to whether there is a suspected fetal demise (I HSB 35585). The value is True when Preterm Labor is documented as True. The value is False when Preterm Labor is documented as false.

Observation.valueCodeableConcept | CodeableConcept | False | | | Returns the value based on one of the following mappings: The mapped SNOMED code for outcome of pregnancy. Uses mapping table to map the documented sex to M(ale)/F(emale)/Un(known). The mapped SNOMED code for obstetric delivery method. Unmapped values are an Epic category-list-based code. Uses SNOMED mapping tables and LSD settings for the fetal vital status. All unmapped values are considered deceased.

Observation.valueDateTime | dateTime | False | | | The date and time of one of the following: The estimated delivery date. The date and time of the delivery.

Observation.valueInteger | integer | True | | | Returns the count of one of the following: Prenatal visits for the patient. Pregnancies (gravida) for the patient. The parity value for the patient. The number of term pregnancies for the patient. The number of preterm pregnancies for the patient. The total number of abortions for the patient. The number of spontaneous abortions for the patient. The number of induced abortions for the patient. The number of ectopic pregnancies for the patient. The number of molar pregnancies for the patient. The number of children that are still living for the patient. The number of live births for the patient. The number of pregnancies that had multiple fetuses. Apgar 1M value. Apgar 5M value.

Observation.valueQuantity | Quantity | False | | | Returns one of the following values in its corresponding unit as defined below: Age at Menarche (in years) Age at First Pregnancy (in years) Months Breastfeeding (in months) Age at Menopause (in years) Age at First Live Birth (in years) Number of Fetuses (Before delivery, this represents the number of fetuses in the pregnancy. After delivery has occurred, however, this might be updated to the number of viable deliveries in the pregnancy (I HSB 35061).) Pregravid Weight (in ounces) Pregravid Height (in inches) Gestational Age (in days) First Stage of Labor (in minutes) Second Stage of Labor (in minutes) Birth Weight (in ounces)

Observation.valueString | string | False | | | Returns one of the following comments written by the clinician: The gynecology history comment. The general obstetrics history summary comment. Or name information: Name of newborn. Name of the delivery location.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24t0bwYrD.nOaXEndldp2kuJ3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24t0bwYrD.nOaXEndldp2kuJ3",
  "status": "final",
  "category": [
```

```
  {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
        "code": "obstetrics-gynecology",
        "display": "Obstetrics and Gynecology"
      }
    ],
    "text": "Obstetrics and Gynecology"
  }
],
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "118215003",
      "display": "Delivery Finding"
    }
  ],
  "text": "Delivery Finding"
},
"subject": {
  "reference": "Patient/ePOxt68XCxQVi3.zsQknn455oPwx6KFx2tVyK0BN-Azo3",
  "display": "Hsu, Trinh Malena"
},
"focus": [
  {
    "reference": "EpisodeOfCare/eY.v47.qYCe7-phtYEkaQUQ3",
    "display": "Pregnancy (Historical)"
  }
],
"effectiveDateTime": "2019-05-23",
"issued": "2021-10-19T05:00:00Z",
"performer": [
  {
    "reference": "Practitioner/eR21-ui08.7aMdcQcFuMzfw3",
    "display": "Physician Obstetrics, MD"
  }
],
"note": [
  {
    "text": "standard birth"
  }
],
"hasMember": [
  {
    "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24s1sYYO5UrBmFj0Q1U2JRrB3",
    "display": "Outcome of pregnancy"
  },
  {
    "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24u6e9qD-
  xXJsPjGD1QRe10elAsEx8VftDiF.K.i6SBpRA3",
    "display": "Date and time of obstetric delivery"
  },
  {
    "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24udYdtH8I4WjCUyfYVPQ6Nk3",
    "display": "Gestational age"
  },
  {
    "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24uOTCSUvtjcIq949ZkeucNe3",
    "display": "Duration of first stage of labor"
  },
  {
    "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24tBEUAx0QkVebC4bmE3xJGa3",
    "display": "Duration of second stage of labor"
  },
  {
    "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24ueVy5rqzIrIbsQGWiTIEdg3",
    "display": "Body weight at birth"
  },
  {
    "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24ugnLJKfpF5k0I9l9BxY5hK3",
    "display": "Baby sex"
  },
  {
    "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24unValHhHFBJxLTMP2Y6VnV3",
    "display": "Obstetric delivery method"
  },
  {
    "reference":
  "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24sTTQC0lPSCrvfH0Cb.KMp9SboWGuF0ptkUyJuqBRhO2A3",
    "display": "Surgical operation note anesthesia"
  },
  {
    "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24v8FtYsYYjq5JguxP-kkZ-f3",
    "display": "Preterm labor symptoms"
  },
  {
    "reference":
```

```
    "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24tS5IFHmY2LfTkmU36Qus.yfW9456bPa00Seu5clBqzew3",
      "display": "Vital status"
    },
    {
      "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24sfCRCoF85OlIUeKPKjpqZx3",
      "display": "APGAR 1M"
    },
    {
      "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24vc1zFJP0XG6wupc5ca.reh3",
      "display": "APGAR 5M"
    },
    {
      "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24us.iHuqdbF4iQYj03lLbjp3",
      "display": "Baby name"
    },
    {
      "reference":
    "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24tdlT8GwaualdJWhHdmuZPS289ZFZUap9Emmj9Na.kQlw3",
      "display": "Complication occurring during labor and delivery"
    },
    {
      "reference":
    "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24vwf5g.iwKFBZfDfqjoLcdibAiNu3i7h-bwPn62P0SbAw3",
      "display": "Other complications of labor and delivery"
    },
    {
      "reference": "Observation/eHphidnV8ADQOleJUxTHMwpxKoeBzf7qwInSq4xS-24uBpdIsbOrNMJbdXJTrrZyl3",
      "display": "Place of birth"
    }
  ],
  "component": [
    {
      "code": {
        "text": "Delivery order"
      },
      "valueInteger": 1
    }
  ]
}
```


13.1.18 Observation.Read (Periodontal) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10070-Observation-Read-Periodontal-R4.html
Epic API ID | 10070
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10070

Description

The periodontal Observation sub-resource can return all of the periodontal readings documented on a patient's tooth during a periodontal exam. They include probing depths, gingival margins, and clinical attachment levels.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | An instance of the R4 Observation resource.
Observation.bodySite | CodeableConcept | True | | | CodeableConcept value.
Observation.category | CodeableConcept | True | Yes | | CodeableConcept value.
Observation.code | CodeableConcept | False | | | CodeableConcept value.
Observation.component | BackboneElement | False | Yes | | Populated only for patient-entered flowsheets.
Observation.component.code | CodeableConcept | False | | | CodeableConcept value.
Observation.component.interpretation | CodeableConcept | True | Yes | | CodeableConcept value.
Observation.component.referenceRange | BackboneElement | True | Yes
Observation.component.valueQuantity | Quantity | True | | | Quantity value.

Observation.effectiveDateTime | dateTime | True
Observation.encounter | Reference | True | | | A FHIR reference.
Observation.focus | Reference | True | Yes | | A FHIR reference.
Observation.performer | Reference | True | Yes | | A FHIR reference.
Observation.status | code | False
Observation.subject | Reference | True | | | A FHIR reference.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/ e6fALeOD7ffdnkI-
    BibQjvi.XAIz1Yz8AAoqPeeEJpiSrPa32OUUT8ea7W6yvmrlecv341jOJaSsCNaxFyGkJMw3
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
<id value="e6fALeOD7ffdnkI-BibQjvi.XAIz1Yz8AAoqPeeEJpiSrPa32OUUT8ea7W6yvmrlecv341jOJaSsCNaxFyGkJMw3" />
<status value="final" />
<category>
<coding>
<system value="http://open.epic.com/FHIR/StructureDefinition/observation-category" />
<code value="periodontal" />
<display value="Periodontal" />
</coding>
<text value="Periodontal" />
</category>
<code>
<coding>
<system value="urn:oid:2.16.840.1.113883.6.96" />
<code value="307536006" />
</coding>
<text value="Examination of periodontium (procedure)" />
</code>
<subject>
<reference value="Patient/e6DDE5.4tu-jjWXY8CtuVKw3/">
<display value="Lacroix, Diego " />
</subject>
<focus>
<reference value="BodyStructure/ec.bWXnOuxgyRAN8CLHCBxWlQPYNAfGid39k5uR9FFnY3" />
<type value="BodyStructure" />
</focus>
<encounter>
<reference value="Encounter/ejsLjQ9QeqL3FtFg09AuzXnouJzKaY8k1vzQkzZKyww43" />
<type value="Encounter" />
</encounter>
<effectiveDateTime value="2021-01-05" />
<performer>
<reference value="Practitioner/eNzt.GkOzCBShLeRDkLPGfw3" />
<type value="Practitioner" />
<display value="Gordon Lofgren, DDS" />
</performer>
<bodySite>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.3.866976.94010" />
<code value="1" />
</coding>
<text value="Tooth (Universal): 1" />
</bodySite>
<component>
<code>
<coding>
<display value="Probing Depth - Facial Distal" />
</coding>
</code>
<valueQuantity>
<value value="2" />
<unit value="mm" />
</valueQuantity>
<interpretation>
<coding>
<system value="urn:oid:2.16.840.1.113883.5.83" />
<code value="N" />
<display value="Normal" />
</coding>
<text value="Normal" />
</interpretation>
<referenceRange>
<high>
<value value="3" />
<unit value="mm" />
</high>
</referenceRange>
<component>
<code>
<coding>
<display value="Clinical Attachment Level - Lingual Distal" />
</coding>
</code>
```

```
<valueQuantity>
<value value="2" />
<unit value="mm" />
</valueQuantity>
</component>
</Observation>
```

13.1.19 Observation.Search (Assessments) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11052-Observation-Search-Assessments-R4.html
Epic API ID | 11052
FHIR version | R4
Categories | Clinical Specialty
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11052

Description

This web service retrieves assessment data from a patient's chart. Assessment data includes structured clinical evaluation results, such as screening instruments (PHQ-9, GAD-7).

Assessments are returned as either a panel observation referencing member items via hasMember, or as item observations that reference the source panel via derivedFrom.

Panel example:

the full PHQ-9 (all questions)

Item example:

the PHQ-9 total score

The following types of assessment data within Epic can be returned by a search with this API:

* Social Determinant of Health (SDOH)

* Functional Status

* Disability Status

* Mental/Cognitive Status

* Physical Activity

* Alcohol Use

* Substance Use

Your application can also search for specific assessments for a patient using the corresponding panel or item LOINC codes.

Observation.Search (Assessments) can be used with organizations on August 2024 or later. The assessments available through this API depend on organizational configuration and assessment usage.

If your application is only interested in searching for SDOH assessments, use

Observation.Search (SDOH Assessments) (R4)

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Conditional | Yes | | Use "survey" to search for all assessments. Otherwise, use one of the following values to search for a specific type of assessments: * functional-status * disability-status * cognitive-status * sdoh * activity * social-history Note: Category assignments are subjective and may differ across organizations. When possible, searching by a specific code provides more consistent results. Alcohol Use and Substance use can be found by searching via social-history.
code | token | Conditional | Yes | | The Observation identifier. Either category or code search parameters must be provided, but not both. The format for the code parameter is <system>|<code>. The accepted <code> values are listed below. Note: The <system> value is optional in each case, but providing it can improve performance. Use case: Search for results by LOINC code System: http://loinc.org Code: <LOINC code>
date | date | Optional | Yes | | The date or date range when the observation was taken.
patient | reference | Conditional | Yes | | Reference to a patient resource the observation is about. Either this element or subject must be provided. If both are provided, they must match.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
combo-code | token | Optional | Yes | | The LOINC code that corresponds to the measured assessment.
combo-data-absent-reason | token | Optional | Yes | | The reason an observation value is missing.
combo-value-concept | token | Optional | Yes | | CodeableConcept value associated with the assessment
data-absent-reason | token | Optional | Yes | | The reason an observation value is missing.
derived-from | String | Optional | Yes | | An observation that is related to the returned observation.
has-member | reference | Optional | Yes | | Panel members the observation contains.
performer | reference | Optional | Yes | | Reference to whomever performed the observation.
status | token | Optional | Yes | | The status of the observation. Possible values include: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted
subject | reference | Optional | Yes | | Reference to a patient resource the observation is about. Either this element or patient must be provided. If both are provided, they must match.
value-concept | token | Optional | Yes | | CodeableConcept value associated with the assessment
value-string | string | Optional | Yes | | String value associated with the assessment

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation
Observation (R4).category | CodeableConcept | | Yes | 1..* | Always returns "survey". Also returns additional assessment-specific categories from the following list of categories: * functional-status * disability-status * cognitive-status * sdoh * activity * social-history
Observation (R4).code | CodeableConcept | | | 1..* | The LOINC code or encoded flowsheet ID that corresponds to the assessment item or panel.
Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an observation value is missing.
Observation (R4).derivedFrom | Reference | | Yes | 0..* | Displays any observations that are related to the returned observation. This is the observation panel for the observation item. Example: PHQ 9 Score -> PHQ 9 Panel
Observation (R4).effectiveDateTime | dateTime | | | 1..1 | The date and time the observation was taken.
Observation (R4).encounter | Reference | | | 1..1 | Reference to an encounter resource that this observation belongs to.
Observation (R4).extension (supportingInfo) (also: extension (supportingInfo)) | Extension | | Yes | 0..* | If the observation is linked to a note written by a clinician, this returns a reference to the DocumentReference resource. URL: http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo
Observation (R4).hasMember | Reference | | Yes | 0..* | The FHIR IDs for any observations related to the returned observation. This includes all documented observation items for an observation panel. Example: PHQ 9 Panel -> PHQ 9 (Question 1), PHQ 9 (Question 2) ...
Observation (R4).issued | instant | | | 1..1 | The date and time the observation was documented.
Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.
Observation (R4).performer | Reference | | Yes | 1..* | Reference to a FHIR resource for who filed the Observation. If the assessment was filed by a patient, this references a Patient resource. Otherwise, it references a Practitioner resource.
Observation (R4).status | code | | | 1..1 | The following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted
Observation (R4).subject | Reference | | | 1..1 | The FHIR ID of the patient associated with the observation.
Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | The value if the data type is a category or custom list. Empty otherwise. Only one of the value* elements is returned.
Observation (R4).valueDateTime | dateTime | | | 0..1 | The value if the data type is date. Only one of the value* elements is returned.
Observation (R4).valueQuantity | Quantity | | | 0..1 | The value if the data type has units, is numeric, or percentile. Only one of the value* elements is returned.
Observation (R4).valueString | string | | | 0..1 | The value if the data type is string or the value doesn't have units. Only one of the value* elements is returned.
Observation (R4).valueTime | time | | | 0..1 | The value if the data type is time. Only one of the value* elements is returned.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation?patient=emhgWtSiGmquOpTKbBQtPFQ3&category=survey
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 4,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?category=survey&patient=emhgWtSiGmquOpTKbBQtPFQ3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/eF.LUIbOTJ9SNpETeKMwESxW-
    JIHOziIE82tBxFetstZTwmr92EKoV.11Hvab66Zd6PpwIFV9QBV9a2FP35fWag3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eF.LUIbOTJ9SNpETeKMwESxW-
    JIHOziIE82tBxFetstZTwmr92EKoV.11Hvab66Zd6PpwIFV9QBV9a2FP35fWag3",
      "resource": {
        "resourceType": "Observation",
        "id": "eF.LUIbOTJ9SNpETeKMwESxW-JIHOziIE82tBxFetstZTwmr92EKoV.11Hvab66Zd6PpwIFV9QBV9a2FP35fWag3",
```

```
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "survey",
            "display": "Survey"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
            "code": "disability-status",
            "display": "Disability Status"
          }
        ]
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
          "code": "tEu4HmVtV48oFUIw46QINYA0",
          "display": "How soon after you wake up do/did you smoke your first cigarette?"
        },
        {
          "system": "http://loinc.org",
          "code": "63636-5",
          "display": "How soon after you wake up do/did you smoke your first cigarette?"
        }
      ]
    },
    "subject": {
      "reference": "Patient/e2ioq1uJXXxLIHq-8Wx4WHw3",
      "display": "Patient, Test"
    },
    "encounter": {
      "reference": "Encounter/eqth4R7xj1WnKOQyjsAdXGI2qcA2aVC1vTa9qmky4b2M3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10011718547"
      },
      "display": "Office Visit"
    },
    "effectiveDateTime": "2025-06-25T21:00:00Z",
    "issued": "2025-06-25T21:55:37Z",
    "performer": [
      {
        "reference": "Practitioner/eoGCVeKi.9z.jLKV2dSWufQ3",
        "display": "FHIR, Test Provider"
      }
    ],
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "LA15792-7",
          "display": "Within 5 minutes (3 points)"
        }
      ]
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eF.LUIbOTJ9SNpETeKMwESxW-
JIHOziIE82tBxFetstaKOeav08Cs4az27cbpTymPvMhMZmeD1utIAja747L0qA3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eF.LUIbOTJ9SNpETeKMwESxW-
JIHOziIE82tBxFetstaKOeav08Cs4az27cbpTymPvMhMZmeD1utIAja747L0qA3",
  "resource": {
    "resourceType": "Observation",
    "id": "eF.LUIbOTJ9SNpETeKMwESxW-JIHOziIE82tBxFetstaKOeav08Cs4az27cbpTymPvMhMZmeD1utIAja747L0qA3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
```

```
            "code": "survey",
            "display": "Survey"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
            "code": "functional-status",
            "display": "Functional Status"
          }
        ]
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
          "code": "tijeKcVSPxJBt4VeWqwfuhQ0",
          "display": "Which cigarette would, or did, you hate most to give up?"
        },
        {
          "system": "http://loinc.org",
          "code": "63639-9",
          "display": "Which cigarette would you hate most to give up?"
        }
      ]
    },
    "subject": {
      "reference": "Patient/e2ioq1uJXXxLIHq-8Wx4WHw3",
      "display": "Patient, Test"
    },
    "encounter": {
      "reference": "Encounter/eqth4R7xj1WnKOQyjsAdXGI2qcA2aVC1vTa9qmky4b2M3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10011718547"
      },
      "display": "Office Visit"
    },
    "effectiveDateTime": "2025-06-25T21:00:00Z",
    "issued": "2025-06-25T21:55:37Z",
    "performer": [
      {
        "reference": "Practitioner/eoGCVeKi.9z.jLKV2dSWufQ3",
        "display": "FHIR, Test Provider"
      }
    ],
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "LA15796-8",
          "display": "The first one in the morning (1 point)"
        }
      ]
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eF.LUIbOTJ9SNpETeKMwESxW-
JIHOziIE82tBxFetstZlug0sq3WjyjPta1WYOMc0GcHoxbw5WIiNH.4dM-2Euw3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eF.LUIbOTJ9SNpETeKMwESxW-
JIHOziIE82tBxFetstZlug0sq3WjyjPta1WYOMc0GcHoxbw5WIiNH.4dM-2Euw3",
  "resource": {
    "resourceType": "Observation",
    "id": "eF.LUIbOTJ9SNpETeKMwESxW-JIHOziIE82tBxFetstZlug0sq3WjyjPta1WYOMc0GcHoxbw5WIiNH.4dM-2Euw3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "survey",
            "display": "Survey"
          }
        ]
      },
      {
```

```
        "coding": [
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
            "code": "cognitive-status",
            "display": "Cognitive Status"
          }
        ]
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
          "code": "eEu4HmVtV48RFULw46QINYA0",
          "display": "How many cigarettes per day do, or did, you smoke"
        },
        {
          "system": "http://loinc.org",
          "code": "63640-7",
          "display": "How many cigarettes per day do, or did, you smoke"
        }
      ]
    },
    "subject": {
      "reference": "Patient/e2ioq1uJXXxLIHq-8Wx4WHw3",
      "display": "Patient, Test"
    },
    "encounter": {
      "reference": "Encounter/eqth4R7xj1WnKOQyjsAdXGI2qcA2aVC1vTa9qmky4b2M3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10011718547"
      },
      "display": "Office Visit"
    },
    "effectiveDateTime": "2025-06-25T21:00:00Z",
    "issued": "2025-06-25T21:55:37Z",
    "performer": [
      {
        "reference": "Practitioner/eoGCVeKi.9z.jLKV2dSWufQ3",
        "display": "FHIR, Test Provider"
      }
    ],
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "LA15914-7",
          "display": "21-30 (2 points)"
        }
      ]
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eF.LUIbOTJ9SNpETeKMwESxW-
JIHOziIE82tBxFetstasQOGQWzehl8RxZ4aXWB.6POGrdv7Se.Qu9XY8czWE7A3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eF.LUIbOTJ9SNpETeKMwESxW-
JIHOziIE82tBxFetstasQOGQWzehl8RxZ4aXWB.6POGrdv7Se.Qu9XY8czWE7A3",
  "resource": {
    "resourceType": "Observation",
    "id": "eF.LUIbOTJ9SNpETeKMwESxW-JIHOziIE82tBxFetstasQOGQWzehl8RxZ4aXWB.6POGrdv7Se.Qu9XY8czWE7A3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "survey",
            "display": "Survey"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-category",
            "code": "sdoh",
            "display": "SDOH"
          }
```

```
            ]
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
              "code": "REu4HmVtV93RBCLw46QINYA0",
              "display": "Do/Did you find it difficult to refrain from smoking in places where it is forbidden, e.g., in
    church, at the library, in a cinema, etc.?"
            },
            {
              "system": "http://loinc.org",
              "code": "63637-3",
              "display": "Do/Did you find it difficult to refrain from smoking in places where it is forbidden, e.g., in
    church, at the library, in a cinema, etc.?"
            }
          ]
        },
        "encounter": {
          "reference": "Encounter/eqth4R7xj1WnKOQyjsAdXGI2qcA2aVC1vTa9qmky4b2M3",
          "identifier": {
            "use": "usual",
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
            "value": "10011718547"
          },
          "display": "Office Visit"
        },
        "effectiveDateTime": "2025-06-25T21:00:00Z",
        "issued": "2025-06-25T21:55:37Z",
        "performer": [
          {
            "reference": "Practitioner/eoGCVeKi.9z.jLKV2dSWufQ3",
            "display": "FHIR, Test Provider"
          }
        ],
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "LA15798-4",
              "display": "Yes (1 point)"
            }
          ]
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


13.1.20 Observation.Search (Family Situation) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10456-Observation-Search-Family-Situation-R4.html
Epic API ID | 10456
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?category={category}&code={code}&issued={issued}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10456

Description

This resource can only be used with healthcare organizations in the Netherlands. This resource cannot be used in any other locales at this time.

The FHIR Observation Family Situation resource searches for patient observations which include all forms of cohabitation that form a recognizable social unit, of people who are or who are not related and who have long-lasting, affective bonds and provide each other with support and care.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | True | | | Use "social-history" to search for family situation observations.
code | token | True | | | SNOMED code 365470003 for family situation observation.

issued | date | True | | | The date ranges the observation is obtained.
patient | reference | Either this or subject must be provided. | | | Reference to a patient resource.
subject | reference | Either this or patient must be provided | | | Reference to a patient resource.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Unsupported
combo-code | token | True | | | The code of the observation type or component type
combo-data-absent-reason | token | True | | | The reason why the expected value in the element Observation.value[x] or Observation.component.value[x] is missing.
combo-value-concept | token | True | | | The value or component value of the observation, if available
component-code | token | True | | | The component code of the observation type
component-data-absent-reason | token | True | | | The reason why the expected value in the element Observation.component.value[x] is missing.
component-value-concept | token | True | | | The value of the component observation, if available.
data-absent-reason | token | True | | | Unsupported
date | date | True | | | Unsupported
derived-from | String | True | | | Unsupported
device | String | True | | | Unsupported
encounter | reference | True | | | Unsupported
focus | reference | True | | | Unsupported
has-member | reference | True | | | Unsupported
identifier | token | True | | | Unsupported
method | token | True | | | Unsupported
part-of | String | True | | | Unsupported
performer | reference | True | | | Unsupported
specimen | String | True | | | Unsupported
status | token | True | | | The status of the observation
value-concept | token | True | | | Unsupported
value-date | date | True | | | Unsupported
value-string | string | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | This is always "social-history".
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | Hard coded to SNOMED code 36547003
Bundle.entry.resource.Observation.component | BackboneElement | False | Yes | | Can return the following when reported data is available: Care responsibility: The activities the patient carries out to care for a dependent family member, composed of a valueString (SNOMED 406192005). Child: Record of a patient's child, including their birth date expressed as a valueDateTime. Only appears if a child's birth was documented with Stork (SNOMED 67822003). Family composition: Make up of a patient's household, which can include non-family members (SNOMED 224130005). Mapped to the values in GezinssamenstellingCodelijst. Number of children: The number of children a patient has, expressed as a valueQuantity (SNOMED 224118004). Number of children living at home: The number of children a patient has that live with them, expressed as a valueQuantity (SNOMED 55811000146107).
Bundle.entry.resource.Observation.dataAbsentReason | CodeableConcept | False | | | Always "unknown," appears only if not data is retrieved.
Bundle.entry.resource.Observation.note | Annotation | False | Yes | | Comments on family situation.
Bundle.entry.resource.Observation.status | code | False | | | This value is always "final".
Bundle.entry.resource.Observation.subject | Reference | False | | | The patient FHIR ID.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?code=http://snomed.info/sct|365470003&patient=eEvB0tF1AgkK4A2cbxByNpQ3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?code=http://snomed.info/sct%7C365470003&patient=eEvB0tF1
    AgkK4A2cbxByNpQ3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/fH6C48xiz47Lxa9k1rwsFPA4"
        }
      ],
```

```
"fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/fH6C48xiz47Lxa9k1rwsFPA4",
"resource": {
  "resourceType": "Observation",
  "id": "fH6C48xiz47Lxa9k1rwsFPA4",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "social-history",
          "display": "Social History"
        }
      ],
      "text": "Social History"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "365470003",
        "display": "Finding of family details and household composition (finding)"
      }
    ],
    "text": "Finding of family details and household composition (finding)"
  },
  "subject": {
    "reference": "Patient/eEvB0tF1AgkK4A2cbxByNpQ3",
    "display": "Mayer, M."
  },
  "note": [
    {
      "text": "Supportive family environment, no concerns\n"
    }
  ],
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "224130005",
            "display": "Household composition (observable entity)"
          }
        ],
        "text": "Household composition (observable entity)"
      },
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "408821002",
            "display": "Lives with partner"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "8871000146101",
            "display": "Lives with underage children"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "224137008",
            "display": "Lives with parents"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "160756002",
            "display": "Lives with relatives"
          }
        ],
        "text": "spouse, children, mother, other family member"
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "224118004",
            "display": "Number of offspring (observable entity)"
          }
        ],
        "text": "Number of offspring (observable entity)"
      },
      "valueQuantity": {
        "value": 2
      }
    },
```

```
          {
            "code": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "55811000146107",
                  "display": "Number of children living at home (observable entity)"
                }
              ],
              "text": "Number of children living at home (observable entity)"
            },
            "valueQuantity": {
              "value": 2
            }
          },
          {
            "code": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "406192005",
                  "display": "Caregiver behavior (observable entity)"
                }
              ],
              "text": "Caregiver behavior (observable entity)"
            },
            "valueString": "Ensures mother has transportation to appointments"
          },
          {
            "code": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "67822003",
                  "display": "Child (person)"
                }
              ],
              "text": "Child (person)"
            },
            "valueDateTime": "2016-06-07"
          },
          {
            "code": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "67822003",
                  "display": "Child (person)"
                }
              ],
              "text": "Child (person)"
            },
            "valueDateTime": "2019-06"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


13.1.21 Observation.Search (Labor and Delivery) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10301-Observation-Search-Labor-and-Delivery-R4.html
Epic API ID | 10301
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?category={category}&code={code}&date={date}&patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10301

Description

This web service returns delivery documentation for the delivering patient. It includes general information about the delivery encounter as well as specific documentation captured during the delivery. See

code

for details on what delivery documentation can be included.

This web service requires system-level configuration to determine which details within the delivery note should be available as documented delivery points.

Without a

code

specified, all documentation within the delivery note that is configured to share will be included in the response, with references to the Observations for each documented delivery point included within the

hasMember

element of the response.

Observation.Search (Newborn Delivery) (R4)

is a complimentary web service that returns delivery documentation for the newborn child.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

_count | number | Optional | Yes | | Number of results per page.

_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.

_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.

_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.

category | token | Required | Yes | | Always "labor-delivery".

code | token | Optional | Yes | | A valid code supported for the Observation. This could either be a LOINC code, SNOMED code, or free-text code that represents a specific point for the delivery. Supported SNOMED codes include: 169821004 - Duration of first stage of labor 169822006 - Duration of second stage of labor 169823001 - Duration of third stage of labor 199745000 - Complication occurring during labor and delivery 200158002 - Other complications of labor and delivery 249120008 - Onset of labor first stage 249160009 - Onset of second stage of labor 249223000 - Apgar heart rate score 249224006 - Apgar respiratory effort score 249225007 - Apgar muscle tone score 249226008 - Apgar response to stimululs score 249227004 - Apgar color score 364297003 - Female perineum observable 397955000 - Analgesia in labor 443848000 - Apgar score at 15 minutes 443849008 - Apgar score at 20 minutes Supported LOINC codes include: 12116-0 - Fetal Umbilical cord Study observation US 21112-8 - Birth date 32401-2 - 10 minute Apgar Color 32402-0 - 10 minute Apgar Heart rate 32403-8 - 10 minute Apgar Muscle tone 32404-6 - 10 minute Apgar Reflex irritability 32405-3 - 10 minute Apgar Respiratory effort 32406-1 - 1 minute Apgar Color 32407-9 - 1 minute Apgar Heart rate 32408-7 - 1 minute Apgar Muscle tone 32409-5 - 1 minute Apgar Reflex irritability 32410-3 - 1 minute Apgar Respiratory effort 32411-1 - 5 minute Apgar Color 32412-9 - 5 minute Apgar Heart rate 32413-7 - 5 minute Apgar Muscle tone 32414-5 - 5 minute Apgar Reflex irritability 32415-2 - 5 minute Apgar Respiratory effort 38386-9 - Color of Amniotic fluid 48767-8 - Annotation comment [Interpretation] Narrative 55281-0 - Number of fetuses 56829-5 - Time membranes ruptured 57069-7 - Preterm labor symptoms 73760-1 - If cesarean, a trial of labor was attempted [US Standard Certificate of Live Birth] 73761-9 - Fetal presentation--at birth [US Standard Certificate of Live Birth] 73762-7 - Final route and method of delivery [US Standard Certificate of Live Birth] 73771-8 - Birth order 75186-7 - Vital status [CDC.CS] 8290-9 - Head Occipital-frontal circumference --at birth- by Tape measure 8293-3 - Chest Circumference --inspiration 8339-4 - Birth weight Measured 8722-1 - Surgical operation note anesthesia 89269-5 - Body height Measured --at birth 9271-8 - 10 minute Apgar Score 9272-6 - 1 minute Apgar Score 9274-2 - 5 minute Apgar Score Supported free-text values include: Abdominal girth Adoption type Analgesia comments Antenatal steroid Augmentation start Augmentation type Breastfeeding initiated instant Breech type Cervical ripening instant Complications comments Contraction start instant Cord blood disposition Cord clamping instant Cord complication Decision instant Delivery comments Episiotomy Established respiration instant Estimated blood loss Feeding intention Forceps attempted Forceps location Gases sent Hide birth mother's info Induction instant Induction type Labor antibiotics Labor identifier Labor type Placenta appearance Placenta instant Placenta removal Placental weight Presentation AP Presentation LR Presentation reference Primary reason for induction Procedure Reason for augmentation Reason for c-section Reason for episiotomy Reason for forceps Reason for induction Reason for vacuum Repair packets Repair suture Resuscitation type Ripening type Rupture odor Rupture of membranes to delivery Rupture type Skin to skin end instant Skin to skin exemption Skin to skin participant Skin to skin start instant Start pushing instant Vacuum attempted Vacuum location Codings specified in External Concept Mappings are also included.

date | date | Optional | Yes | | The date or date range for the delivery encounter. Refer to this document on how to format a date or dateTime range.

patient | reference | Required | Yes | | The Patient FHIR ID of the delivering patient.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description

combo-code | token | Optional | Yes | | The code of the delivery point supported for the Observation.

combo-value-concept | token | Optional | Yes | | The value of the delivery point supported for the Observation when that delivery point is a CodeableConcept. This type of value is returned for: Breech type Gases sent Feeding intention Labor identifier Labor type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Cord vessels Final route and method of delivery Vital status Fetal presentation--at birth If cesarean, a trial of labor was attempted 1 minute Apgar score 5 minute Apgar score 10 minute Apgar Score 15 minute Apgar score 20 minute Apgar score Primary reason for induction

component-code | token | Optional | Yes | | The code of the delivery point supported for the Observation.

component-value-concept | token | Optional | Yes | | The value of the delivery point supported for the Observation when that delivery point is a CodeableConcept. This type of value is returned for: Breech type Gases sent Feeding intention Labor identifier Labor type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Cord vessels Final route and method of delivery Vital status Fetal presentation--at birth If cesarean, a trial of labor was attempted 1 minute Apgar score 5 minute Apgar score 10 minute Apgar Score 15 minute Apgar score 20 minute Apgar score Primary reason for induction Ripening type

data-absent-reason | token | Optional | Yes | | Indicates when documentation has been deleted or was never entered for a specific point within the delivery documentation. The value should be set to "error".

encounter | reference | Optional | Yes | | The Encounter FHIR ID for the labor and delivery encounter.

focus | reference | Optional | Yes | | The Patient FHIR ID for the child.

has-member (also: has-member) | reference | Optional | Yes | | The Observation FHIR ID for one of the specific delivery points or documentation associated with the delivery episode.

identifier | token | Optional | Yes | | The delivery record ID (unique identifier for the birth record).

issued | date | Optional | Yes | | The instant at which the documentation for the delivery was updated.

performer | reference | Optional | Yes | | The user who entered the documentation for the delivery, input as a Practitioner FHIR ID.

status | token | Optional | Yes | | The status of the delivery documentation.

value-concept | token | Optional | Yes | | The CodeableConcept for the documented delivery point. The types of values that can be filtered: Breech type Gases sent Feeding intention Labor identifier Labor type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Cord vessels Final route and method of delivery Vital status Fetal presentation--at birth If cesarean, a trial of labor was attempted 1 minute Apgar score 5 minute Apgar score 10 minute Apgar Score 15 minute Apgar score 20 minute Apgar score Primary reason for induction

value-date | date | Optional | Yes | | The date value for the documented delivery point. The types of dates that can be filtered: Augmentation start Cervical ripening instant Contraction start instant Dilation start instant Induction instant Onset of labor Time membranes ruptured Skin to skin start instant Skin to skin end instant Birth date Breastfeeding initiated instant Cord clamping instant Decision instant Established respiration instant Placenta instant Start pushing instant Dilation complete instant

value-string | string | Optional | Yes | | Filters on Delivery or Birth comments.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation

Observation (R4).category | CodeableConcept | | Yes | 1..* | This is always "labor-delivery".

Observation (R4).code | CodeableConcept | | | 1..1 | Supported SNOMED codes include: 249160009 (Onset of Second Stage of Labor) 169821004 (Duration of first stage of labor) 169822006 (Duration of second stage of labor) 169823001 (Duration of third stage of labor 249120008 (Onset of labor first stage) 397955000 (Analgesia in labor) 200158002 (Other complications of labor and delivery) 199745000 (Complication occurring during labor and delivery) 364297003 (Lacerations) 443848000 (15 minute Apgar score) 443849008 (20 minute Apgar score) Supported LOINC codes include: 48767-8 (Birth comments) 21112-8 (Birth date) 8293-3 (Chest circumference --inspiration) 89269-5 (Body height @ birth measured 8339-4 (Birth weight measured) 8290-9 (Head occipital-frontal circumference --at birth- by Tape measure) 73771-8 (Birth order) 55281-0 (Number of fetuses) 57075-4 (Newborn delivery information) 56829-5 (Time membranes ruptured) 57069-7 (Preterm labor symptoms) 8722-1 (Anesthesia method) 12116-0 (Cord vessels) 73762-7 (Final route and method of delivery) 75186-7 (Vital status) 73761-9 (Fetal presentation--at birth) 38386-9 (Color of amniotic fluid) 73760-1 (If cesarean, a trial of labor was attempted) 9272-6 (1 minute Apgar score) 9274-2 (5 minute Apgar score) 9271-8 (10 minute Apgar score) Other supported values include: Delivery comments Cord clamping instant Forceps attempted Vacuum attempted Start pushing instant Established respiration instant Placenta instant Breastfeeding initiated instant Decision instant Placental weight Rupture of membranes to delivery Repair packets Abdominal girth Estimated blood loss Augmentation start Cervical ripening instant Contraction start instant Dilation start instant Induction instant Skin to skin start instant Skin to skin end instant Augmentation type Breech type Ripening type Cord blood disposition Cord complication Gases sent Episiotomy Feeding intention Forceps location Induction type Labor identifier Labor Type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Primary reason for induction Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Antenatal steroid Vacuum location Labor antibiotics Adoption type Codings specified in External Concept Mappings are also included.

Observation (R4).component | BackboneElement | | Yes | 1..* | Populated only for patient-entered flowsheets.

Observation (R4).component.code | CodeableConcept | | | 1..1 | Returns the values documented for each component below: Augmentation type Ripening type Cord blood disposition Cord complication Episiotomy Forceps location Induction type Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Vacuum location Adoption type Hide birth mother's info Supported LOINC Codes include: 8722-1 (Anesthesia method) 38386-9 (Color of amniotic fluid) 32406-1 (1 minute Apgar color) 32407-9 (1 minute Apgar heart rate) 32409-5 (1 minute Apgar reflex irritability) 32408-7 (1 minute Apgar muscle tone) 32410-3 (1 minute Apgar respiratory effort) 32411-1 (5 minute Apgar color) 32412-9 (5 minute Apgar heart rate) 32414-5 (5 minute Apgar reflex irritability) 32413-7 (5 minute Apgar muscle tone) 32415-2 (5 minute Apgar respiratory effort) 32401-2 (10 minute Apgar color) 32402-0 (10 minute Apgar heart rate) 32404-6 (10 minute Apgar reflex irritability) 32403-8 (10 minute Apgar muscle tone) 32405-3 (10 minute Apgar respiratory effort) Supported SNOMED codes include: 397955000 (Analgesia in labor) 200158002 (Other complications of labor and delivery) 199745000 (Complication occurring during labor and delivery) 364297003 (Lacerations) 249227004 (15 minute Apgar color) 249223000 (15 minute Apgar heart rate) 249226008 (15 minute Apgar reflex irritability) 249225007 (15 minute Apgar muscle tone) 249224006 (15 minute Apgar respiratory effort) 249227004 (20 minute Apgar color) 249223000 (20 minute Apgar heart rate) 249226008 (20 minute Apgar reflex irritability) 249225007 (20 minute Apgar muscle tone) 249224006 (20 minute Apgar respiratory effort)

Observation (R4).component.valueBoolean | boolean | | | 0..1 | If set to true, the newborn name is not displayed. This is driven off of the adoption status for the newborn.

Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 1..1 | Returns the values documented for each component below: Augmentation type Ripening type Cord blood disposition Cord complication Episiotomy Forceps location Induction type Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Vacuum location Anesthesia method Other complications of labor and delivery Lacerations Rupture of amniotic fluid 1 minute Apgar color 1 minute Apgar heart rate 1 minute Apgar reflex irritability 1 minute Apgar muscle tone 1 minute Apgar respiratory effort 5 minute Apgar color 5 minute Apgar heart rate 5 minute Apgar reflex irritability 5 minute Apgar muscle tone 5 minute Apgar respiratory effort 10 minute Apgar color 10 minute Apgar heart rate 10 minute Apgar reflex irritability 10 minute Apgar muscle tone 10 minute Apgar respiratory effort 15 minute Apgar color 15 minute Apgar heart rate 15 minute Apgar reflex irritability 15 minute Apgar muscle tone 15 minute Apgar respiratory effort 20 minute Apgar color 20 minute Apgar heart rate 20 minute Apgar reflex irritability 20 minute Apgar muscle tone 20 minute Apgar respiratory effort Adoption type

Observation (R4).component.valueString | string | | | 1..1 | Returns analgesia comments or complication comments documented in the delivery documentation.

Observation (R4).dataAbsentReason | CodeableConcept | | | 1..1 | Indicates when documentation has been deleted or was never entered for a specific point within the delivery documentation. If the data has been removed for the delivery point, dataAbsentReason will indicate "Data for specified Observation no longer exists." If the data was never completed for the delivery point, dataAbsentReason will indicate "Specified Observation has never had data."

Observation (R4).effectiveDateTime | dateTime | | | 1..1 | Newborn birth date.

Observation (R4).encounter | Reference | | | 1..1 | The Encounter FHIR ID for the labor and delivery encounter.

Observation (R4).extension (delivery-personnel) (also: extension (delivery-personnel)) | Extension | | Yes | 1..* | Reference to the CareTeam resource, which will export this delivery's personnel. URL: http://open.epic.com/FHIR/StructureDefinition/extension/delivery-personnel

Observation (R4).extension (observation-focusCode) (also: extension (observation-focusCode)) | Extension | | Yes | 1..* | Identifies the relationship between the subject and the focus. This extension is included within the focus element. The relationship will always be "Child" when the relationship is allowed to be printed or shared. URL: http://hl7.org/fhir/StructureDefinition/observation-focusCode

Observation (R4).extension (workflow-reasonCode) (also: extension (workflow-reasonCode)) | Extension | | Yes | 0..* | Supports reasoning for other documentation: Reasons for augmentation Reasons for episiotomy Reasons for induction Reasons for use of forceps Reasons for use of vacuum Reasons for c-section URL: http://hl7.org/fhir/StructureDefinition/workflow-reasonCode

Observation (R4).focus | Reference | | Yes | 1..* | The Patient FHIR ID for the child.

Observation (R4).hasMember | Reference | | Yes | 1..* | Includes the Observation FHIR IDs for the delivery points or documentation associated with the delivery episode.

Observation (R4).id | id | | | | The Observation FHIR ID for the Labor and Delivery observation.

Observation (R4).identifier | Identifier | | Yes | 1..* | The delivery record ID (unique identifier for the birth record).

Observation (R4).issued | instant | | | 1..1 | The instant at which the documentation for the delivery was updated.

Observation (R4).note | Annotation | | Yes | 1..* | Returns strings for the following: Comment about analgesics used during labor Comment about complications occurring during labor

Observation (R4).performer | Reference | | Yes | 0..1 | The user who entered the documentation for the delivery, as a Practitioner FHIR ID. Only included in the response when the code request element is used to filter to a specific delivery point.

Observation (R4).status | code | | | 1..1 | The status of the delivery documentation.

Observation (R4).subject | Reference | | | 1..1 | The Patient FHIR ID.

Observation (R4).valueBoolean | boolean | | | 0..1 | Returns yes/no for the following: Labor antibiotics Forceps attempted Vacuum attempted Preterm labor symptoms

Observation (R4).valueCodeableConcept | CodeableConcept | | | 1..1 | Returns a value for the following elements in the delivery documentation: Breech type Gases sent Feeding intention Labor identifier Labor type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Cord vessels Final route and method of delivery Vital status Fetal presentation--at birth If cesarean, a trial of labor was attempted 1 minute Apgar score 5 minute Apgar score 10 minute Apgar Score 15 minute Apgar score 20 minute Apgar score Primary reason for induction

Observation (R4).valueDateTime | dateTime | | | 1..1 | A date or time associated with the component. Applies to the following component types: Augmentation start Cervical ripening instant Contraction start instant Dilation start instant Induction instant Onset of labor Time membranes ruptured Skin to skin start instant Skin to skin end instant Birth date Breastfeeding initiated instant Cord clamping instant Decision instant Established respiration instant Placenta instant Start pushing instant Dilation complete instant

Observation (R4).valueInteger | integer | | | 0..1 | Returns the count for the following: Number of fetuses - before delivery, this represents the number of fetuses in the pregnancy. After delivery has occurred, however, this may be updated to the number of viable deliveries in the pregnancy. Repair packets Birth position

Observation (R4).valueQuantity | Quantity | | | 1..1 | Returns measurements of the following: Abdominal girth Estimated blood loss Chest circumference --inspiration Body height @ birth measured Birth weight measured Head occipital-frontal circumference at birth - by tape measure Rupture of membranes to delivery Placental weight Duration of first stage of labor Duration of second stage of labor Duration of third stage of labor

Observation (R4).valueString | string | | | 1..1 | Returns the comments below: Delivery comments Birth comments

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation?patient=eg1u2l0bPjqy6PDCP1j9RwA3&category=labor-delivery
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?patient=eT-EM40KXOqnnxricZJ-
    gQyHsZYYqAQfaBrw3-xYq0dY3&category=labor-delivery"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/ejm2jERBBTcMHcbMNm-
    pEqwkAxxxwaR4iOpnRaH3QYCMINmB84gupepKJP0l4tSUsNK-1siJmj7OPVM7HIPXhEQ3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/ejm2jERBBTcMHcbMNm-
    pEqwkAxxxwaR4iOpnRaH3QYCMINmB84gupepKJP0l4tSUsNK-1siJmj7OPVM7HIPXhEQ3",
      "resource": {
        "resourceType": "Observation",
        "id": "ejm2jERBBTcMHcbMNm-pEqwkAxxxwaR4iOpnRaH3QYCMINmB84gupepKJP0l4tSUsNK-1siJmj7OPVM7HIPXhEQ3",
        "extension": [
          {
            "valueReference": {
              "reference": "CareTeam/eOBJQvjvE-Hwf.QIEWCVJFk366llNYGR4aXqWgxhS0t83"
            },
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/delivery-personnel"
          }
        ],
        "identifier": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.728366",
            "value": "1046176"
          }
        ],
        "status": "preliminary",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "labor-delivery",
                "display": "Labor and Delivery"
              }
```

```
            ],
            "text": "Labor and Delivery"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "57075-4",
              "display": "Newborn delivery information"
            }
          ],
          "text": "Newborn delivery information"
        },
        "subject": {
          "reference": "Patient/eT-EM40KXOqnnxricZJ-gQyHsZYYqAQfaBrw3-xYq0dY3",
          "display": "Fhir, Jane"
        },
        "focus": [
          {
            "extension": [
              {
                "valueCodeableConcept": {
                  "coding": [
                    {
                      "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
                      "code": "CHILD",
                      "display": "Child"
                    }
                  ],
                  "text": "Child"
                },
                "url": "http://hl7.org/fhir/StructureDefinition/observation-focusCode"
              }
            ],
            "reference": "Patient/evRHk.d80nuSYo7tggVEP02w3DkDs3pVx82VbiERpY.Q3",
            "display": "Fhir, JaneBaby1"
          }
        ],
        "encounter": {
          "reference": "Encounter/eImo5bSMk.sB4u0HBO9gwEOrT2-t.ij55DL6kdPOeqT03",
          "display": "Hospital Encounter"
        },
        "effectiveDateTime": "2021-12-29",
        "hasMember": [
          {
            "reference": "Observation/ejm2jERBBTcMHcbMNm-
    pEqwkAxxxwaR4iOpnRaH3QYCMINmB84gupepKJP0l4tSUs5mRGqEoLFGzEM4gOgCPPyA3",
            "display": "Birth date"
          },
          {
            "reference": "Observation/ejm2jERBBTcMHcbMNm-
    pEqwkAxxxwaR4iOpnRaH3QYCMINmB84gupepKJP0l4tSUsp69g9dKW6Z4lPXJ0JCm3cg3",
            "display": "Vital status"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


13.1.22 Observation.Search (Newborn Delivery) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10307-Observation-Search-Newborn-Delivery-R4.html
Epic API ID | 10307
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?category={category}&code={code}&patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10307

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | False | | | Always "newborn-delivery" for this API.
code | token | True | | | Codes of the Observation, such as from the LOINC or SNOMED systems. If specified, category must also be specified.
patient | reference | False | | | FHIR patient ID of newborn patient.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Unsupported
combo-code | token | True | | | The code of the observation type or component type
combo-data-absent-reason | token | True | | | Unsupported
combo-value-concept | token | True | | | The value or component value of the observation, if the value is a CodeableConcept.
component-code | token | True | | | The component code of the observation type.
component-data-absent-reason | token | True | | | Unsupported
component-value-concept | token | True | | | The value of the component observation, if the value is a CodeableConcept.
data-absent-reason | token | True | | | Unsupported
date | date | True | | | Newborn birth date.
derived-from | String | True | | | Unsupported
device | String | True | | | Unsupported
encounter | reference | True | | | The newborn FHIR encounter ID.
focus | reference | True | | | The patient who delivered the newborn.
has-member | reference | True | | | FHIR observation ID.
identifier | token | True | | | Delivery record HSB ID (unique identifier for the birth record).
issued | date | True | | | Documentation instant.
method | token | True | | | Unsupported
part-of | String | True | | | Unsupported
performer | reference | True | | | User who entered the documentation.
specimen | String | True | | | Unsupported
status | token | True | | | The status of the observation. Supported values are "final", "preliminary", or "cancelled".
value-concept | token | True | | | The value of the observation, if the value is a CodeableConcept.
value-date | date | True | | | The value of the observation, if the value is a date or period of time.
value-string | string | True | | | The value of the observation, if the value is a string, and also searches in CodeableConcept.text

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Observation | Observation | False | | | An instance of the R4 Observation resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | Always "newborn-delivery"
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | Supported SNOMED codes include: 249160009-Onset of Second Stage of Labor 169821004-Duration of first stage of labor 169822006-Duration of second stage of labor 169823001-Duration of third stage of labor 249120008-Onset of labor first stage 397955000-Analgesia in labor 200158002-Other complications of labor and delivery 199745000-Complication occurring during labor and delivery 364297003-Lacerations 443848000-15 minute Apgar score 443849008-20 minute Apgar score Supported LOINC codes include: 48767-8 Birth comments 21112-8 Birth date 8293-3 Chest circumference --inspiration 89269-5 Body height @ birth measured 8339-4 Birth weight measured 8290-9 Head occipital-frontal circumference --at birth- by Tape measure 73771-8 Birth order 55281-0 Number of fetuses 57075-4 Delivery personnel 56829-5 Time membranes ruptured 57069-7 Preterm labor symptoms 8722-1 Anesthesia method 12116-0 Cord vessels 73762-7 Final route and method of delivery 75186-7 Vital status 73761-9 Fetal presentation--at birth 38386-9 Color of amniotic fluid 73760-1 If cesarean, a trial of labor was attempted 9272-6 1 minute Apgar score 9274-2 5 minute Apgar score 9271-8 10 minute Apgar score Other supported values include: Delivery comments Cord clamping instant Forceps attempted Vacuum attempted Start pushing instant Established respiration instant Placenta instant Breastfeeding initiated instant Decision instant Placental weight Rupture of membranes to delivery Repair packets Abdominal girth Estimated blood loss Augmentation start Cervical ripening instant Contraction start instant Dilation start instant Induction instant Skin to skin start instant Skin to skin end instant Augmentation type Breech type Ripening type Cord blood disposition Cord complication Gases sent Episiotomy Feeding intention Forceps location Induction type Labor identifier Labor Type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Antenatal steroid Vacuum location Labor antibiotics Adoption type Codings specified in External Concept Mappings are also included.
Bundle.entry.resource.Observation.component | BackboneElement | False | Yes | | Populated only for patient-entered flowsheets.
Bundle.entry.resource.Observation.component.code | CodeableConcept | False | | | Returns the values documented for each component below: Augmentation type Ripening type Cord blood disposition Cord complication Episiotomy Forceps location Induction type Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Vacuum location Adoption type Hide birth mother's info Supported LOINC Codes include: 8722-1 (Anesthesia method) 38386-9 (Color of amniotic fluid) 32406-1 (1 minute Apgar color) 32407-9 (1 minute Apgar heart rate) 32409-5 (1 minute Apgar reflex irritability) 32408-7 (1 minute Apgar muscle tone) 32410-3 (1 minute Apgar respiratory effort) 32411-1 (5 minute Apgar color) 32412-9 (5 minute Apgar heart rate) 32414-5 (5 minute Apgar reflex irritability) 32413-7 (5 minute Apgar muscle tone) 32415-2 (5 minute Apgar respiratory effort) 32401-2 (10 minute Apgar color) 32402-0 (10 minute Apgar heart rate) 32404-6 (10 minute Apgar reflex irritability) 32403-8 (10 minute Apgar muscle tone) 32405-3 (10 minute Apgar respiratory effort) Supported SNOMED codes include: 397955000 (Analgesia in labor) 200158002 (Other complications of labor and delivery) 199745000 (Complication occurring during labor and delivery) 364297003 (Lacerations) 249227004 (15 minute Apgar color) 249223000 (15 minute Apgar heart rate) 249226008 (15 minute Apgar reflex irritability) 249225007 (15 minute Apgar muscle tone) 249224006 (15 minute Apgar respiratory effort) 249227004 (20 minute Apgar color) 249223000 (20 minute Apgar heart rate) 249226008 (20 minute Apgar reflex irritability) 249225007 (20 minute Apgar muscle tone) 249224006 (20 minute Apgar respiratory effort)
Bundle.entry.resource.Observation.component.valueBoolean | boolean | True | | | Returns true/false for the following in the event of an adoption. If set to true, the newborn name is not displayed. Hide birth mother's info.

Bundle.entry.resource.Observation.component.valueCodeableConcept | CodeableConcept | False | | | Returns the values documented for each component below: occurring during labor and delivery Augmentation type Ripening type Cord blood disposition Cord complication Episiotomy Forceps location Induction type Procedure Skin to skin exemption Repair suture Resuscitation type Rupture odor Rupture type Skin to skin participant Vacuum location Anesthesia method Other complications of labor and delivery Lacerations Rupture of amniotic fluid 1 minute Apgar color 1 minute Apgar heart rate 1 minute Apgar reflex irritability 1 minute Apgar muscle tone 1 minute Apgar respiratory effort 5 minute Apgar color 5 minute Apgar heart rate 5 minute Apgar reflex irritability 5 minute Apgar muscle tone 5 minute Apgar respiratory effort 10 minute Apgar color 10 minute Apgar heart rate 10 minute Apgar reflex irritability 10 minute Apgar muscle tone 10 minute Apgar respiratory effort 15 minute Apgar color 15 minute Apgar heart rate 15 minute Apgar reflex irritability 15 minute Apgar muscle tone 15 minute Apgar respiratory effort 20 minute Apgar color 20 minute Apgar heart rate 20 minute Apgar reflex irritability 20 minute Apgar muscle tone 20 minute Apgar respiratory effort Adoption type

Bundle.entry.resource.Observation.component.valueString | string | False | | | Returns analgesia values documented in the Delivery Summary: Analgesia in labor

Bundle.entry.resource.Observation.extension (also: delivery-personnel) | Extension | True | Yes | | Reference to the CareTeam resource, which will export this delivery's personnel.

Bundle.entry.resource.Observation.effectiveDateTime | dateTime | False | | | Newborn birth date.

Bundle.entry.resource.Observation.encounter | Reference | False | | | The newborn FHIR encounter ID.

Bundle.entry.resource.Observation.focus | Reference | False | Yes | | The patient who delivered the newborn.

Bundle.entry.resource.Observation.hasMember | Reference | False | Yes | | FHIR observation ID. Appears in place of the value*, component, issued, and performer properties when searching on all data points. SmartData elements values are exported with this element.

Bundle.entry.resource.Observation.identifier | Identifier | False | Yes | | Delivery record HSB ID (unique identifier for the birth record).

Bundle.entry.resource.Observation.issued | instant | False | | | Documentation instant.

Bundle.entry.resource.Observation.note | Annotation | False | Yes | | Returns strings for the following: Comment about analgesics used during labor Comment about complications occurring during labor

Bundle.entry.resource.Observation.extension (also: observation-focusCode) | Extension | False | Yes | | Identifies the relationship between the subject and the focus. This extension is included within the focus element.

Bundle.entry.resource.Observation.performer | Reference | False | Yes | | User who entered the documentation.

Bundle.entry.resource.Observation.status | code | False | | | Delivery Summary documentation is "final" if the Delivery Summary is signed. If the Delivery Summary is not currently signed or is open for addendum, it has a status of "preliminary." If documentation has been blanked out: "cancelled."

Bundle.entry.resource.Observation.subject | Reference | False | | | The newborn patient

Bundle.entry.resource.Observation.valueBoolean | boolean | True | | | Returns yes/no for the following: Labor antibiotics Forceps attempted Vacuum attempted Preterm labor symptoms

Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | False | | | Returns a value for the following elements in Delivery Summary: Breech type Gases sent Feeding intention Labor identifier Labor type Placenta appearance Placenta removal Presentation anterior-posterior Presentation left-right Presentation reference Cord vessels Final route and method of delivery Vital status Fetal presentation--at birth If cesarean, a trial of labor was attempted 1 minute Apgar score 5 minute Apgar score 10 minute Apgar Score 15 minute Apgar score 20 minute Apgar score Primary reason for induction

Bundle.entry.resource.Observation.valueDateTime | dateTime | False | | | A date or time associated with the component. Applies to the following component types: Augmentation start Cervical ripening instant Contraction start instant Dilation start instant Induction instant Onset of labor first stage Time membranes ruptured Skin to skin start instant Skin to skin end instant Birth date Breastfeeding initiated instant Cord clamping instant Decision instant Established respiration instant Placenta instant Start pushing instant

Bundle.entry.resource.Observation.valueInteger | integer | True | | | Returns the count for the following: Number of fetuses - before delivery, this represents the number of fetuses in the pregnancy. After delivery has occurred, however, this may be updated to the number of viable deliveries in the pregnancy. Repair packets

Bundle.entry.resource.Observation.valueQuantity | Quantity | False | | | Returns measurements of the following: Abdominal girth Estimated blood loss Chest circumference --inspiration Body height @ birth measured Birth weight measured Head occipital-frontal circumference --at birth- by Tape measure Rupture of membranes to delivery Placental weight Duration of first stage of labor Duration of second stage of labor Duration of third stage of labor

Bundle.entry.resource.Observation.valueString | string | False | | | Returns the comments below: Delivery comments Birth comments

Bundle.entry.resource.Observation.extension (also: workflow-reasonCode) | Extension | True | Yes | | Supports reasoning for other documentation: Reasons for augmentation Reasons for episiotomy Reasons for induction Reasons for use of forceps Reasons for use of vacuum Reasons for c-section

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?patient=eg1u2l0bPjqy6PDCP1j9RwA3&category=newborn-delivery
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?patient=evRHk.d80nuSYo7tggVEP02w3DkDs3pVx82VbiERpY.Q3&ca
    tegory=newborn-delivery&_include=Observation:hasMember"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/eDKo3D-Zpnp0V1-
    01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJrz6QEJAM05okUqGRbhBxHEw3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eDKo3D-Zpnp0V1-
    01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJrz6QEJAM05okUqGRbhBxHEw3",
```

```
  "resource": {
    "resourceType": "Observation",
    "id": "eDKo3D-Zpnp0V1-01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJrz6QEJAM05okUqGRbhBxHEw3",
    "extension": [
      {
        "valueReference": {
          "reference": "CareTeam/e-gJ-GXuSbEa1c2nI.NvFo6fX6aKeDXWg88FeVTN1vi43"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/delivery-personnel"
      }
    ],
    "status": "preliminary",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
            "code": "newborn-delivery",
            "display": "Newborn Delivery"
          }
        ],
        "text": "Newborn Delivery"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "57075-4",
          "display": "Newborn delivery information"
        }
      ],
      "text": "Newborn delivery information"
    },
    "subject": {
      "reference": "Patient/evRHk.d80nuSYo7tggVEP02w3DkDs3pVx82VbiERpY.Q3",
      "display": "Cheny, LilaBaby1"
    },
    "focus": [
      {
        "extension": [
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
                  "code": "MTH",
                  "display": "Mother"
                }
              ],
              "text": "Mother"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/observation-focusCode"
          }
        ],
        "reference": "Patient/eT-EM40KXOqnnxricZJ-gQyHsZYYqAQfaBrw3-xYq0dY3",
        "display": "Cheny, Lila"
      }
    ],
    "encounter": {
      "reference": "Encounter/e60GsidVkWiQC9uoYcT7w0sqcDD1cmSGm-FxakXOiHAQ3",
      "display": "Hospital Encounter"
    },
    "effectiveDateTime": "2021-12-29",
    "hasMember": [
      {
        "reference":
"Observation/eDKo3D-Zpnp0V1-01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJrZCKg488O5goVMwYTgGqY1Q3",
        "display": "Birth date"
      },
      {
        "reference":
"Observation/eDKo3D-Zpnp0V1-01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJreMnminM5UyDhRmLkbdl.6w3",
        "display": "Vital status"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eDKo3D-Zpnp0V1-
01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJrZCKg488O5goVMwYTgGqY1Q3"
    }
  ],
```

```
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eDKo3D-Zpnp0V1-
01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJrZCKg488O5goVMwYTgGqY1Q3",
  "resource": {
    "resourceType": "Observation",
    "id": "eDKo3D-Zpnp0V1-01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJrZCKg488O5goVMwYTgGqY1Q3",
    "status": "preliminary",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
            "code": "newborn-delivery",
            "display": "Newborn Delivery"
          }
        ],
        "text": "Newborn Delivery"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "21112-8",
          "display": "Birth date"
        }
      ],
      "text": "Birth date"
    },
    "subject": {
      "reference": "Patient/evRHk.d80nuSYo7tggVEP02w3DkDs3pVx82VbiERpY.Q3",
      "display": "Cheny, LilaBaby1"
    },
    "focus": [
      {
        "extension": [
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
                  "code": "MTH",
                  "display": "Mother"
                }
              ],
              "text": "Mother"
            },
            "url": "http://hl7.org/fhir/StructureDefinition/observation-focusCode"
          }
        ],
        "reference": "Patient/eT-EM40KXOqnnxricZJ-gQyHsZYYqAQfaBrw3-xYq0dY3",
        "display": "Cheny, Lila"
      }
    ],
    "encounter": {
      "reference": "Encounter/e60GsidVkWiQC9uoYcT7w0sqcDD1cmSGm-FxakXOiHAQ3",
      "display": "Hospital Encounter"
    },
    "issued": "2021-12-29T20:31:39Z",
    "performer": [
      {
        "reference": "Practitioner/emGEw7qNXw.MtevUAAAHb7g3",
        "type": "Practitioner",
        "display": "Obstetrics,Physician"
      }
    ],
    "valueDateTime": "2021-12-29T20:31:37Z"
  },
  "search": {
    "mode": "include"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eDKo3D-Zpnp0V1-
01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJreMnminM5UyDhRmLkbdl.6w3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eDKo3D-Zpnp0V1-
01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJreMnminM5UyDhRmLkbdl.6w3",
  "resource": {
    "resourceType": "Observation",
    "id": "eDKo3D-Zpnp0V1-01u2Khpn7wZrGEEtpabIn760ZpXN.AjGDrvBss3daLOk1rgBJreMnminM5UyDhRmLkbdl.6w3",
    "status": "preliminary",
    "category": [
      {
        "coding": [
          {
```

```
              "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
              "code": "newborn-delivery",
              "display": "Newborn Delivery"
            }
          ],
          "text": "Newborn Delivery"
        }
      ],
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "75186-7",
            "display": "Vital status"
          }
        ],
        "text": "Vital status"
      },
      "subject": {
        "reference": "Patient/evRHk.d80nuSYo7tggVEP02w3DkDs3pVx82VbiERpY.Q3",
        "display": "Cheny, LilaBaby1"
      },
      "focus": [
        {
          "extension": [
            {
              "valueCodeableConcept": {
                "coding": [
                  {
                    "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
                    "code": "MTH",
                    "display": "Mother"
                  }
                ],
                "text": "Mother"
              },
              "url": "http://hl7.org/fhir/StructureDefinition/observation-focusCode"
            }
          ],
          "reference": "Patient/eT-EM40KXOqnnxricZJ-gQyHsZYYqAQfaBrw3-xYq0dY3",
          "display": "Cheny, Lila"
        }
      ],
      "encounter": {
        "reference": "Encounter/e60GsidVkWiQC9uoYcT7w0sqcDD1cmSGm-FxakXOiHAQ3",
        "display": "Hospital Encounter"
      },
      "issued": "2021-12-29T20:31:43Z",
      "performer": [
        {
          "reference": "Practitioner/emGEw7qNXw.MtevUAAAHb7g3",
          "type": "Practitioner",
          "display": "Obstetrics,Physician"
        }
      ],
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "LA4247-8",
            "display": "Alive"
          }
        ],
        "text": "Living"
      }
    },
    "search": {
      "mode": "include"
    }
  }
 ]
}
```


13.1.23 Observation.Search (Obstetrics and Gynecology) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10294-Observation-Search-Obstetrics-and-Gynecology-R4.html
Epic API ID | 10294
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?&category={category}&code={code}&date={date}&focus={focus}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10294

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | False | | | Use "obstetrics-gynecology" to search for OB/Gyn history details. Either this or code is required, but both are allowed.
code | token | False | | | The LOINC code of the observation. The following codes can be searched on with the mother's patient ID: Age at Menarche (63888-2) Age at First Pregnancy (42797-1) Age at Menopause (42802-9) Gynecology History Comment (89471-7) Gravida (11996-6) Para (11977-6) Abortion (11612-9) SAB (11614-5) IAB (11613-7) Ectopic (33065-4) Living (11638-4) Live Births (11636-8) Delivery Record (118215003) Number of Fetuses (55281-0) Obstetrics Summary Note or General Perinatal Comment (89470-9) Pregravid Weight (8348-5) Pregravid Height (8302-2) Prenatal Visits Count (68493-6) General Estimated Delivery Date (11778-8) Delivery Date Estimated from Last Menstrual Period (11779-6) Delivery Date Estimated from Conception Date (53692-0) Delivery Date US Composite Estimate (11781-2) Delivery Date Estimated from Prior Gestational Age Assessment (Patient Reported) (53694-6) Delivery Date Estimated from Quickening Date (57063-0) Delivery Date Estimated from Date Fundal Height Reaches Umbilicus (57064-8) Fetal Sex US (11882-8)
date | date | False | | | Date range to search for deliveries. All deliveries with an effective date within the specified date range are returned. Deliveries with no effective date are returned if the search range has no start date (return everything before the specified date). They are not returned if there is a start date but no end date. Deliveries with fuzzy dates (only a year or month/year), are returned if there is any overlap between the date range and the fuzziness range. For example, if the outcome date is 2009 and the start of the search is 6/5/2009, the 2009 delivery is returned.
focus | reference | False | | | Reference to a pregnancy EpisodeOfCare resource. If patient or subject is also provided, they must match the patient attached to the EpisodeOfCare resource. If not provided, observations associated with all pregnancy episodes for the patient are included.
patient | reference | False | | | Reference to a patient resource. If both patient and subject are provided, they must match.
subject | reference | False | | | The FHIR Patient ID.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Unsupported
combo-code | token | True | | | The code of the observation type or component type. See the component/code response parameter for more information.
combo-data-absent-reason | token | True | | | Unsupported
combo-value-concept | token | True | | | See valueCodeableConcept response parameter for more information.
component-code | token | True | | | See the component code response parameter for more details.
component-data-absent-reason | token | True | | | Unsupported
component-value-concept | token | True | | | The value based on the corresponding component. See the component parameter in the response for more information.
data-absent-reason | token | True | | | Unsupported
derived-from | String | True | | | If this EDD observation is the main selected EDD for the pregnancy, then derivedFrom contains a reference to every EDD event that has been documented on that episode, regardless of whether it was selected as the main EDD to use for the pregnancy.
device | String | True | | | Unsupported
encounter | reference | True | | | Unsupported
has-member | reference | True | | | A hasMember reference is included for each supported data point that contains data.
identifier | token | True | | | Unsupported
method | token | True | | | The basis for this EDD. For example, the EDD could be based on an ultrasound.
part-of | String | True | | | Unsupported
performer | reference | True | | | The clinician who documented the observation or the delivery clinician.
specimen | String | True | | | Unsupported
status | token | True | | | The status of the observation. Set to “final” except for special logic for the EDD.
value-concept | token | True | | | The value of the observation, if the value is a CodeableConcept. See the valueCodeableConcept response parameter for more information.
value-date | date | True | | | Estimated date of conception, EDD Basis Event Date, or Last Menstrual Period Date. See the valueDateTime response parameter for more details.
value-string | string | True | | | Searches for one of the following comments written by the clinician: The gynecology history comment. The general obstetrics history summary comment. Or name information: Name of newborn. Name of the delivery location.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | This is always "obstetrics-gynecology."
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | The code or string of one of the following Observation Types: Age at Menarche (63888-2) Age at First Pregnancy (42797-1) Months Breastfeeding ("# of months breastfeeding”) Age at Menopause (42802-9) Age at First Live Birth (“Age at first live birth”) Gynecology History Comment (89471-7) Number of Fetuses (55281-0) Delivery Date Estimated (11778-8, 11779-6, 53692-0, 11781-2, 57063-0, 57064-8, 53694-6) Fetal Sex US (11882-8) Suspected Fetal Demise (“Suspected Fetal Demise”) Prenatal Visits Count (68493-6) Obstetrics Summary Note (89470-9) Gravida (11996-6) Para (11977-6) Term (no code) Preterm (no code) Abortion (11612-9) SAB (11614-5) IAB (11613-7) Ectopic (33065-4) Molar (no code) Living (11638-4) Live Births (11636-8) Multiple (no code) General Perinatal Comment (89470-9) Delivery Record (118215003) Outcome (63893-2) Outcome Date/Time (93857-1) GA (18185-9) Length of Labor Stage 1 (169821004) Length of Labor Stage 2 (169822006) Birth Weight (8339-4) Baby Sex (76689-9) Delivery Method (73762-7) Anesthesia (8722-1) Preterm Labor (57069-7) Living Status – Current (75186-7) Apgar 1 (9272-6) Apgar 5 (9274-2) Baby Name (no code) Labor Complications (199745000) Additional Complications (200158002) Delivery Location (169812000) Pregravid Weight (8348-5) Pregravid Height (8302-2)

Bundle.entry.resource.Observation.component | BackboneElement | False | Yes | | Returns the value based on the corresponding components: Components of of Fetuses Fetal chorion Fetal amnion Number of fetuses code Components of Estimated Delivery Date Last menstrual period date LMP precision On birth control. Estimated date of conception Cycle lngth Luteal length EDD basis event date GA at event. Either this or days to EDD from event is included, but not both. Days to EDD from event. Either this or GA at event is included, but not both. Components of Sex of Fetus A component is added for each fetal sex that has been documented.

Components of of Fetuses | Fetal chorion

Fetal amnion

Number of fetuses code

Components of Estimated Delivery Date | Last menstrual period date

LMP precision

On birth control.

Estimated date of conception

Cycle lngth

Luteal length

EDD basis event date

GA at event. Either this or days to EDD from event is included, but not both.

Days to EDD from event. Either this or GA at event is included, but not both.

Components of Sex of Fetus | A component is added for each fetal sex that has been documented.

Bundle.entry.resource.Observation.component.code | CodeableConcept | False | | | Fetal Chorion - 92568-5 Fetal Amnion Last menstrual period start date - 8665-2 LMP Precision On Birth Control Conception date - 33067-0 Cycle Length Luteal Length EDD Basis Event Date GA at Event Days to EDD from Event Fetal Sex US - 11882-8 (Fetal Sex US)

Bundle.entry.resource.Observation.component.valueBoolean | boolean | True | | | On Birth Control - Whether the patient was on birth control at the time of this LMP.

Bundle.entry.resource.Observation.component.valueCodeableConcept | CodeableConcept | False | | | Fetal Chorion [1] Monochorionic: LA30047-7 [2] Dichorionic: LA30048-5 [3] Trichorionic: (no LOINC code) [99] Other: (no LOINC code) Value derived from I HSB 35571 – first part Fetal Amnion [1] Monoamniotic [2] Diamniotic [3] Triamniotic [99] Other Value derived from I HSB 35571 – second part. Number of Fetuses Code The number of fetuses and chorionicity/amnionicity expressed as SNOMED codes. LMP Precision How precise the associated LMP date is, according to the patient. Sex of Individual Fetus The fetal sex. Surprise (I HSB 35574 code) Female (LA21135-1) Male (LA21134-4) See overview (I HSB 35574 code) The mapped SNOMED code for each surgical operation note anesthesia. Unmapped values are an Epic-based category code with the category value and display name. The living status at birth for the delivery. The current living status for the delivery. The mapped SNOMED code for each complication occurring during labor and delivery. Unmapped values show the Epic-based category code with the category value and display name. Uses EDG to SNOMED mappings for other complications of labor and delivery.

Fetal Chorion | [1] Monochorionic: LA30047-7 [2] Dichorionic: LA30048-5 [3] Trichorionic: (no LOINC code) [99] Other: (no LOINC code) Value derived from I HSB 35571 – first part

Fetal Amnion | [1] Monoamniotic [2] Diamniotic [3] Triamniotic [99] Other Value derived from I HSB 35571 – second part.

Number of Fetuses Code | The number of fetuses and chorionicity/amnionicity expressed as SNOMED codes.

LMP Precision | How precise the associated LMP date is, according to the patient.

Sex of Individual Fetus | The fetal sex. Surprise (I HSB 35574 code) Female (LA21135-1) Male (LA21134-4) See overview (I HSB 35574 code)

Bundle.entry.resource.Observation.component.valueDateTime | String | False | | | Estimated Date of Conception - The estimated date of conception. EDD Basis Event Date - The date of the event. This, combined with either the “GA at Event” component or the “Days to EDD from Event” component, is used to calculate the EDD. For example, this could be the date than an ultrasound was performed. Last Menstrual Period Date – The first day of the patient’s last menstrual period.

Bundle.entry.resource.Observation.component.valueInteger | Int32 | True | | | This is the order of the delivery within OB History, which is not necessarily the same as calculated “birth order” based on outcome date/time.

Bundle.entry.resource.Observation.component.valueQuantity | Quantity | False | | | Cycle Length - The cycle length in number of days. Luteal Length - The luteal length in number of days. Gestational Age at Event - The Gestational age the patient was at the time of the event, represented in number of days. This, combined with the event date, is used to calculate the EDD.For example, this could be the gestational age determined during an ultrasound. Gestational Age is represented here in number of days Days to EDD from Event - The defined number of days to the EDD from the time of the event. This, combined with the event date, is used to calculate the EDD.

Bundle.entry.resource.Observation.derivedFrom | Reference | False | Yes | | If this EDD observation is the main selected EDD for the pregnancy, then derivedFrom contains a reference to every EDD event that has been documented on that episode, regardless of whether it was selected as the main EDD to use for the pregnancy.

Bundle.entry.resource.Observation.effectiveDateTime | dateTime | False | | | This is the outcome date/time of the delivery.

Bundle.entry.resource.Observation.effectivePeriod | Period | False | | | The date range over which there have been prenatal visits.

Bundle.entry.resource.Observation.focus | Reference | False | Yes | | The reference ID of one of the following: The pregnancy episode. The Delivery Resource Observation.

Bundle.entry.resource.Observation.hasMember | Reference | False | Yes | | A hasMember reference is included for each supported data point that contains data.

Bundle.entry.resource.Observation.interpretation | CodeableConcept | False | Yes | | Populated using SNOMED code 900000000000548007 (Preferred) when the entry represents the working expected delivery date. Otherwise, this property is absent.

Bundle.entry.resource.Observation.issued | instant | False | | | Returns the instant of one of the following: When this observation was last edited. The Contact date of the eptDAT (which is a history contact). The instant of entry for this EDD.

Bundle.entry.resource.Observation.method | CodeableConcept | False | | | The basis for this EDD. For example, the EDD could be based on an ultrasound.

Bundle.entry.resource.Observation.note | Annotation | False | Yes | | The comment for one of the following: The comment associated with the EDD. Free-text birth comment for the delivery. Living status comment. Delivery location comment.

Bundle.entry.resource.Observation.performer | Reference | False | Yes | | Returns the clinician’s name for one of the following: The clinician who documented the observation. The delivery clinician.

Bundle.entry.resource.Observation.status | code | False | | | Status is set to “final” except for special logic for the EDD: If the EDD is attached to the current pregnancy and the patient is marked as pregnant, this transmits as “preliminary.” Otherwise (the EDD is attached to a pregnancy and that pregnancy has been resolved or the patient is not pregnant), this transmits as “final.” If the pregnancy is deleted, then the Status is entered-in-error.

Bundle.entry.resource.Observation.subject | Reference | False | | | The patient

Bundle.entry.resource.Observation.extension (also: supportingInfo) | Extension | False | Yes | | The L&D Observation for this delivery. This is included only for real delivery records; historical delivery records do not have this focus.

Bundle.entry.resource.Observation.valueBoolean | boolean | True | | | Yes/No answer to whether there is a suspected fetal demise (I HSB 35585). The value is True when Preterm Labor is documented as True. The value is False when Preterm Labor is documented as false.

Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | False | | | Returns the value based on one of the following mappings: The mapped SNOMED code for outcome of pregnancy. Uses mapping table to map the documented sex to M(ale)/F(emale)/Un(known). The mapped SNOMED code for obstetric delivery method. Unmapped values are an Epic category-list-based code. Uses SNOMED mapping tables and LSD settings for the fetal vital status. All unmapped values are considered deceased.

Bundle.entry.resource.Observation.valueDateTime | dateTime | False | | | The date and time of one of the following: The estimated delivery date. The date and time of the delivery.

Bundle.entry.resource.Observation.valueInteger | integer | True | | | Returns the count of one of the following: Prenatal visits for the patient. Pregnancies (gravida) for the patient. The parity value for the patient. The number of term pregnancies for the patient. The number of preterm pregnancies for the patient. The total number of abortions for the patient. The number of spontaneous abortions for the patient. The number of induced abortions for the patient. The number of ectopic pregnancies for the patient. The number of molar pregnancies for the patient. The number of children that are still living for the patient. The number of live births for the patient. The number of pregnancies that had multiple fetuses. Apgar 1M value. Apgar 5M value.

Bundle.entry.resource.Observation.valueQuantity | Quantity | False | | | Returns one of the following values in its corresponding unit as defined below: Age at Menarche (in years) Age at First Pregnancy (in years) Months Breastfeeding (in months) Age at Menopause (in years) Age at First Live Birth (in years) Number of Fetuses (Before delivery, this represents the number of fetuses in the pregnancy. After delivery has occurred, however, this might be updated to the number of viable deliveries in the pregnancy (I HSB 35061).) Pregravid Weight (in ounces) Pregravid Height (in inches) Gestational Age (in days) First Stage of Labor (in minutes) Second Stage of Labor (in minutes) Birth Weight (in ounces)

Bundle.entry.resource.Observation.valueString | string | False | | | Returns one of the following comments written by the clinician: The gynecology history comment. The general obstetrics history summary comment. Or name information: Name of newborn. Name of the delivery location.

Bundle.entry.response | Response | False | | | A single resource returned as part of this bundle.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?patient=e80B2-qUU60-
    cEtsVdQ9VE4.p4bi6qYpMOeLfZaHcPgE3&category=obstetrics-gynecology
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 5,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?patient=e80B2-qUU60-
    cEtsVdQ9VE4.p4bi6qYpMOeLfZaHcPgE3&category=obstetrics-gynecology&code=11977-6,89470-9,55281-0,118215003"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwikTRrZ7fA1PovzAmiPxQ6M3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwikTRrZ7fA1PovzAmiPxQ6M3",
      "resource": {
        "resourceType": "Observation",
        "id": "eMssDN9DYPei0pJJo1EgTwikTRrZ7fA1PovzAmiPxQ6M3",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "obstetrics-gynecology",
                "display": "Obstetrics and Gynecology"
              }
            ],
            "text": "Obstetrics and Gynecology"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "11977-6",
              "display": "[#] Parity"
            }
          ],
          "text": "[#] Parity"
        },
        "subject": {
          "reference": "Patient/e80B2-qUU60-cEtsVdQ9VE4.p4bi6qYpMOeLfZaHcPgE3",
          "display": "Garcia, Sol"
        },
        "issued": "2021-12-20T06:00:00Z",
        "valueInteger": 2
      },
      "search": {
        "mode": "match"
      }
    },
    {
```

```
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwrQG-
jj5yzwBcR4IVheeObkvs6q5807K0QnFdpVcQbHL3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwrQG-
jj5yzwBcR4IVheeObkvs6q5807K0QnFdpVcQbHL3",
  "resource": {
    "resourceType": "Observation",
    "id": "eMssDN9DYPei0pJJo1EgTwrQG-jj5yzwBcR4IVheeObkvs6q5807K0QnFdpVcQbHL3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
            "code": "obstetrics-gynecology",
            "display": "Obstetrics and Gynecology"
          }
        ],
        "text": "Obstetrics and Gynecology"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "89470-9",
          "display": "Obstetrics Summary note"
        }
      ],
      "text": "Obstetrics Summary note"
    },
    "subject": {
      "reference": "Patient/e80B2-qUU60-cEtsVdQ9VE4.p4bi6qYpMOeLfZaHcPgE3",
      "display": "Garcia, Sol"
    },
    "issued": "2021-12-20T06:00:00Z",
    "valueString": "Patient has history of substance abuse"
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-
th5B7XaNt5hQ96ITxm2XJLjSYmZOuv7gGNq0GKA0zL8vVbrlQ3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-
th5B7XaNt5hQ96ITxm2XJLjSYmZOuv7gGNq0GKA0zL8vVbrlQ3",
  "resource": {
    "resourceType": "Observation",
    "id": "eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-th5B7XaNt5hQ96ITxm2XJLjSYmZOuv7gGNq0GKA0zL8vVbrlQ3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
            "code": "obstetrics-gynecology",
            "display": "Obstetrics and Gynecology"
          }
        ],
        "text": "Obstetrics and Gynecology"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "118215003",
          "display": "Delivery Finding"
        }
      ],
      "text": "Delivery Finding"
    },
    "subject": {
      "reference": "Patient/e80B2-qUU60-cEtsVdQ9VE4.p4bi6qYpMOeLfZaHcPgE3",
      "display": "Garcia, Sol"
    },
    "focus": [
      {
        "reference": "EpisodeOfCare/eWMewRsBhKsBNCCYGe9yqzvH6tjN1xgFWfpcmMPfyL5E3",
```

```
        "display": "Pregnancy (Historical)"
      }
    ],
    "effectiveDateTime": "2016-12-17",
    "issued": "2021-12-20T06:00:00Z",
    "performer": [
      {
        "display": "OBSTETRICS, PHYSICIAN"
      }
    ],
    "hasMember": [
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-th5B7VGcjufgeZAxvYwd11abrI33",
        "display": "Outcome of pregnancy"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-
th5B7VQ.o9fya7C39-lYfYtjjkJTIIePk6rrU44jPYWWiAkDQ3",
        "display": "Date and time of obstetric delivery"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-th5B7Xz73jNgla1UeSejSHibl8B3",
        "display": "Gestational age"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-th5B7WQfsRUNtUWeSEum1iWGCzX3",
        "display": "Baby sex"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-th5B7VYAo8-M2edPJorR5QxKuJN3",
        "display": "Preterm labor symptoms"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-
th5B7VrSvK5WCM7SRwOVK4hr0ORST4Mhw9Tw7RaPcX6HMFCxw3",
        "display": "Vital status"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-th5B7UyijbTxnBVEeNcblDjQsZN3",
        "display": "Baby name"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwvPFvW0qQ8IGHWVO-th5B7ULzBFEqJzXlp-TDh0uiO.A3",
        "display": "Place of birth"
      }
    ],
    "component": [
      {
        "code": {
          "text": "Delivery order"
        },
        "valueInteger": 1
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwsnWm1N9spdsIT8ZzJlR0vfFCNvaWSL
1G7IvV6lKGlrqp9r8DceONtRSCq-zH6-DaQ3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwsnWm1N9spdsIT8ZzJlR0vfFCNvaWSL
1G7IvV6lKGlrqp9r8DceONtRSCq-zH6-DaQ3",
  "resource": {
    "resourceType": "Observation",
    "id": "eMssDN9DYPei0pJJo1EgTwsnWm1N9spdsIT8ZzJlR0vfFCNvaWSL1G7IvV6lKGlrqp9r8DceONtRSCq-zH6-DaQ3",
    "extension": [
      {
        "valueReference": {
          "reference":
"Observation/efA7VQikwVCwfflTk6WQR0TC0XVFHyvEGbzYco7hzpLCLpx5xPXBDSTQxfJbd4ClitIHOKbVSkc58GV53fkV-9w3",
          "display": "Newborn delivery information"
        },
        "url": "http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo"
      }
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
            "code": "obstetrics-gynecology",
```

```
            "display": "Obstetrics and Gynecology"
          }
        ],
        "text": "Obstetrics and Gynecology"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "118215003",
          "display": "Delivery Finding"
        }
      ],
      "text": "Delivery Finding"
    },
    "subject": {
      "reference": "Patient/e80B2-qUU60-cEtsVdQ9VE4.p4bi6qYpMOeLfZaHcPgE3",
      "display": "Garcia, Sol"
    },
    "focus": [
      {
        "reference": "EpisodeOfCare/eMT3ThSb1l7eubKEVJbV7NjdKR2irBWt..4XvveNh4CY3",
        "display": "Pregnancy (Pregnancy)"
      }
    ],
    "effectiveDateTime": "2021-12-20T21:26:54Z",
    "issued": "2021-12-20T06:00:00Z",
    "performer": [
      {
        "reference": "Practitioner/eR21-ui08.7aMdcQcFuMzfw3",
        "display": "Physician Obstetrics, MD"
      }
    ],
    "hasMember": [
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwsnWm1N9spdsIT8ZzJlR0vdatAdhuz8-TBHgd778eJ5X3",
        "display": "Outcome of pregnancy"
      },
      {
        "reference":
"Observation/eMssDN9DYPei0pJJo1EgTwsnWm1N9spdsIT8ZzJlR0vcwU6S0NFi3AgkKGmBGzCEXGY.MfA53rTpn6lgI4fu8yg3",
        "display": "Date and time of obstetric delivery"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwsnWm1N9spdsIT8ZzJlR0vdslKXOxDi9O-IdeZqt5UBv3",
        "display": "Gestational age"
      },
      {
        "reference":
"Observation/eMssDN9DYPei0pJJo1EgTwsnWm1N9spdsIT8ZzJlR0vcv86.iwexysCesuyIoz0nSiTVZTSnVFRJVw4dkiGq-Fg3",
        "display": "Vital status"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwsnWm1N9spdsIT8ZzJlR0vdGmTik.oFMz8.Grw2OcZ6z3",
        "display": "Baby name"
      },
      {
        "reference": "Observation/eMssDN9DYPei0pJJo1EgTwsnWm1N9spdsIT8ZzJlR0vcsPfk09IAHtlAIgUgm-mdg3",
        "display": "Place of birth"
      }
    ],
    "component": [
      {
        "code": {
          "text": "Delivery order"
        },
        "valueInteger": 1
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwhp---
YUyCSpbp1DeiBYXqixr3y9mvZL7HP9x0JPQ6wJrTKnb6sK9UkOdA-.il064A3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eMssDN9DYPei0pJJo1EgTwhp---
YUyCSpbp1DeiBYXqixr3y9mvZL7HP9x0JPQ6wJrTKnb6sK9UkOdA-.il064A3",
  "resource": {
    "resourceType": "Observation",
    "id": "eMssDN9DYPei0pJJo1EgTwhp---YUyCSpbp1DeiBYXqixr3y9mvZL7HP9x0JPQ6wJrTKnb6sK9UkOdA-.il064A3",
    "status": "final",
```

```
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "obstetrics-gynecology",
                "display": "Obstetrics and Gynecology"
              }
            ],
            "text": "Obstetrics and Gynecology"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "55281-0",
              "display": "Number of fetuses"
            }
          ],
          "text": "Number of fetuses"
        },
        "subject": {
          "reference": "Patient/e80B2-qUU60-cEtsVdQ9VE4.p4bi6qYpMOeLfZaHcPgE3",
          "display": "Garcia, Sol"
        },
        "focus": [
          {
            "reference": "EpisodeOfCare/eMT3ThSb1l7eubKEVJbV7NjdKR2irBWt..4XvveNh4CY3",
            "display": "Pregnancy"
          }
        ],
        "issued": "2021-12-20T21:28:03Z",
        "performer": [
          {
            "reference": "Practitioner/eZGHFkrvIU05w7jV3oi5bIA3",
            "type": "Practitioner",
            "display": "Physician Obstetrics, MD"
          }
        ],
        "valueQuantity": {
          "value": 1
        },
        "component": [
          {
            "code": {
              "coding": [
                {
                  "system": "http://loinc.org",
                  "code": "55281-0",
                  "display": "Number of fetuses"
                }
              ],
              "text": "Number of fetuses"
            },
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "237244005",
                  "display": "Single pregnancy"
                }
              ],
              "text": "Single pregnancy"
            }
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


13.1.24 Observation.Search (Periodontal) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10071-Observation-Search-Periodontal-R4.html
Epic API ID | 10071
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021

HTTP method | GET
URL template | /api/FHIR/R4/Observation?category={category}&date={date}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10071

Description

The periodontal Observation sub-resource can return all of the periodontal readings documented on a patient's tooth during a periodontal exam. They include probing depths, gingival margins, and clinical attachment levels.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | True | | | Always set to "periodontal" to return dental data.
date | date | True | | | Accepts a single date or a date range to search for periodontal data. Date corresponds to the marked date of the periodontal exam. No date means readings from the last exam will be listed.
patient | reference | True | | | The patient to search data for.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Unsupported
code | token | True | | | If provided, should always be set to SNOMED code 307536006: "Examination of periodontium (procedure)."
combo-code | token | True | | | Matches to the code or component.code response element. Refer to those response elements for additional information.
combo-data-absent-reason | token | True | | | Unsupported
combo-value-concept | token | True | | | Unsupported
component-code | token | True | | | Reference to a specific reading type and location. Types are Probing Depth, Gingival Margin, or Clinical Attachment Level. Location is the gum site where the measurement was taken for a particular tooth, such as lingual distal.
component-data-absent-reason | token | True | | | Unsupported
component-value-concept | token | True | | | Unsupported
data-absent-reason | token | True | | | Unsupported
derived-from | String | True | | | Unsupported
device | String | True | | | Unsupported
encounter | reference | True | | | The encounter this periodontal reading was noted on.
focus | reference | True | | | A link to the tooth structure where this periodontal reading was documented.
has-member | reference | True | | | Unsupported
identifier | token | True | | | Unsupported
issued | date | True | | | Unsupported
method | token | True | | | Unsupported
part-of | String | True | | | Unsupported
performer | reference | True | | | The provider who documented the periodontal reading.
specimen | String | True | | | Unsupported
status | token | True | | | Always "final" for periodontal observations.
value-concept | token | True | | | Unsupported
value-date | date | True | | | Unsupported
value-string | string | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Observation | Observation | False | | | An instance of the R4 Observation resource.
Bundle.entry.resource.Observation.bodySite | CodeableConcept | True | | | CodeableConcept value.
Bundle.entry.resource.Observation.category | CodeableConcept | True | Yes | | CodeableConcept value.
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | CodeableConcept value.
Bundle.entry.resource.Observation.component | BackboneElement | True | Yes | | Populated only for patient-entered flowsheets.
Bundle.entry.resource.Observation.component.code | CodeableConcept | False | | | CodeableConcept value.
Bundle.entry.resource.Observation.component.interpretation | CodeableConcept | True | Yes | | CodeableConcept value.
Bundle.entry.resource.Observation.component.referenceRange | BackboneElement | True | Yes
Bundle.entry.resource.Observation.component.valueQuantity | Quantity | True | | | Quantity value.
Bundle.entry.resource.Observation.effectiveDateTime | dateTime | True
Bundle.entry.resource.Observation.encounter | Reference | True | | | The encounter this periodontal reading was noted on.
Bundle.entry.resource.Observation.focus | Reference | True | Yes | | A link to the tooth structure where this periodontal reading was documented.
Bundle.entry.resource.Observation.performer | Reference | True | Yes | | A FHIR reference.
Bundle.entry.resource.Observation.status | code | False | | | Returns "final."
Bundle.entry.resource.Observation.subject | Reference | True | | | A FHIR reference.

Sample request

https://hostname/instance/api/FHIR/R4/Observation?category=periodontal&subject=e6DDE5.4tu-jjWXY8CtuVKw3&date=2021-01

Sample response


```
<Bundle xmlns="http://hl7.org/fhir">
<type value="searchset" />
<total value="8" />
<link>
<relation value="self" />
<url value="https://hostname/instance/api/FHIR/R4/Observation?category=periodontal&amp;subject=e6DDE5.4tu-
    jjWXY8CtuVKw3&amp;date=2021-01" />
</link>
<entry>
<link>
<relation value="self" />
<url value="https://hostname/instance/api/FHIR/R4/Observation/e6fALeOD7ffdnkI-
    BibQjvi.XAIz1Yz8AAoqPeeEJpiSrPa32OUUT8ea7W6yvmrlecv341jOJaSsCNaxFyGkJMw3" />
</link>
<fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/e6fALeOD7ffdnkI-
    BibQjvi.XAIz1Yz8AAoqPeeEJpiSrPa32OUUT8ea7W6yvmrlecv341jOJaSsCNaxFyGkJMw3" />
<resource>
<Observation>
<id value="some_id />
<status value="final" />
<category>
<coding>
<system value="http://open.epic.com/FHIR/StructureDefinition/observation-category" />
<code value="periodontal" />
<display value="Periodontal" />
</coding>
<text value="Periodontal" />
</category>
<code>
<coding>
<system value="urn:oid:2.16.840.1.113883.6.96" />
<code value="307536006" />
</coding>
<text value="Examination of periodontium (procedure)" />
</code>
<subject>
<reference value="Patient/ e6DDE5.4tu-jjWXY8CtuVKw3" />
<display value="Lacroix, Diego" />
</subject>
<focus>
<reference value="BodyStructure/ec.bWXnOuxgyRAN8CLHCBxWlQPYNAfGid39k5uR9FFnY3" />
<type value="BodyStructure" />
</focus>
<encounter>
<reference value="Encounter/ejsLjQ9QeqL3FtFg09AuzXnouJzKaY8k1vzQkzZKyww43" />
<type value="Encounter" />
</encounter>
<effectiveDateTime value="2021-01-05" />
<performer>
<reference value="Practitioner/eNzt.GkOzCBShLeRDkLPGfw3" />
<type value="Practitioner" />
<display value="Gordon Lofgren, DDS" />
</performer>
<bodySite>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.3.866976.94010" />
<code value="1" />
</coding>
<text value="Tooth (Universal): 1" />
</bodySite>
<component>
<code>
<coding>
<display value="Probing Depth - Facial Distal" />
</coding>
</code>
<valueQuantity>
<value value="2" />
<unit value="mm" />
</valueQuantity>
<interpretation>
<coding>
<system value="urn:oid:2.16.840.1.113883.5.83" />
<code value="N" />
<display value="Normal" />
</coding>
<text value="Normal" />
</interpretation>
<referenceRange>
<high>
<value value="3" />
<unit value="mm" />
</high>
</referenceRange>
<component>
```

```
<code>
<coding>
<display value="Clinical Attachment Level - Lingual Distal" />
</coding>
</code>
<valueQuantity>
<value value="2" />
<unit value="mm" />
</valueQuantity>
</component>
</Observation>
```

13.1.25 Procedure.Create (External Radiotherapy Summary) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11048-Procedure-Create-External-Radiotherapy-Summary-R4.html
Epic API ID | 11048
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | POST
URL template | /api/FHIR/R4/Procedure
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11048

Description

The Procedure resource is used to represent radiotherapy delivery progress at all levels (courses, phases, and plans). A Procedure for a course, phase, or plan indicates that the course, phase, or plan has progressed from prescribed/planned to treated.

There are three types of Procedure resources: Course Summary, Treated Phase, and Treated Plan. The table below shows the profile, category, and code used in each.

Type | Profile | Category | Code
Course Summary | Radiotherapy Course Summary - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1217123003 (Course)
Treated Phase | Radiotherapy Treated Phase - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1222565005 (Phase)
Treated Plan | Radiotherapy Treated Plan - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1255724003 (Plan)

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Procedure (R4) | Procedure
Procedure (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Course Summary: Can reference a ServiceRequest course prescription or planned course that this course summary fulfills. Treated Phase: Can reference a ServiceRequest phase prescription or planned phase that this treated phase fulfills. Treated Plan: Can reference a ServiceRequest plan prescription or treatment plan that this treated plan fulfills.
Procedure (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.
Procedure (R4).bodySite.coding | Coding | | Yes | 1..* | The coded representation of the anatomical site.
Procedure (R4).bodySite.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-treatment-location-vs.
Procedure (R4).bodySite.coding.display | String | | | 0..1 | The textual representation of the coded value in the given system.
Procedure (R4).bodySite.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
Procedure (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, the laterality of the body site (left, right, bilateral).
Procedure (R4).bodySite.extension (lateralityQualifier).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier
Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept | CodeableConcept | | | 0..1 | CodeableConcept container for the locationQualifier.
Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the lateralityQualifier.
Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the lateralityQualifier. The value must be a member of https://hl7.org/fhir/us/mcode/STU4/ValueSet-mcode-laterality-qualifier-vs.html.
Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The display name of the SNOMED code.
Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.
Procedure (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, additional qualifiers specifying the body site being treated.

Procedure (R4).bodySite.extension (locationQualifier).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept | CodeableConcept | | | 0..* | CodeableConcept container for the locationQualifier.

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the locationQualifier.

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the locationQualifier. The value must be a member of https://hl7.org/fhir/us/mcode/STU4/ValueSet-mcode-body-location-qualifier-vs.html.

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The display name of the SNOMED code.

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.

Procedure (R4).category | CodeableConcept | | | 1..* | Categories that apply to this Procedure.

Procedure (R4).category.coding | Coding | | Yes | 0..1 | The coded representation of the category.

Procedure (R4).category.coding.code | String | | | 1..1 | Always the SNOMED code for Radiotherapy (procedure): 1287742003.

Procedure (R4).category.coding.display | String | | | 0..1 | The textual representation of the coded value. Only allowed value: "Radiotherapy (procedure)".

Procedure (R4).category.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).category.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).code | CodeableConcept | | | 1..1 | The coded representation of the Procedure.

Procedure (R4).code.coding | Coding | | Yes | 1..1 | The coded representation of the concept.

Procedure (R4).code.coding.code | String | | | 1..1 | The code of the service being requested. Possible values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)

Procedure (R4).code.coding.display | String | | | 1..1 | The textual representation of the coded value. For course types, the value must be "Radiotherapy course of treatment (regime/therapy)". For phase types, the value must be "Radiotherapy treatment phase (regime/therapy)". For plan types, the value must be "Radiotherapy treatment plan (regime/therapy)".

Procedure (R4).code.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).code.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | Indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before.

Procedure (R4).extension (courseInvolvesReirradiation).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation

Procedure (R4).extension (courseInvolvesReirradiation).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

Procedure (R4).extension (doseDeliveredToVolume) (also: extension (doseDeliveredToVolume)) | Extension | | Yes | 0..* | Specifies doses delivered to one or more volumes as part of this regimen.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | Specifies the fractional dose prescribed for this volume.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).url | uri | | | 1..1 | Must be: fractionDose

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity | Quantity | | | 1..1 | Quantity of the dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity.code | String | | | 1..1 | Must be: cGy

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | Specifies how many fractions are prescribed for this volume in the course.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractions).url | uri | | | 1..1 | Must be: fractions

Procedure (R4).extension (doseDeliveredToVolume).extension (fractions).valuePositiveInt | positiveInt | | | 1..1 | The number of fractions prescribed.

Procedure (R4).extension (doseDeliveredToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen.

Procedure (R4).extension (doseDeliveredToVolume).extension (pointDose).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose

Procedure (R4).extension (doseDeliveredToVolume).extension (pointDose).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

Procedure (R4).extension (doseDeliveredToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan.

Procedure (R4).extension (doseDeliveredToVolume).extension (primaryPlanDose).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose

Procedure (R4).extension (doseDeliveredToVolume).extension (primaryPlanDose).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | Provides a more tightly specified metric for the given dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric).valueQuantity | Quantity | | | 1..1 | The quantity of the radiobiologic metric.

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric).valueQuantity.unit | String | | | 1..1 | Must be a value as described in http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric.

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric).valueQuantity.value | Decimal | | | 1..1 | The number representing the dose according to the given radiobiologic metric.

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | Specifies the total dose prescribed for this volume.

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).url | uri | | | 1..1 | Must be: totalDose

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity | Quantity | | | 1..1 | Quantity of the total dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity.code | String | | | 1..1 | Must be: cGy

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | Identifies a volume to be treated with radiation.

Procedure (R4).extension (doseDeliveredToVolume).extension (volume).url | uri | | | 1..1 | Must be: volume

Procedure (R4).extension (doseDeliveredToVolume).extension (volume).valueReference | Reference | | | 1..1 | A reference to the radiotherapy volume (BodyStructure) whose dose is being described.

Procedure (R4).extension (doseDeliveredToVolume).extension(uniformFractionation) (also: extension(uniformFractionation)) | Extension | | Yes | 0..1 | Indicates whether or not every fraction for this volume was the same dose for the entire course.

Procedure (R4).extension (doseDeliveredToVolume).extension(uniformFractionation).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-uniform-fractionation

Procedure (R4).extension (doseDeliveredToVolume).extension(uniformFractionation).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

Procedure (R4).extension (doseDeliveredToVolume).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume

Procedure (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, defines what free-breathing method is used.

Procedure (R4).extension (freeBreathingMotionManagementTechnique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept | CodeableConcept | | | 1..1 | The CodeableConcept container for the free-breathing method.

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the free-breathing method.

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-free-breathing-motion-mgmt-technique-vs

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.display | String | | | 0..1 | The textual representation of the coded value in the given system.

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used.

Procedure (R4).extension (imageGuidedRadiotherapyModality).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the image guidance.

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the imaging guidance.

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/STU2/ValueSet-codexrt-image-guided-radiotherapy-modality-vs.

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification.

Procedure (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | Defines the imaging procedure used for intrafraction verification.

Procedure (R4).extension (intrafractionVerification).extension (procedure).url | uri | | | 1..1 | Must be: procedure

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the procedure.

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the procedure.

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-intrafraction-verification-procedure-vs

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | Further describes the imaging used for intrafraction verification.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).url | uri | | | 1..1 | Must be: qualifier

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the qualifier.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the qualifier.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-intrafraction-verification-qualifier-vs.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (intrafractionVerification).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification

Procedure (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation.

Procedure (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | Specifies the modality of the radiation used.

Procedure (R4).extension (modalityAndTechnique).extension (modality).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the modality.

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the modality.

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-modality-vs

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, giving either the beam energy (for external-beam modalities) or source isotope (for brachytherapy modalities).

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept | CodeableConcept | | | 0..1 | If specifying an isotope, a CodeableConcept representation of the isotope as a SNOMED code.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the isotope.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-isotope-vs.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity | Quantity | | | 0..1 | If specifying a field energy, a Quantity representing the energy in kV, MV, or MeV.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.code | String | | | 1..1 | Must be from the value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-energy-unit-vs.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.value | Decimal | | | 1..1 | Value of the energy.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | Defines what type of applicator is used for a brachytherapy source.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the type of applicator.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the type of applicator.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-brachytherapy-applicator-type-vs.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | Specifies the technique of the radiation used, which is more specific than the modality.

Procedure (R4).extension (modalityAndTechnique).extension (technique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the technique.

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the technique.

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-technique-vs.

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.system | String | | | 1..1 | Must be http://snomed.info/sct

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (modalityAndTechnique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique

Procedure (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, defines what method is used.

Procedure (R4).extension (respiratoryMotionManagement).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept | CodeableConcept | | | 1..1 | The CodeableConcept container for the motion management.

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the motion management.

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-respiratory-motion-management-vs

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.

Procedure (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.

Procedure (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).

Procedure (R4).identifier.value | String | | | 1..1 | The value of the identifier.

Procedure (R4).meta | Meta

Procedure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).

Procedure (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this Procedure resource. The following profiles are supported: Profile Name Profile Value Code Radiotherapy Course Summary http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary 1217123003 Radiotherapy Treated Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase 1222565005 Radiotherapy Treated Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan 1255724003

Profile Name | Profile Value | Code

Radiotherapy Course Summary | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary | 1217123003

Radiotherapy Treated Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase | 1222565005

Radiotherapy Treated Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan | 1255724003

Procedure (R4).partOf | Reference | | Yes | 0..* | Reference to a Procedure resource of which this Procedure is a component. For a treated phase Procedure, this can reference a course summary Procedure. For a treated plan Procedure, this can reference a treated phase Procedure.

Procedure (R4).performedPeriod | Period | | | 0..1 | The time during which this procedure took place.

Procedure (R4).performedPeriod.end | String | | | 0..1 | The end of the period.

Procedure (R4).performedPeriod.start | String | | | 0..1 | The start of the period.

Procedure (R4).performer | BackboneElement | | Yes | 0..1 | The person that performed the procedure.

Procedure (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.

Procedure (R4).reasonCode.coding | Coding | | Yes | 0..1 | Coded representation of the problem.

Procedure (R4).reasonCode.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-cancer-disorder-vs.

Procedure (R4).reasonCode.coding.display | String | | | 0..1 | The textual representation of the coded value.

Procedure (R4).reasonCode.coding.system | String | | | 0..1 | The URI for the system in which this coded value is assigned. Preferred values: http://snomed.info/sct http://hl7.org/fhir/sid/icd-10-cm

Procedure (R4).reasonCode.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.

Procedure (R4).status | code | | | 1..1 | The status of the course/phase/plan. The value should be from the value set http://hl7.org/fhir/R4/ValueSet/event-status

Procedure (R4).subject | Reference | | | 1..1 | Reference to the Patient FHIR resource that this radiotherapy course/plan/phase is planned or prescribed to.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Procedure (R4) | Procedure

Procedure (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Course Summary: Can reference a ServiceRequest course prescription or planned course that this course summary fulfills. Treated Phase: Can reference a ServiceRequest phase prescription or planned phase that this treated phase fulfills. Treated Plan: Can reference a ServiceRequest plan prescription or treatment plan that this treated plan fulfills.

Procedure (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.

Procedure (R4).bodySite.coding | Coding | | Yes | 1..* | The SNOMED code for this anatomical site.

Procedure (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, the laterality of the anatomical site (left, right, bilateral). URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier

Procedure (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, additional qualifiers specifying the anatomical site being treated. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer

Procedure (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.

Procedure (R4).category | CodeableConcept | | | 1..* | Always the SNOMED code for Radiotherapy (procedure): 1287742003.

Procedure (R4).code | CodeableConcept | | | 1..1 | The code of the service being requested. Possible values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)

Procedure (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | Indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation

Procedure (R4).extension (doseDeliveredToVolume) (also: extension (doseDeliveredToVolume)) | Extension | | Yes | 0..* | Specifies doses delivered to one or more volumes as part of this regimen. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionsDelivered) (also: extension (fractionsDelivered)) | Extension | | Yes | 0..1 | Specifies the fractions delivered for the volume. URL: fractionsDelivered

Procedure (R4).extension (doseDeliveredToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose

Procedure (R4).extension (doseDeliveredToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | Provides a more tightly specified metric for the given dose. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDoseDelivered) (also: extension (totalDoseDelivered)) | Extension | | Yes | 1..1 | Specifies the total dose delivered to this volume so far. URL: totalDoseDelivered

Procedure (R4).extension (doseDeliveredToVolume).extension (uniformFractionation) (also: extension (uniformFractionation)) | Extension | | Yes | 0..1 | Indicates whether or not every fraction for this volume was the same dose for the entire course. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-uniform-fractionation

Procedure (R4).extension (doseDeliveredToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | A reference to the radiotherapy volume (BodyStructure FHIR resource) whose dose is being described. URL: volume

Procedure (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, defines what free-breathing method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique

Procedure (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality

Procedure (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification

Procedure (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | Defines the imaging procedure used for intrafraction verification. URL: procedure

Procedure (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | Further describes the imaging used for intrafraction verification. URL: qualifier

Procedure (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique

Procedure (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | Specifies the modality of the radiation used. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, giving either the beam energy (for external-beam modalities) or source isotope (for brachytherapy modalities). URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | Defines what type of applicator is used for a brachytherapy source. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

Procedure (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | Specifies the technique of the radiation used, which is more specific than the modality. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

Procedure (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, defines what method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management

Procedure (R4).id | id | | | 0..1 | The Procedure FHIR ID.

Procedure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.

Procedure (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.

Procedure (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).

Procedure (R4).identifier.value | String | | | 1..1 | The value of the identifier.

Procedure (R4).meta | Meta

Procedure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).

Procedure (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this Procedure resource. The following profiles are supported: Profile Name Profile Value Code Radiotherapy Course Summary http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary 1217123003 Radiotherapy Treated Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase 1222565005 Radiotherapy Treated Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan 1255724003

Profile Name | Profile Value | Code

Radiotherapy Course Summary | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary | 1217123003

Radiotherapy Treated Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase | 1222565005

Radiotherapy Treated Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan | 1255724003

Procedure (R4).partOf | Reference | | Yes | 0..* | Reference to a Procedure resource of which this Procedure is a component. For a treated phase Procedure, this can reference a course summary Procedure. For a treated plan Procedure, this can reference a treated phase Procedure.

Procedure (R4).performedPeriod | Period | | | 0..1 | The time during which this procedure took place.

Procedure (R4).performer | BackboneElement | | Yes | 0..1 | The person that performed the procedure.

Procedure (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.

Procedure (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.

Procedure (R4).status | code | | | 1..1 | The status of the course/phase/plan. Must be a value defined in http://hl7.org/fhir/R4/ValueSet/event-status

Procedure (R4).statusReason | CodeableConcept | | | 0..1 | Only supported for Course Summary profiles. Reason for the current status as defined in SNOMED URL: http://hl7.org/fhir/us/mcode/ValueSet/mcode-treatment-termination-reason-vs

Procedure (R4).subject | Reference | | | 1..1 | A FHIR reference.

Sample request

```
POST https://hostname/instance/api/FHIR/R4/Procedure
{
  "resourceType": "Procedure",
  "meta": {
    "lastUpdated": "2026-02-01T09:09:51-06:00",
    "profile": [
      "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary",
      "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-course-summary"
    ]
  },
  "category": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "1287742003"
      }
    ]
  },
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "1217123003"
      }
    ]
  },
  "basedOn": [
    {
      "reference": "ServiceRequest/BZi6oojEVTlpWA7JJYw3-fJVBYgnhYSRQ1Cx1qK8JWius"
    },
    {
      "reference": "ServiceRequest/e214EEAa7RY8csEhIEJW-qOpDYFRKCusulwMwwzo333g3",
      "display": "Bilateral Breast"
    }
  ],
  "identifier": [
    {
      "use": "usual",
      "system": "urn:hand:display",
      "value": "Bilateral Breast"
    },
    {
      "use": "official",
      "system": "urn:dicom:uid",
      "value": "22542ccf-6d72-4745-a9b0-79fc87ed513e"
    }
  ],
  "status": "in-progress",
  "subject": {
    "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
    "display": "Test, Patient"
  },
  "performedPeriod": {
    "start": "2026-02-01T09:09:51-06:00",
    "end": ""
  },
  "reasonCode": [
    {
```

```
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "C50",
        "display": "Malignant neoplasm of breast"
      }
    ],
    "text": "Cancer Problem"
  }
],
"reasonReference": [
  {
    "reference": "Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3"
  }
],
"bodySite": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "Breast"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangent"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "7771000",
              "display": "Left"
            }
          ]
        }
      }
    ]
  },
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "Breast"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangent"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "24028007",
              "display": "Right"
            }
          ]
        }
      }
    ]
  },
  {
    "coding": [
      {
```

```
        "system": "http://snomed.info/sct",
        "code": "68171009",
        "display": "Axillary lymph node"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "7771000",
              "display": "Left"
            }
          ]
        }
      }
    ]
  }
],
"location": {
  "reference": "Location/e-iz4EXUrNFfCKJwdsZPd9Q3 "
},
"extension": [
  {
    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": 129428001,
          "display": "Preventive"
        }
      ]
    }
  },
  {
    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique",
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "1156506007",
              "display": "Photons"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": 1162782007,
              "display": "3D"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-
isotope",
        "valueQuantity": {
          "value": "18",
          "code": "MeV",
          "system": "http://unitsofmeasure.org"
        }
      }
    ]
  },
  {
    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-sessions",
    "valueUnsignedInt": "1"
  },
  {
    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume",
    "extension": [
      {
        "url": "volume",
        "valueReference": {
          "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
        }
      },
      {
```

```
          "url": "totalDoseDelivered",
          "valueQuantity": {
            "value": "266",
            "system": "http://unitsofmeasure.org",
            "code": "cGy"
          }
        },
        {
          "url": "fractionsDelivered",
          "valueUnsignedInt": "1"
        },
        {
          "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-
    dose",
          "valueBoolean": true
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume",
      "extension": [
        {
          "url": "volume",
          "valueReference": {
            "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
          }
        },
        {
          "url": "totalDoseDelivered",
          "valueQuantity": {
            "value": "266",
            "system": "http://unitsofmeasure.org",
            "code": "cGy"
          }
        },
        {
          "url": "fractionsDelivered",
          "valueUnsignedInt": "1"
        },
        {
          "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-
    dose",
          "valueBoolean": true
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume",
      "extension": [
        {
          "url": "volume",
          "valueReference": {
            "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
          }
        },
        {
          "url": "totalDoseDelivered",
          "valueQuantity": {
            "value": "0",
            "system": "http://unitsofmeasure.org",
            "code": "cGy"
          }
        },
        {
          "url": "fractionsDelivered",
          "valueUnsignedInt": "0"
        },
        {
          "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-
    dose",
          "valueBoolean": true
        }
      ]
    }
  ]
}
```


Sample response

```
Status code: Created (201)
Content-Type: application/json; charset=utf-8
Location: Procedure/edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3
```

13.1.26 Procedure.Read (External Radiotherapy Summary) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11049-Procedure-Read-External-Radiotherapy-Summary-R4.html
Epic API ID | 11049

FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
HTTP method | GET
URL template | /api/FHIR/R4/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11049

Description

The Procedure resource is used to represent radiotherapy delivery progress at all levels (courses, phases, and plans). A Procedure for a course, phase, or plan indicates that the course, phase, or plan has progressed from prescribed/planned to treated.

There are three types of Procedure resources: Course Summary, Treated Phase, and Treated Plan. The table below shows the profile, category, and code used in each.

Type | Profile | Category | Code
Course Summary | Radiotherapy Course Summary - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1217123003 (Course)
Treated Phase | Radiotherapy Treated Phase - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1222565005 (Phase)
Treated Plan | Radiotherapy Treated Plan - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1255724003 (Plan)

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | Procedure FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure (R4) | Procedure
Procedure (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Course Summary: Can reference a ServiceRequest course prescription or planned course that this course summary fulfills. Treated Phase: Can reference a ServiceRequest phase prescription or planned phase that this treated phase fulfills. Treated Plan: Can reference a ServiceRequest plan prescription or treatment plan that this treated plan fulfills.
Procedure (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.
Procedure (R4).bodySite.coding | Coding | | Yes | 1..* | The SNOMED code for this anatomical site.
Procedure (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, the laterality of the anatomical site (left, right, bilateral). URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier
Procedure (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, additional qualifiers specifying the anatomical site being treated. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer
Procedure (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.
Procedure (R4).category | CodeableConcept | | | 1..* | Always the SNOMED code for Radiotherapy (procedure): 1287742003.
Procedure (R4).code | CodeableConcept | | | 1..1 | The code of the service being requested. Possible values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)
Procedure (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | Indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation
Procedure (R4).extension (doseDeliveredToVolume) (also: extension (doseDeliveredToVolume)) | Extension | | Yes | 0..* | Specifies doses delivered to one or more volumes as part of this regimen. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume
Procedure (R4).extension (doseDeliveredToVolume).extension (fractionsDelivered) (also: extension (fractionsDelivered)) | Extension | | Yes | 0..1 | Specifies the fractions delivered for the volume. URL: fractionsDelivered
Procedure (R4).extension (doseDeliveredToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose
Procedure (R4).extension (doseDeliveredToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose
Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | Provides a more tightly specified metric for the given dose. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric
Procedure (R4).extension (doseDeliveredToVolume).extension (totalDoseDelivered) (also: extension (totalDoseDelivered)) | Extension | | Yes | 1..1 | Specifies the total dose delivered to this volume so far. URL: totalDoseDelivered
Procedure (R4).extension (doseDeliveredToVolume).extension (uniformFractionation) (also: extension (uniformFractionation)) | Extension | | Yes | 0..1 | Indicates whether or not every fraction for this volume was the same dose for the entire course. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-uniform-fractionation
Procedure (R4).extension (doseDeliveredToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | A reference to the radiotherapy volume (BodyStructure FHIR resource) whose dose is being described. URL: volume

Procedure (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, defines what free-breathing method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique

Procedure (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality

Procedure (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification

Procedure (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | Defines the imaging procedure used for intrafraction verification. URL: procedure

Procedure (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | Further describes the imaging used for intrafraction verification. URL: qualifier

Procedure (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique

Procedure (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | Specifies the modality of the radiation used. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, giving either the beam energy (for external-beam modalities) or source isotope (for brachytherapy modalities). URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | Defines what type of applicator is used for a brachytherapy source. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

Procedure (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | Specifies the technique of the radiation used, which is more specific than the modality. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

Procedure (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, defines what method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management

Procedure (R4).id | id | | | 0..1 | The Procedure FHIR ID.

Procedure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.

Procedure (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.

Procedure (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).

Procedure (R4).identifier.value | String | | | 1..1 | The value of the identifier.

Procedure (R4).meta | Meta

Procedure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).

Procedure (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this Procedure resource. The following profiles are supported: Profile Name Profile Value Code Radiotherapy Course Summary http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary 1217123003 Radiotherapy Treated Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase 1222565005 Radiotherapy Treated Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan 1255724003

Profile Name | Profile Value | Code

Radiotherapy Course Summary | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary | 1217123003

Radiotherapy Treated Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase | 1222565005

Radiotherapy Treated Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan | 1255724003

Procedure (R4).partOf | Reference | | Yes | 0..* | Reference to a Procedure resource of which this Procedure is a component. For a treated phase Procedure, this can reference a course summary Procedure. For a treated plan Procedure, this can reference a treated phase Procedure.

Procedure (R4).performedPeriod | Period | | | 0..1 | The time during which this procedure took place.

Procedure (R4).performer | BackboneElement | | Yes | 0..1 | The person that performed the procedure.

Procedure (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.

Procedure (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.

Procedure (R4).status | code | | | 1..1 | The status of the course/phase/plan. Must be a value defined in http://hl7.org/fhir/R4/ValueSet/event-status

Procedure (R4).statusReason | CodeableConcept | | | 0..1 | Only supported for Course Summary profiles. Reason for the current status as defined in SNOMED URL: http://hl7.org/fhir/us/mcode/ValueSet/mcode-treatment-termination-reason-vs

Procedure (R4).subject | Reference | | | 1..1 | A FHIR reference.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Procedure/edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3
```

Sample response


```
{
  "resourceType": "Procedure",
  "id": "edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3",
  "meta": {
    "lastUpdated": "2026-02-01T09:09:51-06:00",
    "profile": [
      "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary",
      "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-course-summary"
    ]
  },
  "category": {
    "coding": [
```

```
    {
      "system": "http://snomed.info/sct",
      "code": "1287742003"
    }
  ]
},
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "1217123003"
    }
  ]
},
"basedOn": [
  {
    "reference": "ServiceRequest/BZi6oojEVTlpWA7JJYw3-fJVBYgnhYSRQ1Cx1qK8JWius"
  },
  {
    "reference": "ServiceRequest/e214EEAa7RY8csEhIEJW-qOpDYFRKCusulwMwwzo333g3",
    "display": "Bilateral Breast"
  }
],
"identifier": [
  {
    "use": "usual",
    "system": "urn:hand:display",
    "value": "Bilateral Breast"
  },
  {
    "use": "official",
    "system": "urn:dicom:uid",
    "value": "22542ccf-6d72-4745-a9b0-79fc87ed513e"
  }
],
"status": "in-progress",
"subject": {
  "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
  "display": "Test, Patient"
},
"performedPeriod": {
  "start": "2026-02-01T09:09:51-06:00",
  "end": ""
},
"reasonCode": [
  {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "C50",
        "display": "Malignant neoplasm of breast"
      }
    ],
    "text": "Cancer Problem"
  }
],
"reasonReference": [
  {
    "reference": "Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3"
  }
],
"bodySite": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "Breast"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangent"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
```

```
              "code": "7771000",
              "display": "Left"
            }
          ]
        }
      }
    ]
  },
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "Breast"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangent"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "24028007",
              "display": "Right"
            }
          ]
        }
      }
    ]
  },
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "68171009",
        "display": "Axillary lymph node"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "7771000",
              "display": "Left"
            }
          ]
        }
      }
    ]
  }
],
"location": {
  "reference": "Location/e-iz4EXUrNFfCKJwdsZPd9Q3"
},
"extension": [
  {
    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": 129428001,
          "display": "Preventive"
        }
      ]
    }
  },
  {
    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique",
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality",
        "valueCodeableConcept": {
```

```
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "1156506007",
            "display": "Photons"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": 1162782007,
            "display": "3D"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-
isotope",
      "valueQuantity": {
        "value": "18",
        "code": "MeV",
        "system": "http://unitsofmeasure.org"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-sessions",
  "valueUnsignedInt": "1"
},
{
  "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume",
  "extension": [
    {
      "url": "volume",
      "valueReference": {
        "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
      }
    },
    {
      "url": "totalDoseDelivered",
      "valueQuantity": {
        "value": "266",
        "system": "http://unitsofmeasure.org",
        "code": "cGy"
      }
    },
    {
      "url": "fractionsDelivered",
      "valueUnsignedInt": "1"
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-
dose",
      "valueBoolean": true
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume",
  "extension": [
    {
      "url": "volume",
      "valueReference": {
        "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
      }
    },
    {
      "url": "totalDoseDelivered",
      "valueQuantity": {
        "value": "266",
        "system": "http://unitsofmeasure.org",
        "code": "cGy"
      }
    },
    {
      "url": "fractionsDelivered",
      "valueUnsignedInt": "1"
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-
dose",
      "valueBoolean": true
```

```
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume",
      "extension": [
        {
          "url": "volume",
          "valueReference": {
            "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
          }
        },
        {
          "url": "totalDoseDelivered",
          "valueQuantity": {
            "value": "0",
            "system": "http://unitsofmeasure.org",
            "code": "cGy"
          }
        },
        {
          "url": "fractionsDelivered",
          "valueUnsignedInt": "0"
        },
        {
          "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-
    dose",
          "valueBoolean": true
        }
      ]
    }
  ]
}
```


13.1.27 Procedure.Search (External Radiotherapy Summary) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11050-Procedure-Search-External-Radiotherapy-Summary-R4.html
Epic API ID | 11050
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Procedure?category={category}&code={code}&date={date}&identifier={identifier}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11050

Description

The Procedure resource is used to represent radiotherapy delivery progress at all levels (courses, phases, and plans). A Procedure for a course, phase, or plan indicates that the course, phase, or plan has progressed from prescribed/planned to treated.

There are three types of Procedure resources: Course Summary, Treated Phase, and Treated Plan. The table below shows the profile, category, and code used in each.

Type | Profile | Category | Code
Course Summary | Radiotherapy Course Summary - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1217123003 (Course)
Treated Phase | Radiotherapy Treated Phase - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1222565005 (Phase)
Treated Plan | Radiotherapy Treated Plan - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1255724003 (Plan)

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others are ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Optional | Yes | | Procedure category, this is always a fixed code for all CodeX RT profiles.
code | token | Optional | Yes | | Procedure code, these are fixed values for the CodeX RT profiles to distinguish between courses, phases, and plans.
date | date | Optional | Yes | | The date for a radiotherapy procedure. This is matched against the start/end dates in the radiotherapy summary, any overlap between the searched date and the date range in the resource is considered a match.

identifier | token | Optional | Yes | | Identifiers associated with the Procedure. They should be formatted with system|value.
patient | reference | Conditional | Yes | | The FHIR ID for the patient associated with the ServiceRequest. Either patient or subject must be provided.
subject | reference | Conditional | Yes | | The FHIR ID for the patient associated with the ServiceRequest. Either patient or subject must be provided.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | reference | Optional | Yes | | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Course Summary: Can reference a ServiceRequest course prescription or planned course that this course summary fulfills. Treated Phase: Can reference a ServiceRequest phase prescription or planned phase that this treated phase fulfills. Treated Plan: Can reference a ServiceRequest plan prescription or treatment plan that this treated plan fulfills.
part-of | reference | Optional | Yes | | Reference to a Procedure resource of which this Procedure is a component. For a treated phase Procedure, this can reference a course summary Procedure. For a treated plan Procedure, this can reference a treated phase Procedure.
performer | reference | Optional | Yes | | The person that performed the procedure.
reason-code | token | Optional | Yes | | The cancer problem to be treated by radiotherapy.
reason-reference | reference | Optional | Yes | | A reference to a cancer problem Condition resource.
status | token | Optional | Yes | | The status of the course/phase/plan. The value should come form the value set http://hl7.org/fhir/R4/ValueSet/event-status

Response elements

Name | Type | Optional | Array | Cardinality | Description
Procedure (R4) | Procedure
Procedure (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Course Summary: Can reference a ServiceRequest course prescription or planned course that this course summary fulfills. Treated Phase: Can reference a ServiceRequest phase prescription or planned phase that this treated phase fulfills. Treated Plan: Can reference a ServiceRequest plan prescription or treatment plan that this treated plan fulfills.
Procedure (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.
Procedure (R4).bodySite.coding | Coding | | Yes | 1..* | The SNOMED code for this anatomical site.
Procedure (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, the laterality of the anatomical site (left, right, bilateral). URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier
Procedure (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, additional qualifiers specifying the anatomical site being treated. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer
Procedure (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.
Procedure (R4).category | CodeableConcept | | | 1..* | Always the SNOMED code for Radiotherapy (procedure): 1287742003.
Procedure (R4).code | CodeableConcept | | | 1..1 | The code of the service being requested. Possible values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)
Procedure (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | Indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation
Procedure (R4).extension (doseDeliveredToVolume) (also: extension (doseDeliveredToVolume)) | Extension | | Yes | 0..* | Specifies doses delivered to one or more volumes as part of this regimen. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume
Procedure (R4).extension (doseDeliveredToVolume).extension (fractionsDelivered) (also: extension (fractionsDelivered)) | Extension | | Yes | 0..1 | Specifies the fractions delivered for the volume. URL: fractionsDelivered
Procedure (R4).extension (doseDeliveredToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose
Procedure (R4).extension (doseDeliveredToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose
Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | Provides a more tightly specified metric for the given dose. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric
Procedure (R4).extension (doseDeliveredToVolume).extension (totalDoseDelivered) (also: extension (totalDoseDelivered)) | Extension | | Yes | 1..1 | Specifies the total dose delivered to this volume so far. URL: totalDoseDelivered
Procedure (R4).extension (doseDeliveredToVolume).extension (uniformFractionation) (also: extension (uniformFractionation)) | Extension | | Yes | 0..1 | Indicates whether or not every fraction for this volume was the same dose for the entire course. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-uniform-fractionation
Procedure (R4).extension (doseDeliveredToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | A reference to the radiotherapy volume (BodyStructure FHIR resource) whose dose is being described. URL: volume
Procedure (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, defines what free-breathing method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique
Procedure (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality
Procedure (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification
Procedure (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | Defines the imaging procedure used for intrafraction verification. URL: procedure
Procedure (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | Further describes the imaging used for intrafraction verification. URL: qualifier
Procedure (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique

Procedure (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | Specifies the modality of the radiation used. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, giving either the beam energy (for external-beam modalities) or source isotope (for brachytherapy modalities). URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | Defines what type of applicator is used for a brachytherapy source. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

Procedure (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | Specifies the technique of the radiation used, which is more specific than the modality. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

Procedure (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, defines what method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management

Procedure (R4).id | id | | | 0..1 | The Procedure FHIR ID.

Procedure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.

Procedure (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.

Procedure (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).

Procedure (R4).identifier.value | String | | | 1..1 | The value of the identifier.

Procedure (R4).meta | Meta

Procedure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).

Procedure (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this Procedure resource. The following profiles are supported: Profile Name Profile Value Code Radiotherapy Course Summary http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary 1217123003 Radiotherapy Treated Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase 1222565005 Radiotherapy Treated Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan 1255724003

Profile Name | Profile Value | Code

Radiotherapy Course Summary | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary | 1217123003

Radiotherapy Treated Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase | 1222565005

Radiotherapy Treated Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan | 1255724003

Procedure (R4).partOf | Reference | | Yes | 0..* | Reference to a Procedure resource of which this Procedure is a component. For a treated phase Procedure, this can reference a course summary Procedure. For a treated plan Procedure, this can reference a treated phase Procedure.

Procedure (R4).performedPeriod | Period | | | 0..1 | The time during which this procedure took place.

Procedure (R4).performer | BackboneElement | | Yes | 0..1 | The person that performed the procedure.

Procedure (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.

Procedure (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.

Procedure (R4).status | code | | | 1..1 | The status of the course/phase/plan. Must be a value defined in http://hl7.org/fhir/R4/ValueSet/event-status

Procedure (R4).statusReason | CodeableConcept | | | 0..1 | Only supported for Course Summary profiles. Reason for the current status as defined in SNOMED URL: http://hl7.org/fhir/us/mcode/ValueSet/mcode-treatment-termination-reason-vs

Procedure (R4).subject | Reference | | | 1..1 | A FHIR reference.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Procedure?category=108290001&code=1217123003&patient=eP-BmFG0190obAH1wt4wjUw3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Procedure?patient=eP-BmFG0190obAH1wt4wjUw3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Procedure/edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Procedure/edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3",
      "resource": {
        "resourceType": "Procedure",
        "id": "edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2025-12-17T01:39:14Z",
          "profile": [
            "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary"
          ]
        },
        "extension": [
```

```
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "129428001",
              "display": "Preventive - procedure intent"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent"
      },
      {
        "extension": [
          {
            "valueReference": {
              "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
            },
            "url": "volume"
          },
          {
            "valueUnsignedInt": 1,
            "url": "fractionsDelivered"
          },
          {
            "valueQuantity": {
              "value": 100,
              "unit": "cGy",
              "system": "http://unitsofmeasure.org",
              "code": "cGy"
            },
            "url": "totalDoseDelivered"
          },
          {
            "valueBoolean": true,
            "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-
plan-dose"
          }
        ],
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume"
      },
      {
        "extension": [
          {
            "valueReference": {
              "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
            },
            "url": "volume"
          },
          {
            "valueUnsignedInt": 1,
            "url": "fractionsDelivered"
          },
          {
            "valueQuantity": {
              "value": 100,
              "unit": "cGy",
              "system": "http://unitsofmeasure.org",
              "code": "cGy"
            },
            "url": "totalDoseDelivered"
          },
          {
            "valueBoolean": true,
            "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-
plan-dose"
          }
        ],
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume"
      },
      {
        "extension": [
          {
            "valueReference": {
              "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
            },
            "url": "volume"
          },
          {
            "valueUnsignedInt": 0,
            "url": "fractionsDelivered"
          },
          {
            "valueQuantity": {
              "value": 0,
              "unit": "cGy",
              "system": "http://unitsofmeasure.org",
              "code": "cGy"
            },
```

```
            "url": "totalDoseDelivered"
          },
          {
            "valueBoolean": true,
            "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-
plan-dose"
          }
        ],
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume"
      },
      {
        "extension": [
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "1156506007",
                  "display": "External beam radiation therapy using photons"
                }
              ]
            },
            "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality"
          },
          {
            "valueCodeableConcept": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "1162782007",
                  "display": "Three dimensional external beam radiation therapy"
                }
              ]
            },
            "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique"
          },
          {
            "valueQuantity": {
              "value": 18,
              "unit": "MeV",
              "system": "http://unitsofmeasure.org",
              "code": "MeV"
            },
            "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-
or-isotope"
          }
        ],
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique"
      },
      {
        "valueUnsignedInt": 1,
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-sessions"
      }
    ],
    "identifier": [
      {
        "use": "usual",
        "system": "urn:hand:display",
        "value": "Bilateral Breast"
      },
      {
        "use": "official",
        "system": "urn:dicom:uid",
        "value": "a00ac862-a954-4896-a1f8-86e207d0f120"
      }
    ],
    "basedOn": [
      {
        "reference": "ServiceRequest/BZi6oojEVTlpWA7JJYw3-fJVBYgnhYSRQ1Cx1qK8JWius",
        "type": "ServiceRequest",
        "display": "Radiotherapy Course Summary - Planned"
      },
      {
        "reference": "ServiceRequest/e214EEAa7RY8csEhIEJW-qOpDYFRKCusulwMwwzo333g3",
        "type": "ServiceRequest",
        "display": "Radiotherapy Course Summary - Prescribed"
      }
    ],
    "status": "in-progress",
    "category": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "108290001",
          "display": "Radiation Oncology And/Or Radiotherapy"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "1287742003",
```

```
      "display": "Radiotherapy"
    }
  ]
},
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "1217123003",
      "display": "Radiotherapy course of treatment"
    }
  ]
},
"subject": {
  "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
  "display": "Test, Patient"
},
"performedPeriod": {
  "start": "2025-11-17T19:37:03Z"
},
"location": {
  "reference": "Location/e-iz4EXUrNFfCKJwdsZPd9Q3"
},
"reasonCode": [
  {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "C50",
        "display": "Malignant neoplasm of breast"
      }
    ],
    "text": "Cancer Problem"
  }
],
"reasonReference": [
  {
    "reference": "Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3"
  }
],
"bodySite": [
  {
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "7771000",
              "display": "Left"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangential projection"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier"
      }
    ],
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "Breast structure"
      }
    ]
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    "extension": [
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        "valueCodeableConcept": {
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              "system": "http://snomed.info/sct",
              "code": "24028007",
              "display": "Right"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier"
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            },
            {
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                    "system": "http://snomed.info/sct",
                    "code": "399110001",
                    "display": "Tangential projection"
                  }
                ]
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              "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier"
            }
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              "system": "http://snomed.info/sct",
              "code": "76752008",
              "display": "Breast structure"
            }
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                    "system": "http://snomed.info/sct",
                    "code": "7771000",
                    "display": "Left"
                  }
                ]
              },
              "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier"
            }
          ],
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "68171009",
              "display": "Axillary lymph node structure"
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  }
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}
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13.1.28 Procedure.Update (External Radiotherapy Summary) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11051-Procedure-Update-External-Radiotherapy-Summary-R4.html
Epic API ID | 11051
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | PUT
URL template | /api/FHIR/R4/Procedure/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11051

Description

The Procedure resource is used to represent radiotherapy delivery progress at all levels (courses, phases, and plans). A Procedure for a course, phase, or plan indicates that the course, phase, or plan has progressed from prescribed/planned to treated.

There are three types of Procedure resources: Course Summary, Treated Phase, and Treated Plan. The table below shows the profile, category, and code used in each.

Type | Profile | Category | Code
Course Summary | Radiotherapy Course Summary - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1217123003 (Course)

Treated Phase | Radiotherapy Treated Phase - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1222565005 (Phase)

Treated Plan | Radiotherapy Treated Plan - CodeX Radiation Therapy v2.0.0 | SNOMED Code 108290001 (Radiotherapy) | SNOMED Code 1255724003 (Plan)

Request parameters

Name | Type | Optional | Array | Cardinality | Description

Procedure (R4) | Procedure

Procedure (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Course Summary: Can reference a ServiceRequest course prescription or planned course that this course summary fulfills. Treated Phase: Can reference a ServiceRequest phase prescription or planned phase that this treated phase fulfills. Treated Plan: Can reference a ServiceRequest plan prescription or treatment plan that this treated plan fulfills.

Procedure (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.

Procedure (R4).bodySite.coding | Coding | | Yes | 1..* | The coded representation of the anatomical site.

Procedure (R4).bodySite.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-treatment-location-vs.

Procedure (R4).bodySite.coding.display | String | | | 0..1 | The textual representation of the coded value in the given system.

Procedure (R4).bodySite.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, the laterality of the body site (left, right, bilateral).

Procedure (R4).bodySite.extension (lateralityQualifier).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier

Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept | CodeableConcept | | | 0..1 | CodeableConcept container for the locationQualifier.

Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the lateralityQualifier.

Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the lateralityQualifier. The value must be a member of https://hl7.org/fhir/us/mcode/STU4/ValueSet-mcode-laterality-qualifier-vs.html.

Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The display name of the SNOMED code.

Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, additional qualifiers specifying the body site being treated.

Procedure (R4).bodySite.extension (locationQualifier).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept | CodeableConcept | | | 0..* | CodeableConcept container for the locationQualifier.

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the locationQualifier.

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the locationQualifier. The value must be a member of https://hl7.org/fhir/us/mcode/STU4/ValueSet-mcode-body-location-qualifier-vs.html.

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The display name of the SNOMED code.

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).bodySite.extension (locationQualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.

Procedure (R4).category | CodeableConcept | | | 1..* | Categories that apply to this Procedure.

Procedure (R4).category.coding | Coding | | Yes | 0..1 | The coded representation of the category.

Procedure (R4).category.coding.code | String | | | 1..1 | Always the SNOMED code for Radiotherapy (procedure): 1287742003.

Procedure (R4).category.coding.display | String | | | 0..1 | The textual representation of the coded value. Only allowed value: "Radiotherapy (procedure)".

Procedure (R4).category.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).category.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).code | CodeableConcept | | | 1..1 | The coded representation of the Procedure.

Procedure (R4).code.coding | Coding | | Yes | 1..1 | The coded representation of the concept.

Procedure (R4).code.coding.code | String | | | 1..1 | The code of the service being requested. Possible values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)

Procedure (R4).code.coding.display | String | | | 1..1 | The textual representation of the coded value. For course types, the value must be "Radiotherapy course of treatment (regime/therapy)". For phase types, the value must be "Radiotherapy treatment phase (regime/therapy)". For plan types, the value must be "Radiotherapy treatment plan (regime/therapy)".

Procedure (R4).code.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).code.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | Indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before.

Procedure (R4).extension (courseInvolvesReirradiation).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation

Procedure (R4).extension (courseInvolvesReirradiation).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

Procedure (R4).extension (doseDeliveredToVolume) (also: extension (doseDeliveredToVolume)) | Extension | | Yes | 0..* | Specifies doses delivered to one or more volumes as part of this regimen.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | Specifies the fractional dose prescribed for this volume.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).url | uri | | | 1..1 | Must be: fractionDose

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity | Quantity | | | 1..1 | Quantity of the dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity.code | String | | | 1..1 | Must be: cGy

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy

Procedure (R4).extension (doseDeliveredToVolume).extension (fractionDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | Specifies how many fractions are prescribed for this volume in the course.

Procedure (R4).extension (doseDeliveredToVolume).extension (fractions).url | uri | | | 1..1 | Must be: fractions

Procedure (R4).extension (doseDeliveredToVolume).extension (fractions).valuePositiveInt | positiveInt | | | 1..1 | The number of fractions prescribed.

Procedure (R4).extension (doseDeliveredToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen.

Procedure (R4).extension (doseDeliveredToVolume).extension (pointDose).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose

Procedure (R4).extension (doseDeliveredToVolume).extension (pointDose).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

Procedure (R4).extension (doseDeliveredToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan.

Procedure (R4).extension (doseDeliveredToVolume).extension (primaryPlanDose).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose

Procedure (R4).extension (doseDeliveredToVolume).extension (primaryPlanDose).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | Provides a more tightly specified metric for the given dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric).valueQuantity | Quantity | | | 1..1 | The quantity of the radiobiologic metric.

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric).valueQuantity.unit | String | | | 1..1 | Must be a value as described in http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric.

Procedure (R4).extension (doseDeliveredToVolume).extension (radiobiologicMetric).valueQuantity.value | Decimal | | | 1..1 | The number representing the dose according to the given radiobiologic metric.

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | Specifies the total dose prescribed for this volume.

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).url | uri | | | 1..1 | Must be: totalDose

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity | Quantity | | | 1..1 | Quantity of the total dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity.code | String | | | 1..1 | Must be: cGy

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy

Procedure (R4).extension (doseDeliveredToVolume).extension (totalDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.

Procedure (R4).extension (doseDeliveredToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | Identifies a volume to be treated with radiation.

Procedure (R4).extension (doseDeliveredToVolume).extension (volume).url | uri | | | 1..1 | Must be: volume

Procedure (R4).extension (doseDeliveredToVolume).extension (volume).valueReference | Reference | | | 1..1 | A reference to the radiotherapy volume (BodyStructure) whose dose is being described.

Procedure (R4).extension (doseDeliveredToVolume).extension(uniformFractionation) (also: extension(uniformFractionation)) | Extension | | Yes | 0..1 | Indicates whether or not every fraction for this volume was the same dose for the entire course.

Procedure (R4).extension (doseDeliveredToVolume).extension(uniformFractionation).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-uniform-fractionation

Procedure (R4).extension (doseDeliveredToVolume).extension(uniformFractionation).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

Procedure (R4).extension (doseDeliveredToVolume).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume

Procedure (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, defines what free-breathing method is used.

Procedure (R4).extension (freeBreathingMotionManagementTechnique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept | CodeableConcept | | | 1..1 | The CodeableConcept container for the free-breathing method.

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the free-breathing method.

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-free-breathing-motion-mgmt-technique-vs

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.display | String | | | 0..1 | The textual representation of the coded value in the given system.

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used.

Procedure (R4).extension (imageGuidedRadiotherapyModality).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the image guidance.

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the imaging guidance.

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/STU2/ValueSet-codexrt-image-guided-radiotherapy-modality-vs.

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification.

Procedure (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | Defines the imaging procedure used for intrafraction verification.

Procedure (R4).extension (intrafractionVerification).extension (procedure).url | uri | | | 1..1 | Must be: procedure

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the procedure.

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the procedure.

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-intrafraction-verification-procedure-vs

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | Further describes the imaging used for intrafraction verification.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).url | uri | | | 1..1 | Must be: qualifier

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the qualifier.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the qualifier.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-intrafraction-verification-qualifier-vs.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (intrafractionVerification).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification

Procedure (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation.

Procedure (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | Specifies the modality of the radiation used.

Procedure (R4).extension (modalityAndTechnique).extension (modality).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the modality.

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the modality.

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-modality-vs

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, giving either the beam energy (for external-beam modalities) or source isotope (for brachytherapy modalities).

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept | CodeableConcept | | | 0..1 | If specifying an isotope, a CodeableConcept representation of the isotope as a SNOMED code.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the isotope.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-isotope-vs.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity | Quantity | | | 0..1 | If specifying a field energy, a Quantity representing the energy in kV, MV, or MeV.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.code | String | | | 1..1 | Must be from the value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-energy-unit-vs.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.value | Decimal | | | 1..1 | Value of the energy.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | Defines what type of applicator is used for a brachytherapy source.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the type of applicator.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the type of applicator.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-brachytherapy-applicator-type-vs.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | Specifies the technique of the radiation used, which is more specific than the modality.

Procedure (R4).extension (modalityAndTechnique).extension (technique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the technique.

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the technique.

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-technique-vs.

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.system | String | | | 1..1 | Must be http://snomed.info/sct

Procedure (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).extension (modalityAndTechnique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique

Procedure (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, defines what method is used.

Procedure (R4).extension (respiratoryMotionManagement).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept | CodeableConcept | | | 1..1 | The CodeableConcept container for the motion management.

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the motion management.

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-respiratory-motion-management-vs

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

Procedure (R4).extension (respiratoryMotionManagement).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).id | id | | | 1..1 | The Procedure FHIR ID.

Procedure (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.

Procedure (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.

Procedure (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).

Procedure (R4).identifier.value | String | | | 1..1 | The value of the identifier.

Procedure (R4).meta | Meta

Procedure (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).

Procedure (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this Procedure resource. The following profiles are supported: Profile Name Profile Value Code Radiotherapy Course Summary http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary 1217123003 Radiotherapy Treated Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase 1222565005 Radiotherapy Treated Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan 1255724003

Profile Name | Profile Value | Code

Radiotherapy Course Summary | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary | 1217123003

Radiotherapy Treated Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-phase | 1222565005

Radiotherapy Treated Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treated-plan | 1255724003

Procedure (R4).partOf | Reference | | Yes | 0..* | Reference to a Procedure resource of which this Procedure is a component. For a treated phase Procedure, this can reference a course summary Procedure. For a treated plan Procedure, this can reference a treated phase Procedure.

Procedure (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.

Procedure (R4).reasonCode.coding | Coding | | Yes | 0..1 | Coded representation of the problem.

Procedure (R4).reasonCode.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-cancer-disorder-vs.

Procedure (R4).reasonCode.coding.display | String | | | 0..1 | The textual representation of the coded value.

Procedure (R4).reasonCode.coding.system | String | | | 0..1 | The URI for the system in which this coded value is assigned. Preferred values: http://snomed.info/sct http://hl7.org/fhir/sid/icd-10-cm

Procedure (R4).reasonCode.text | String | | | 0..1 | An optional human-readable description of the coded value.

Procedure (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.

Procedure (R4).status | code | | | 1..1 | The status of the course/phase/plan. The value should be from the value set http://hl7.org/fhir/R4/ValueSet/event-status

Procedure (R4).subject | Reference | | | 1..1 | Reference to the Patient FHIR resource that this radiotherapy course/plan/phase is planned or prescribed to.

Sample request

```
PUT https://hostname/instance/api/FHIR/R4/Procedure/edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3
{
    "resourceType": "Procedure",
    "id": "edKALtDcuTHWnRKIEr3Jo0pNpyvR27ZfzRHNVtzFTSVk3"
    "meta": {
        "lastUpdated": "2026-02-01T09:09:51-06:00",
        "profile": [
            "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-summary",
            "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-course-summary"
        ]
    },
    "category": {
        "coding": [
            {
                "system": "http://snomed.info/sct",
                "code": "1287742003"
            }
        ]
    },
    "code": {
        "coding": [
            {
                "system": "http://snomed.info/sct",
                "code": "1217123003"
            }
        ]
    },
    "basedOn": [
        {
            "reference": "ServiceRequest/BZi6oojEVTlpWA7JJYw3-fJVBYgnhYSRQ1Cx1qK8JWius"
        },
        {
            "reference": "ServiceRequest/e214EEAa7RY8csEhIEJW-qOpDYFRKCusulwMwwzo333g3",
            "display": "Bilateral Breast"
        }
    ],
    "identifier": [
        {
            "use": "usual",
            "system": "urn:hand:display",
            "value": "Bilateral Breast"
        },
        {
            "use": "official",
            "system": "urn:dicom:uid",
            "value": "22542ccf-6d72-4745-a9b0-79fc87ed513e"
        }
    ],
    "status": "in-progress",
    "subject": {
        "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
```

```
    "display": "Test, Patient"
},
"performedPeriod": {
    "start": "2026-02-01T09:09:51-06:00",
    "end": ""
},
"reasonCode": [
    {
    "coding": [
        {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "C50",
        "display": "Malignant neoplasm of breast"
        }
    ],
    "text": "Cancer Problem"
    }
],
"reasonReference": [
    {
        "reference": "Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3"
    }
],
"bodySite": [
    {
    "coding": [{
        "system" : "http://snomed.info/sct",
        "code" : "76752008",
        "display" : "Breast"
    }],
    "extension": [
        {
        "url" : "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
            "coding":[
            {
                "system": "http://snomed.info/sct",
                "code": "399110001",
                "display": "Tangent"
            }
            ]
        }
        },
        {
        "url" : "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
            "coding":[
            {
                "system": "http://snomed.info/sct",
                "code": "7771000",
                "display": "Left"
            }
            ]
        }
        }
    ]
    },
    {
    "coding": [{
        "system" : "http://snomed.info/sct",
        "code" : "76752008",
        "display" : "Breast"
    }],
    "extension": [
        {
        "url" : "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
            "coding":[
            {
                "system": "http://snomed.info/sct",
                "code": "399110001",
                "display": "Tangent"
            }
            ]
        }
        },
        {
        "url" : "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
            "coding":[
            {
                "system": "http://snomed.info/sct",
                "code": "24028007",
                "display": "Right"
            }
            ]
        }
        }
    ]
```

```
    },
    {
    "coding": [{
        "system" : "http://snomed.info/sct",
        "code" : "68171009",
        "display" : "Axillary lymph node"
    }],
    "extension": [
        {
        "url" : "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
            "coding":[
            {
                "system": "http://snomed.info/sct",
                "code": "7771000",
                "display": "Left"
            }
            ]
        }
        }
    ]
    }
],
"location": {
    "reference": "Location/e-iz4EXUrNFfCKJwdsZPd9Q3"
},
"extension": [
    {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent",
        "valueCodeableConcept": {
            "coding": [{
            "system": "http://snomed.info/sct",
            "code": 129428001,
            "display": "Preventive"
            }]
        }
    },
    {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique",
        "extension": [
            {
                "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality",
                "valueCodeableConcept": {
                    "coding": [{
                    "system": "http://snomed.info/sct",
                    "code": "1156506007",
                    "display": "Photons"
                    }]
                }
            },
            {
                "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique",
                "valueCodeableConcept": {
                    "coding": [{
                    "system": "http://snomed.info/sct",
                    "code": 1162782007,
                    "display": "3D"
                    }]
                }
            },
            {
                "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-
energy-or-isotope",
                "valueQuantity": {
                    "value": "18",
                    "code": "MeV",
                    "system": "http://unitsofmeasure.org"
                }
            }
        ]
    },
    {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-sessions",
        "valueUnsignedInt": "1"
    },
    {
        "url" : "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume",
        "extension" : [
            {
                "url" : "volume",
                "valueReference" : {
                    "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
                }
            },
            {
                "url" : "totalDoseDelivered",
                "valueQuantity" : {
                    "value" : "266",
                    "system" : "http://unitsofmeasure.org",
```

```
                        "code" : "cGy"
                    }
                },
                {
                    "url" : "fractionsDelivered",
                    "valueUnsignedInt" : "1"
                },
                {
                    "url" : "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-
    primary-plan-dose",
                    "valueBoolean" : true
                }
            ]
        },
        {
            "url" : "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume",
            "extension" : [
                {
                    "url" : "volume",
                    "valueReference" : {
                        "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
                    }
                },
                {
                    "url" : "totalDoseDelivered",
                    "valueQuantity" : {
                        "value" : "266",
                        "system" : "http://unitsofmeasure.org",
                        "code" : "cGy"
                    }
                },
                {
                    "url" : "fractionsDelivered",
                    "valueUnsignedInt" : "1"
                },
                {
                    "url" : "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-
    primary-plan-dose",
                    "valueBoolean" : true
                }
            ]
        },
        {
            "url" : "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-dose-delivered-to-volume",
            "extension" : [
                {
                    "url" : "volume",
                    "valueReference" : {
                        "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
                    }
                },
                {
                    "url" : "totalDoseDelivered",
                    "valueQuantity" : {
                        "value" : "0",
                        "system" : "http://unitsofmeasure.org",
                        "code" : "cGy"
                    }
                },
                {
                    "url" : "fractionsDelivered",
                    "valueUnsignedInt" : "0"
                },
                {
                    "url" : "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-
    primary-plan-dose",
                    "valueBoolean" : true
                }
            ]
        }
    ]
}
```


Sample response

```
Status Code: OK (200)
Content-Type: application/json; charset=utf-8
```

13.1.29 RequestGroup.Read (Oncology Plan Day) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10077-RequestGroup-Read-Oncology-Plan-Day-R4.html
Epic API ID | 10077
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
API group | Oncology Plan Group

USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET
URL template | /api/FHIR/R4/RequestGroup/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10077

Description

The RequestGroup resource represents a group of activities that may be performed for a specific patient or context.

The RequestGroup (Oncology Plan Day) sub-resource returns references to medication and procedure orders that can be performed for a patient.

This read interaction enables the lookup of a specific oncology RequestGroup resource by FHIR ID. You can obtain FHIR IDs for RequestGroup (Oncology Plan Day) through calls to CarePlan (Oncology)

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for a RequestGroup resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
RequestGroup | RequestGroup | False | | | An instance of the R4 RequestGroup FHIR resource.
RequestGroup.action | Action | True | Yes | | Treatment order information.
RequestGroup.action.description | String | True | | | Treatment order display name.
RequestGroup.action.resource | Reference | True | | | Reference to the MedicationRequest or ServiceRequest resource.
RequestGroup.action.title | String | True | | | Treatment order category title.
RequestGroup.basedOn | Reference | True | Yes | | Reference to the cycle CarePlan.
RequestGroup.intent | String | False | | | For oncology RequestGroup, this always equals "plan".
RequestGroup.status | String | False | | | Plan/cycle status value. Supported values include: Planned Scheduled Completed Deleted Started Canceled Given Externally Unscheduled Non-Treatment Orders
RequestGroup.subject | Reference | True | | | The patient or subject of this care plan.
RequestGroup.extension (also: treatment-period) | Extension | True | Yes | | Treatment day start and end dates.

Sample request

```
https://hostname/instance/api/FHIR/R4/RequestGroup/etc2cbk8Ae3VGw9-PKxBiVdyuGJmkcwUId8Gbx72GOyk3
```

Sample response


```
<RequestGroup xmlns="http://hl7.org/fhir">
  <id value="etc2cbk8Ae3VGw9-PKxBiVdyuGJmkcwUId8Gbx72GOyk3"/>
  <basedOn>
    <reference value="CarePlan/eI9U0bYkO3DGp.eizX.fdo5BhOb.C9lmuqXQNGMwbCCD9sUuaah3l7uZrlTS2j-w33"/>
    <display value="Cycle 2"/>
  </basedOn>
  <status value="completed"/>
  <intent value="plan"/>
  <subject>
    <reference value="Patient/eBRPOaY79qqguRBWdO8s98g3"/>
    <display value="Morrissey, Graces"/>
  </subject>
  <action>
    <title value="Labs"/>
    <description value="COMPREHENSIVE METABOLIC PANEL"/>
    <resource>
      <reference value="ServiceRequest/eMFR3HpLUZj-D0Ky57HzhXvDEm-zqjceuCEiiJupraLM3"/>
      <type value="ServiceRequest"/>
    </resource>
  </action>
  <action>
    <title value="Labs"/>
    <description value="CBC WITH DIFFERENTIAL"/>
    <resource>
      <reference value="ServiceRequest/eMFR3HpLUZj-D0Ky57HzhXqAATmIFxTcHPNPAj6dokhQ3"/>
      <type value="ServiceRequest"/>
    </resource>
  </action>
  <action>
    <title value="Nursing Orders"/>
    <description value="TREATMENT CONDITIONS"/>
    <resource>
      <reference value="ServiceRequest/eGKxfVP5NNqA7tWGJTPpPlzWpEzoVgVKZUzVx-5Vi0EI3"/>
      <type value="ServiceRequest"/>
    </resource>
```

```
  </action>
  <action>
    <title value="Pre-Medications"/>
    <description value="dexamethasone (DECADRON) injection 10 mg"/>
    <resource>
      <reference value="MedicationRequest/eatqc4AHv3DOnz2IZo1KtvG8PmZ.BPO8m7mlkNXEunPo3"/>
      <type value="MedicationRequest"/>
    </resource>
  </action>
  <action>
    <title value="Pre-Medications"/>
    <description value="ondansetron (ZOFRAN) tablet 24 mg"/>
    <resource>
      <reference value="MedicationRequest/eatqc4AHv3DOnz2IZo1KtvIhrots9i3O.yvqyLAatE7g3"/>
      <type value="MedicationRequest"/>
    </resource>
  </action>
  <action>
    <title value="Pre-Medications"/>
    <description value="lorazepam (ATIVAN) injection 1 mg"/>
    <resource>
      <reference value="MedicationRequest/eatqc4AHv3DOnz2IZo1KtvBR.pT6C71zLod401UJ5pMY3"/>
      <type value="MedicationRequest"/>
    </resource>
  </action>
  <action>
    <title value="Chemotherapy"/>
    <description value="CARBOplatin (PARAPLATIN) in sodium chloride 0.9 % 150 mL chemo infusion"/>
    <resource>
      <reference value="MedicationRequest/eatqc4AHv3DOnz2IZo1KtvLIMDt-2u.DNvG4bXyioWLE3"/>
      <type value="MedicationRequest"/>
    </resource>
  </action>
  <action>
    <title value="Chemotherapy"/>
    <description value="gemcitabine (GEMZAR) 1,520 mg in sodium chloride 0.9 % 150 mL chemo infusion"/>
    <resource>
      <reference value="MedicationRequest/eatqc4AHv3DOnz2IZo1KtvHMr03G7KHany-2O6RKWPjE3"/>
      <type value="MedicationRequest"/>
    </resource>
  </action>
  <action>
    <title value="Nursing Orders"/>
    <description value="heparin (porcine) injection 500 Units"/>
    <resource>
      <reference value="MedicationRequest/eatqc4AHv3DOnz2IZo1KtvHqNGWxFcpsPFkeAXySBLAM3"/>
      <type value="MedicationRequest"/>
    </resource>
  </action>
  <action>
    <title value="Nursing Orders"/>
    <description value="sodium chloride 0.9 % injection 10 mL"/>
    <resource>
      <reference value="MedicationRequest/eatqc4AHv3DOnz2IZo1KtvAUen3TKheyNcrjGV61OO83"/>
      <type value="MedicationRequest"/>
    </resource>
  </action>
</RequestGroup>
```


13.1.30 RequestGroup.Search (Dental Visit) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10080-RequestGroup-Search-Dental-Visit-R4.html
Epic API ID | 10080
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
API group | Dental Care Plan Group
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET
URL template | /api/FHIR/R4/RequestGroup?_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10080

Description

This web service retrieves information about a patient's dental visits, such as the visit's status and scheduling information.

This search interaction currently supports only the _id search parameter. As a result, it functions similarly to RequestGroup.Read (Dental Visit).

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.RequestGroup | RequestGroup | False | | | An instance of the R4 RequestGroup FHIR resource.
Bundle.entry.resource.RequestGroup.action | Action | True | Yes | | Dental procedure information.
Bundle.entry.resource.RequestGroup.action.resource | Reference | True | | | The procedure for the visit.
Bundle.entry.resource.RequestGroup.author | Reference | True | | | The user who created the dental visit.
Bundle.entry.resource.RequestGroup.authoredOn | String | True | | | The instant the dental visit was created.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-name) | Extension | True | Yes | | The name of the visit.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-number) | Extension | True | Yes | | The relative order of the visit within the treatment.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-requested-date) | Extension | False | Yes | | The date on which the provider requested that the appointment be scheduled.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-requested-department) | Extension | False | Yes | | The requested department for scheduling.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-requested-length) | Extension | False | Yes | | The requested appointment length for the visit.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-requested-provider) | Extension | False | Yes | | The requested visit provider for scheduling.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-requested-providerType) | Extension | False | Yes | | The requested provider type for scheduling.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-requested-spacing-days) | Extension | False | Yes | | The requested number of spacing, in days, between scheduled visits.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-requested-type) | Extension | False | Yes | | The requested visit type for scheduling.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-scheduling-instructions) | Extension | False | Yes | | The scheduling instructions for the visit.
Bundle.entry.resource.RequestGroup.extension (also: dental-visit-scheduling-status) | Extension | True | Yes | | The scheduling status of the visit.
Bundle.entry.resource.RequestGroup.encounter | Reference | True | | | The encounter for the dental visit.
Bundle.entry.resource.RequestGroup.intent | String | False | | | The intention for the dental visit. Can either be "plan", "order", or "proposal". An intention of "plan" or "order" indicate a visit's scheduling status. An intention of "proposal" indicates a potential treatment that does not yet have a corresponding visit.
Bundle.entry.resource.RequestGroup.extension (also: ready-for-scheduling) | Extension | False | Yes | | Whether the visit is ready for scheduling.
Bundle.entry.resource.RequestGroup.status | String | False | | | The status of the dental visit.
Bundle.entry.resource.RequestGroup.subject | Reference | True | | | The patient for the dental visit.

Sample request

```
https://hostname/instance/api/FHIR/R4/RequestGroup?_id=ecCjsC8tcjA8qBSWDjtOYvA3
```

Sample response

```
<RequestGroup xmlns="http://hl7.org/fhir">
  <id value="ecCjsC8tcjA8qBSWDjtOYvA3"/>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-number">
    <valueInteger value="2"/>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-name">
    <valueString value="follow-up"/>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-scheduling-status">
    <valueCodeableConcept>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.798381.94111"/>
        <code value="5"/>
        <display value="Completed"/>
      </coding>
      <text value="Completed"/>
    </valueCodeableConcept>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-requested-length">
    <valueTiming>
      <repeat>
        <duration value="30"/>
        <durationUnit value="min"/>
      </repeat>
    </valueTiming>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-requested-spacing-days">
    <valueQuantity>
      <value value="5"/>
      <unit value="day"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="d"/>
    </valueQuantity>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-scheduling-instructions">
    <valueString value="Follow-up"/>
```

```
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-requested-date">
    <valueDate value="2020-08-22"/>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-requested-type">
    <valueCodeableConcept>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.808267"/>
        <code value="1667"/>
        <display value=""/>
      </coding>
      <text value="DENTAL TREATMENT"/>
    </valueCodeableConcept>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-requested-department">
    <valueReference>
      <reference value="Location/e-iz4EXUrNFfCKJwdsZPd9Q3"/>
      <display value="Dentistry West"/>
    </valueReference>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/dental-visit-requested-provider-type">
    <valueCodeableConcept>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.836982.1040"/>
        <code value="115"/>
        <display value="Elizabeth Robinson"/>
      </coding>
      <text value="Elizabeth Robinson"/>
    </valueCodeableConcept>
  </extension>
  <extension url="http://open.epic.com/FHIR/StructureDefinition/extension/ready-for-scheduling">
    <valueBoolean value="true"/>
  </extension>
  <status value="revoked"/>
  <intent value="order"/>
  <subject>
    <reference value="Patient/eoEDZ-Hn2iFRy8GU04AZo1w3"/>
    <display value="Robishaw, Jeff"/>
  </subject>
  <authoredOn value="2020-08-17T22:36:07Z"/>
  <author>
    <reference value="Practitioner/e0NUFReEifX1FloybAssaXg3"/>
    <display value="Fred Jenkins, MD"/>
  </author>
</RequestGroup>
```


13.1.31 RequestGroup.Search (Oncology Plan Day) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10078-RequestGroup-Search-Oncology-Plan-Day-R4.html
Epic API ID | 10078
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
API group | Oncology Plan Group
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2021
HTTP method | GET
URL template | /api/FHIR/R4/RequestGroup?_id={_id}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10078

Description

The RequestGroup resource represents a group of activities that may be performed for a specific patient or context.

The RequestGroup (Oncology Plan Day) sub-resource returns references to medication and procedure orders that can be performed for a patient.

This search interaction enables the lookup of oncology RequestGroup resources filtered by the specified parameters. Currently, only the _id search parameter is supported, making this interaction similar to RequestGroup.Read. You can obtain FHIR IDs for RequestGroup (Oncology Plan Day) through calls to CarePlan (Oncology)

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.

Bundle.entry.resource.RequestGroup | RequestGroup | False | | | An instance of the R4 RequestGroup FHIR resource.
Bundle.entry.resource.RequestGroup.action | Action | True | Yes | | Treatment order details.
Bundle.entry.resource.RequestGroup.action.description | String | True | | | Treatment order display name.
Bundle.entry.resource.RequestGroup.action.resource | Reference | True | | | Reference to the MedicationRequest or ServiceRequest resource.
Bundle.entry.resource.RequestGroup.action.title | String | True | | | Treatment order category title.
Bundle.entry.resource.RequestGroup.basedOn | Reference | True | Yes | | Reference to cycle CarePlan.
Bundle.entry.resource.RequestGroup.intent | String | False | | | For oncology RequestGroup, this always equals "plan".
Bundle.entry.resource.RequestGroup.status | String | False | | | Plan/cycle status. Supported values include: Planned Scheduled Completed Deleted Started Canceled Given Externally Unscheduled Non-Treatment Orders
Bundle.entry.resource.RequestGroup.subject | Reference | True | | | The patient or subject of this care plan.
Bundle.entry.resource.RequestGroup.extension (also: treatment-period) | Extension | True | Yes | | Treatment day start and end dates.

Sample request

```
https://hostname/instance/api/FHIR/R4/RequestGroup?_id=eWAOVohIDucch6SD-jHjXtYdIkiaTI7XoJFJRItoAXMo3
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/RequestGroup?_id=eWAOVohIDucch6SD-jHjXtYdIkiaTI7XoJFJRItoAXMo3"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/RequestGroup/eWAOVohIDucch6SD-jHjXtYdIkiaTI7XoJFJRItoAXMo3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/RequestGroup/eWAOVohIDucch6SD-jHjXtYdIkiaTI7XoJFJRItoAXMo3"/>
    <resource>
      <RequestGroup>
        <id value="eWAOVohIDucch6SD-jHjXtYdIkiaTI7XoJFJRItoAXMo3"/>
        <basedOn>
          <reference value="CarePlan/eBKvQEjhnGT-kvq4St.KLIaswSUj-l8AvpYKkm1ADQKXTD.xcyB8v3KnB-07kFW7d3"/>
          <display value="Cycle 4"/>
        </basedOn>
        <status value="draft"/>
        <intent value="plan"/>
        <subject>
          <reference value="Patient/eBRPOaY79qqguRBWdO8s98g3"/>
          <display value="Name, Patient"/>
        </subject>
        <action>
          <title value="Labs"/>
          <description value="COMPREHENSIVE METABOLIC PANEL"/>
          <resource>
            <reference value="ServiceRequest/eTnYtJ2f-yL0319wijfWo1A3"/>
            <type value="ServiceRequest"/>
          </resource>
        </action>
        <action>
          <title value="Labs"/>
          <description value="CBC WITH DIFFERENTIAL"/>
          <resource>
            <reference value="ServiceRequest/e-fkJEaxOqOuJUmuOKVAK9g3"/>
            <type value="ServiceRequest"/>
          </resource>
        </action>
        <action>
          <title value="Nursing Orders"/>
          <description value="TREATMENT CONDITIONS"/>
          <resource>
            <reference value="ServiceRequest/eSxHqNSC4.edOGjh3eb1QYQ3"/>
            <type value="ServiceRequest"/>
          </resource>
        </action>
        <action>
          <title value="Nursing Orders"/>
          <description value="SODIUM CHLORIDE 0.9 % IV SOLN"/>
          <resource>
            <reference value="MedicationRequest/eT5BtY7-6uRZ1tvLigXeLWQ3"/>
            <type value="MedicationRequest"/>
          </resource>
        </action>
        <action>
          <title value="Pre-Medications"/>
          <description value="dexamethasone (DECADRON) injection 10 mg"/>
          <resource>
            <reference value="MedicationRequest/epqRNACZwFTETG9eLFVJN4g3"/>
            <type value="MedicationRequest"/>
          </resource>
        </action>
        <action>
          <title value="Pre-Medications"/>
```

```
          <description value="ONDANSETRON HCL 4 MG PO TABS"/>
          <resource>
            <reference value="MedicationRequest/eVUsLI4MQ49-Fht5.q8DDcQ3"/>
            <type value="MedicationRequest"/>
          </resource>
        </action>
        <action>
          <title value="Pre-Medications"/>
          <description value="lorazepam (ATIVAN) injection 0.5 mg"/>
          <resource>
            <reference value="MedicationRequest/eC3Rrvk9HKMY.sekWHuknYQ3"/>
            <type value="MedicationRequest"/>
          </resource>
        </action>
        <action>
          <title value="Pre-Medications"/>
          <description value="SCOPOLAMINE BASE 1.5 MG TD PT72"/>
          <resource>
            <reference value="MedicationRequest/ekpnPib114r7Vd2EyNo3nbQ3"/>
            <type value="MedicationRequest"/>
          </resource>
        </action>
        <action>
          <title value="Chemotherapy"/>
          <description value="DOXOrubicin (ADRIAMYCIN) injection"/>
          <resource>
            <reference value="MedicationRequest/eKSs0q7L5Umg7NCE2omBtWA3"/>
            <type value="MedicationRequest"/>
          </resource>
        </action>
        <action>
          <title value="Chemotherapy"/>
          <description value="cyclophosphamide (CYTOXAN) in sodium chloride 0.9 % chemo infusion"/>
          <resource>
            <reference value="MedicationRequest/ehpGGo-9okYHdsoVyMxDI.g3"/>
            <type value="MedicationRequest"/>
          </resource>
        </action>
        <action>
          <title value="Nursing Orders"/>
          <description value="heparin (porcine) injection 500 Units"/>
          <resource>
            <reference value="MedicationRequest/euoiZtb7n-2h51i3jGDQPfQ3"/>
            <type value="MedicationRequest"/>
          </resource>
        </action>
        <action>
          <title value="Nursing Orders"/>
          <description value="sodium chloride 0.9 % injection 10 mL"/>
          <resource>
            <reference value="MedicationRequest/eu1rZL3oxtZJqOuFNmEH8yA3"/>
            <type value="MedicationRequest"/>
          </resource>
        </action>
      </RequestGroup>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

13.1.32 ServiceRequest.Create (External Radiotherapy Summary) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11044-ServiceRequest-Create-External-Radiotherapy-Summary-R4.html
Epic API ID | 11044
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | POST
URL template | /api/FHIR/R4/ServiceRequest
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11044

Description

ServiceRequest resources represent planning summaries at the course, phase, and plan levels. These resources can be sent either prescribed (oncologist has defined what treatment the patient needs) or planned (intended treatment has been planned

through dosimetry).

There are six types of ServiceRequests: Course Prescription, Planned Course, Phase Prescription, Planned Phase, Plan Prescription, and Treatment Plan. The table below describes the category, code, and intent that indicates the differences between the types.

Type | Profile | Category | SNOMED Code | Intent
Course Prescription | Radiotherapy Course Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1217123003 | original-order
Planned Course | Radiotherapy Planned Course - CodeX Radiation Therapy v2.0.0 | 108290001 | 1217123003 | filler-order
Phase Prescription | Radiotherapy Phase Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1222565005 | original-order
Planned Phase | Radiotherapy Planned Phase - CodeX Radiation Therapy v2.0.0 | 108290001 | 1222565005 | filler-order
Plan Prescription | Radiotherapy Plan Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1255724003 | original-order
Treatment Plan | Radiotherapy Treatment Plan - CodeX Radiation Therapy v2.0.0 | 108290001 | 1255724003 | filler-order

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ServiceRequest (R4) | Resource | | | | An instance of the R4 ServiceRequest resource.
ServiceRequest (R4).authoredOn | dateTime | | | 0..1 | When this ServiceRequest was signed.
ServiceRequest (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Radiotherapy Planned Course: References a ServiceRequest course prescription that this planned course fulfills. Radiotherapy Phase Prescription: References a ServiceRequest course prescription of which this phase prescription is a component. Radiotherapy Planned Phase: Can reference either a ServiceRequest planned course of which this planned phase is a component, or a ServiceRequest phase prescription that this planned phase fulfills. Radiotherapy Plan Prescription: Can reference either a ServiceRequest course prescription of which this plan prescription is a component, or a ServiceRequest phase prescription of which this plan prescription is a component. Radiotherapy Treatment Plan: Can reference either a ServiceRequest planned phase of which this treatment plan is a component, or a ServiceRequest plan prescription that this treatment plan fulfills.
ServiceRequest (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.
ServiceRequest (R4).bodySite.coding | Coding | | Yes | 1..* | The coded representation of the anatomical site.
ServiceRequest (R4).bodySite.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-treatment-location-vs.
ServiceRequest (R4).bodySite.coding.display | String | | | 0..1 | The textual representation of the coded value in the given system.
ServiceRequest (R4).bodySite.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, the laterality of the body site (left, right, bilateral).
ServiceRequest (R4).bodySite.extension (lateralityQualifier).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept | CodeableConcept | | | 0..1 | CodeableConcept container for the locationQualifier.
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the lateralityQualifier.
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the lateralityQualifier. The value must be a member of https://hl7.org/fhir/us/mcode/STU4/ValueSet-mcode-laterality-qualifier-vs.html.
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The display name of the SNOMED code.
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, additional qualifiers specifying the body site being treated.
ServiceRequest (R4).bodySite.extension (locationQualifier).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept | CodeableConcept | | | 0..* | CodeableConcept container for the locationQualifier.
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the locationQualifier.
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the locationQualifier. The value must be a member of https://hl7.org/fhir/us/mcode/STU4/ValueSet-mcode-body-location-qualifier-vs.html.
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The display name of the SNOMED code.
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.
ServiceRequest (R4).category | CodeableConcept | | Yes | 1..* | Categories that apply to this ServiceRequest.
ServiceRequest (R4).category.coding | Coding | | Yes | 0..1 | The coded representation of the category.
ServiceRequest (R4).category.coding.code | String | | | 1..1 | Always the SNOMED code for Radiotherapy (procedure): 1287742003.
ServiceRequest (R4).category.coding.display | String | | | 0..1 | The textual representation of the coded value. Only allowed value: "Radiotherapy (procedure)".
ServiceRequest (R4).category.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).category.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).code | CodeableConcept | | | 1..1 | The coded representation of the ServiceRequest.
ServiceRequest (R4).code.coding | Coding | | Yes | 1..1 | The coded representation of the concept.
ServiceRequest (R4).code.coding.code | String | | | 1..1 | Possible code values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)

ServiceRequest (R4).code.coding.display | String | | | 1..1 | The textual representation of the coded value. For course types, the value must be "Radiotherapy course of treatment (regime/therapy)". For phase types, the value must be "Radiotherapy treatment phase (regime/therapy)". For plan types, the value must be "Radiotherapy treatment plan (regime/therapy)".

ServiceRequest (R4).code.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).code.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | Indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before.

ServiceRequest (R4).extension (courseInvolvesReirradiation).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation

ServiceRequest (R4).extension (courseInvolvesReirradiation).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, defines what free-breathing method is used.

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept | CodeableConcept | | | 1..1 | The CodeableConcept container for the free-breathing method.

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the free-breathing method.

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-free-breathing-motion-mgmt-technique-vs

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.display | String | | | 0..1 | The textual representation of the coded value in the given system.

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used.

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the image guidance.

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the imaging guidance.

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/STU2/ValueSet-codexrt-image-guided-radiotherapy-modality-vs.

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification.

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | Defines the imaging procedure used for intrafraction verification.

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).url | uri | | | 1..1 | Must be: procedure

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the procedure.

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the procedure.

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-intrafraction-verification-procedure-vs

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | Further describes the imaging used for intrafraction verification.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).url | uri | | | 1..1 | Must be: qualifier

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the qualifier.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the qualifier.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-intrafraction-verification-qualifier-vs.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (intrafractionVerification).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification

ServiceRequest (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | Specifies the modality of the radiation used.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the modality.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the modality.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-brachytherapy-applicator-type-vs

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, giving either the beam energy (for external-beam modalities) or source isotope (for brachytherapy modalities).

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept | CodeableConcept | | | 0..1 | If specifying an isotope, a CodeableConcept representation of the isotope as a SNOMED code.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the isotope.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-isotope-vs.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity | Quantity | | | 0..1 | If specifying a field energy, a Quantity representing the energy in kV, MV, or MeV.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.code | String | | | 1..1 | Must be from the value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-energy-unit-vs.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.value | Decimal | | | 1..1 | Value of the energy.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | Defines what type of applicator is used for a brachytherapy source.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the type of applicator.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the type of applicator.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-brachytherapy-applicator-type-vs.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | Specifies the technique of the radiation used, which is more specific than the modality.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the technique.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the technique.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the value set

http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-technique-vs.
ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.
ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.system | String | | | 1..1 | Must be http://snomed.info/sct
ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).extension (modalityAndTechnique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume) (also: extension (radiotherapyDosePlannedToVolume)) | Extension | | Yes | 0..* | Specifies the plan of doses to be delivered to one or more volumes as part of this regimen.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | Specifies the fractional dose prescribed for this volume.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).url | uri | | | 1..1 | Must be: fractionDose
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity | Quantity | | | 1..1 | Quantity of the dose.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity.code | String | | | 1..1 | Must be: cGy
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | Specifies how many fractions are prescribed for this volume in the course.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractions).url | uri | | | 1..1 | Must be: fractions
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractions).valuePositiveInt | positiveInt | | | 1..1 | The number of fractions prescribed.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (pointDose).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (pointDose).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (primaryPlanDose).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (primaryPlanDose).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | Provides a more tightly specified metric for the given dose.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric).valueQuantity | Quantity | | | 1..1 | The quantity of the radiobiologic metric.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric).valueQuantity.unit | String | | | 1..1 | Must be a value as described in http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric).valueQuantity.value | Decimal | | | 1..1 | The number representing the dose according to the given radiobiologic metric.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | Specifies the total dose prescribed for this volume.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).url | uri | | | 1..1 | Must be: totalDose
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity | Quantity | | | 1..1 | Quantity of the total dose.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity.code | String | | | 1..1 | Must be: cGy
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | Identifies a volume to be treated with radiation.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (volume).url | uri | | | 1..1 | Must be: volume
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (volume).valueReference | Reference | | | 1..1 | A reference to the radiotherapy volume (BodyStructure) whose dose is being described.
ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-planned-to-volume
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume) (also: extension (radiotherapyDosePrescribedToVolume)) | Extension | | Yes | 0..* | Specifies the prescription of doses to be delivered to one or more volumes as part of this regimen.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | Specifies the fractional dose prescribed for this volume.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).url | uri | | | 1..1 | Must be: fractionDose

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity | Quantity | | | 1..1 | Quantity of the dose.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity.code | String | | | 1..1 | Must be: cGy
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | Specifies how many fractions are prescribed for this volume in the course.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractions).url | uri | | | 1..1 | Must be: fractions
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractions).valuePositiveInt | positiveInt | | | 1..1 | The number of fractions prescribed.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | Specifies the total dose prescribed for this volume.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).url | uri | | | 1..1 | Must be: totalDose
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity | Quantity | | | 1..1 | Quantity of the total dose.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity.code | String | | | 1..1 | Must be: cGy
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | Identifies a volume to be treated with radiation.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (volume).url | uri | | | 1..1 | Must be: volume
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (volume).valueReference | Reference | | | 1..1 | A reference to the radiotherapy volume (BodyStructure) whose dose is being described.
ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-to-volume
ServiceRequest (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, defines what method is used.
ServiceRequest (R4).extension (respiratoryMotionManagement).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management
ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept | CodeableConcept | | | 1..1 | The CodeableConcept container for the motion management.
ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the motion management.
ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-respiratory-motion-management-vs
ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.
ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).extension (treatmentIntent) (also: extension (treatmentIntent)) | Extension | | Yes | 0..1 | Indicates the clinical intent of this radiotherapy regimen.
ServiceRequest (R4).extension (treatmentIntent).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent
ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the treatment intent.
ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.coding | Coding | | Yes | 0..1 | Coded representation of the treatment intent.
ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-procedure-intent-vs.
ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.
ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.coding.system | String | | | 1..1 | Must be http://snomed.info/sct
ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.
ServiceRequest (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.
ServiceRequest (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).
ServiceRequest (R4).identifier.value | String | | | 1..1 | The value of the identifier.
ServiceRequest (R4).intent | code | | | 1..1 | If the course/phase/plan is prescribed, then this value is original-order. If it is planned, then the value is filler-order.
ServiceRequest (R4).locationReference | Reference | | Yes | 0..* | Reference to the Location FHIR resource where radiotherapy has been requested.
ServiceRequest (R4).meta | Meta
ServiceRequest (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).
ServiceRequest (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this ServiceRequest resource. The following profiles are supported. Profile Name Profile Value Code Intent Radiotherapy Course Prescription

http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription 1217123003 original-order Radiotherapy Planned Course http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course 1217123003 filler-order Radiotherapy Phase Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription 1222565005 original-order Radiotherapy Planned Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase 1222565005 filler-order Radiotherapy Plan Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription 1255724003 original-order Radiotherapy Treatment Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan 1255724003 filler-order

Profile Name | Profile Value | Code | Intent

Radiotherapy Course Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription | 1217123003 | original-order

Radiotherapy Planned Course | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course | 1217123003 | filler-order

Radiotherapy Phase Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription | 1222565005 | original-order

Radiotherapy Planned Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase | 1222565005 | filler-order

Radiotherapy Plan Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription | 1255724003 | original-order

Radiotherapy Treatment Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan | 1255724003 | filler-order

ServiceRequest (R4).occurrencePeriod | Period | | | 0..1 | The period during which this course/phase/plan is active.

ServiceRequest (R4).occurrencePeriod.end | String | | | 0..1 | The end of the period.

ServiceRequest (R4).occurrencePeriod.start | String | | | 0..1 | The start of the period.

ServiceRequest (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.

ServiceRequest (R4).reasonCode.coding | Coding | | Yes | 0..1 | Coded representation of the problem.

ServiceRequest (R4).reasonCode.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-cancer-disorder-vs.

ServiceRequest (R4).reasonCode.coding.display | String | | | 0..1 | The textual representation of the coded value.

ServiceRequest (R4).reasonCode.coding.system | String | | | 0..1 | The URI for the system in which this coded value is assigned. Preferred values: http://snomed.info/sct http://hl7.org/fhir/sid/icd-10-cm

ServiceRequest (R4).reasonCode.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.

ServiceRequest (R4).replaces | Reference | | Yes | 0..* | Reference to a ServiceRequest resource that represents the previous version of this course/phase/plan that is replaced by this one.

ServiceRequest (R4).requester | Reference | | | 0..1 | The person or entity responsible for the request.

ServiceRequest (R4).status | code | | | 1..1 | The status of the course/phase/plan. The value should come form the value set http://hl7.org/fhir/R4/ValueSet/request-status

ServiceRequest (R4).subject | Reference | | | 1..1 | Reference to the Patient FHIR resource that this radiotherapy course/plan/phase is planned or prescribed to.

Response elements

Name | Type | Optional | Array | Cardinality | Description

ServiceRequest (R4) | Resource | | | | An instance of the R4 ServiceRequest resource.

ServiceRequest (R4).authoredOn | dateTime | | | 0..1 | When this ServiceRequest was signed.

ServiceRequest (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Radiotherapy Planned Course: References a ServiceRequest course prescription that this planned course fulfills. Radiotherapy Phase Prescription: References a ServiceRequest course prescription of which this phase prescription is a component. Radiotherapy Planned Phase: Can reference either a ServiceRequest planned course of which this planned phase is a component, or a ServiceRequest phase prescription that this planned phase fulfills. Radiotherapy Plan Prescription: Can reference either a ServiceRequest course prescription of which this plan prescription is a component, or a ServiceRequest phase prescription of which this plan prescription is a component. Radiotherapy Treatment Plan: Can reference either a ServiceRequest planned phase of which this treatment plan is a component, or a ServiceRequest plan prescription that this treatment plan fulfills.

ServiceRequest (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.

ServiceRequest (R4).bodySite.coding | Coding | | Yes | 1..* | The SNOMED code for this anatomical site.

ServiceRequest (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, a CodeableConcept for the laterality of the anatomical site (left, right, bilateral). URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier

ServiceRequest (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, a CodeableConcept for additional qualifiers specifying the anatomical site being treated. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer

ServiceRequest (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.

ServiceRequest (R4).category | CodeableConcept | | Yes | 1..* | Always the SNOMED code for Radiotherapy (procedure): 1287742003.

ServiceRequest (R4).code | CodeableConcept | | | 1..1 | The code of the service being requested. Possible values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)

ServiceRequest (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | A Boolean that indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, a CodeableConcept that defines what free-breathing method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality

ServiceRequest (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | A CodeableConcept that defines the imaging procedure used for intrafraction verification. URL: procedure

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | A CodeableConcept that further describes the imaging used for intrafraction verification. URL: qualifier

ServiceRequest (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | A CodeableConcept that specifies the modality of the radiation used. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, as a Quantity for the beam energy (for external-beam modalities) or CodeableConcept for source isotope (for brachytherapy modalities). URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | A CodeableConcept that defines what type of applicator is used for a brachytherapy source. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | A CodeableConcept that specifies the technique of the radiation used, which is more specific than the modality. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume) (also: extension (radiotherapyDosePlannedToVolume)) | Extension | | Yes | 0..* | Specifies the plan of doses to be delivered to one or more volumes as part of this regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-planned-to-volume

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | A Quantity for the fractional dose prescribed for this volume. URL: fractionDose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | An Integer for how many fractions are prescribed for this volume in the course. URL: fractions

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | A Boolean that indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | A Boolean that indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | A Quantity that provides a more tightly specified metric for the given dose. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | A Quantity that specifies the total dose prescribed for this volume. URL: totalDose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | A Reference to the radiotherapy volume (BodyStructure FHIR resource) whose dose is being described. URL: volume

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume) (also: extension (radiotherapyDosePrescribedToVolume)) | Extension | | Yes | 0..* | Specifies the prescription of doses to be delivered to one or more volumes as part of this regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-to-volume

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | A Quantity that specifies the fractional dose prescribed for this volume. URL: fractionDose

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | An Integer for how many fractions are prescribed for this volume in the course. URL: fractions

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | A Quantity for the total dose prescribed for this volume. URL: totalDose

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | A Reference to the radiotherapy volume (BodyStructure FHIR resource) whose dose is being described. URL: volume

ServiceRequest (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, a CodeableConcept that defines what method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management

ServiceRequest (R4).extension (treatmentIntent) (also: extension (treatmentIntent)) | Extension | | Yes | 0..1 | A CodeableConcept that indicates the clinical intent of this radiotherapy regimen. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent

ServiceRequest (R4).id | id | | | 0..1 | The ServiceRequest FHIR ID.

ServiceRequest (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.

ServiceRequest (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.

ServiceRequest (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).

ServiceRequest (R4).identifier.value | String | | | 1..1 | The value of the identifier.

ServiceRequest (R4).intent | code | | | 1..1 | If the course/phase/plan is prescribed, then this value is original-order. If it is planned, then the value is filler-order.

ServiceRequest (R4).locationReference | Reference | | Yes | 0..* | Reference to the Location FHIR resource where radiotherapy has been requested.

ServiceRequest (R4).meta | Meta

ServiceRequest (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).

ServiceRequest (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this ServiceRequest resource. The following profiles are supported. Profile Name Profile Value Code Intent Radiotherapy Course Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription 1217123003 original-order Radiotherapy Planned Course http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course 1217123003 filler-order

Radiotherapy Phase Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription 1222565005 original-order Radiotherapy Planned Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase 1222565005 filler-order Radiotherapy Plan Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription 1255724003 original-order Radiotherapy Treatment Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan 1255724003 filler-order

Profile Name | Profile Value | Code | Intent

Radiotherapy Course Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription | 1217123003 | original-order

Radiotherapy Planned Course | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course | 1217123003 | filler-order

Radiotherapy Phase Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription | 1222565005 | original-order

Radiotherapy Planned Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase | 1222565005 | filler-order

Radiotherapy Plan Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription | 1255724003 | original-order

Radiotherapy Treatment Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan | 1255724003 | filler-order

ServiceRequest (R4).occurrencePeriod | Period | | | 0..1 | The period during which this course/phase/plan is active.

ServiceRequest (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.

ServiceRequest (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.

ServiceRequest (R4).replaces | Reference | | Yes | 0..* | Reference to a ServiceRequest resource that represents the previous version of this course/phase/plan that is replaced by this one.

ServiceRequest (R4).requester | Reference | | | 0..1 | The person or entity responsible for the request.

ServiceRequest (R4).status | code | | | 1..1 | The status of the course/phase/plan. The value should come from the value set http://hl7.org/fhir/R4/ValueSet/request-status

ServiceRequest (R4).subject | Reference | | | 1..1 | Reference to the Patient FHIR resource that this radiotherapy course/plan/phase is planned or prescribed to.

Sample request

```
POST https://hostname/instance/api/FHIR/R4/ServiceRequest
{
  "resourceType": "ServiceRequest",
  "meta": {
    "lastUpdated": "2026-02-01T09:09:51-06:00",
    "profile": [
      "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription"
    ]
  },
  "category": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "1287742003"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "1217123003"
      }
    ]
  },
  "intent": "original-order",
  "identifier": [
    {
      "use": "usual",
      "system": "urn:hand:display",
      "value": "Bilateral Breast"
    },
    {
      "use": "official",
      "system": "urn:dicom:uid",
      "value": "38dc290f-f773-4489-9a46-b0c2075e89d2"
    }
  ],
  "status": "active",
  "requester": {
    "reference": "PractitionerRole/fIidEgs9aUIFFBCsj1TnOXmmbOuEnhxY8XHSG-tyNFpo4"
  },
  "subject": {
    "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
    "display": "Test,Patient"
  },
  "occurrencePeriod": {
    "start": "2026-02-01T09:09:51-06:00",
    "end": ""
  },
```

```
"authoredOn": "2026-02-01T09:09:51-06:00",
"reasonCode": [
  {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "C50",
        "display": "Malignant neoplasm of breast"
      }
    ],
    "text": "Cancer Problem"
  }
],
"reasonReference": [
  {
    "reference": "Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3"
  }
],
"bodySite": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "Breast"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangent"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "7771000",
              "display": "Left"
            }
          ]
        }
      }
    ]
  },
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "Breast"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangent"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "24028007",
              "display": "Right"
            }
          ]
        }
      }
    ]
  },
```

```
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "68171009",
          "display": "Axillary lymph node"
        }
      ],
      "extension": [
        {
          "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "7771000",
                "display": "Left"
              }
            ]
          }
        }
      ]
    }
  ],
  "locationReference": [
    {
      "reference": "Location/e-iz4EXUrNFfCKJwdsZPd9Q3"
    }
  ],
  "extension": [
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": 129428001,
            "display": "Preventive"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification",
      "extension": [
        {
          "url": "procedure",
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "168537006",
                "display": "Plain X-ray"
              }
            ]
          }
        },
        {
          "url": "qualifier",
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "255204007",
                "display": "Multiple"
              }
            ]
          }
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification",
      "extension": [
        {
          "url": "procedure",
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://hl7.org/fhir/v3/NullFlavor",
                "code": "UNC",
                "display": "un-encoded"
              }
            ],
            "text": "IFV procedure"
          }
        }
      ]
    },
```

```
{
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification",
  "extension": [
    {
      "url": "procedure",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "168537006",
            "display": "Plain X-ray"
          }
        ]
      }
    },
    {
      "url": "qualifier",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/v3/NullFlavor",
            "code": "UNC",
            "display": "un-encoded"
          }
        ],
        "text": "IFV X-ray qualifier"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique",
  "extension": [
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "1156506007",
            "display": "Photons"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique",
      "valueCodeableConcept": {
        "coding": [
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            "system": "http://snomed.info/sct",
            "code": 1162782007,
            "display": "3D"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-
isotope",
      "valueQuantity": {
        "value": "18",
        "code": "MeV",
        "system": "http://unitsofmeasure.org"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique",
  "extension": [
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "45643008",
            "display": "Electrons"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
```

```
            "code": 1162782007,
            "display": "3D"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-
isotope",
      "valueQuantity": {
        "value": "18",
        "code": "MeV",
        "system": "http://unitsofmeasure.org"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-
to-volume",
  "extension": [
    {
      "url": "volume",
      "valueReference": {
        "reference": "BodyStructure/fVOwxz38-u4Es-qKSYFEHIQ4",
        "display": "Left Breast"
      }
    },
    {
      "url": "totalDose",
      "valueQuantity": {
        "value": "4256",
        "code": "cGy",
        "unit": "cGy",
        "system": "http://unitsofmeasure.org"
      }
    },
    {
      "url": "fractions",
      "valuePositiveInt": "16"
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric",
      "valueQuantity": {
        "value": 64,
        "unit": "EQD2Gy(a/b=2.5)",
        "system": "http://unitsofmeasure.org",
        "code": "Gy"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-
to-volume",
  "extension": [
    {
      "url": "volume",
      "valueReference": {
        "reference": "BodyStructure/f4AD3BX346M0Nhm2jQpdEOg4",
        "display": "Right Breast"
      }
    },
    {
      "url": "totalDose",
      "valueQuantity": {
        "value": "4256",
        "code": "cGy",
        "unit": "cGy",
        "system": "http://unitsofmeasure.org"
      }
    },
    {
      "url": "fractions",
      "valuePositiveInt": "16"
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric",
      "valueQuantity": {
        "value": 64,
        "unit": "EQD2Gy(a/b=2.5)",
        "system": "http://unitsofmeasure.org",
        "code": "Gy"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-
```

```
    to-volume",
      "extension": [
        {
          "url": "volume",
          "valueReference": {
            "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4",
            "display": "Left Axillary Node"
          }
        },
        {
          "url": "totalDose",
          "valueQuantity": {
            "value": "4256",
            "code": "cGy",
            "unit": "cGy",
            "system": "http://unitsofmeasure.org"
          }
        },
        {
          "url": "fractions",
          "valuePositiveInt": "16"
        },
        {
          "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric",
          "valueQuantity": {
            "value": 64,
            "unit": "EQD2Gy(a/b=2.5)",
            "system": "http://unitsofmeasure.org",
            "code": "Gy"
          }
        }
      ]
    }
  ]
}
```

Sample response

```
Status code: Created (201)
Content-Type: application/json; charset=utf-8
Location: ServiceRequest/fXG74SEeaQkfhgC534YkPkQ4
```

13.1.33 ServiceRequest.Read (External Radiotherapy Summary) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11045-ServiceRequest-Read-External-Radiotherapy-Summary-R4.html
Epic API ID | 11045
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/ServiceRequest/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11045

Description

ServiceRequest resources represent planning summaries at the course, phase, and plan levels. These resources can be sent either prescribed (oncologist has defined what treatment the patient needs) or planned (intended treatment has been planned through dosimetry).

There are six types of ServiceRequests: Course Prescription, Planned Course, Phase Prescription, Planned Phase, Plan Prescription, and Treatment Plan. The table below describes the category, code, and intent that indicates the differences between the types.

Type | Profile | Category | SNOMED Code | Intent
Course Prescription | Radiotherapy Course Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1217123003 | original-order
Planned Course | Radiotherapy Planned Course - CodeX Radiation Therapy v2.0.0 | 108290001 | 1217123003 | filler-order
Phase Prescription | Radiotherapy Phase Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1222565005 | original-order
Planned Phase | Radiotherapy Planned Phase - CodeX Radiation Therapy v2.0.0 | 108290001 | 1222565005 | filler-order
Plan Prescription | Radiotherapy Plan Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1255724003 | original-order
Treatment Plan | Radiotherapy Treatment Plan - CodeX Radiation Therapy v2.0.0 | 108290001 | 1255724003 | filler-order

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | Required | | | The ServiceRequest FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description

ServiceRequest (R4) | Resource | | | | An instance of the R4 ServiceRequest resource.

ServiceRequest (R4).authoredOn | dateTime | | | 0..1 | When this ServiceRequest was signed.

ServiceRequest (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Radiotherapy Planned Course: References a ServiceRequest course prescription that this planned course fulfills. Radiotherapy Phase Prescription: References a ServiceRequest course prescription of which this phase prescription is a component. Radiotherapy Planned Phase: Can reference either a ServiceRequest planned course of which this planned phase is a component, or a ServiceRequest phase prescription that this planned phase fulfills. Radiotherapy Plan Prescription: Can reference either a ServiceRequest course prescription of which this plan prescription is a component, or a ServiceRequest phase prescription of which this plan prescription is a component. Radiotherapy Treatment Plan: Can reference either a ServiceRequest planned phase of which this treatment plan is a component, or a ServiceRequest plan prescription that this treatment plan fulfills.

ServiceRequest (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.

ServiceRequest (R4).bodySite.coding | Coding | | Yes | 1..* | The SNOMED code for this anatomical site.

ServiceRequest (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, a CodeableConcept for the laterality of the anatomical site (left, right, bilateral). URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier

ServiceRequest (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, a CodeableConcept for additional qualifiers specifying the anatomical site being treated. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer

ServiceRequest (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.

ServiceRequest (R4).category | CodeableConcept | | Yes | 1..* | Always the SNOMED code for Radiotherapy (procedure): 1287742003.

ServiceRequest (R4).code | CodeableConcept | | | 1..1 | The code of the service being requested. Possible values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)

ServiceRequest (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | A Boolean that indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, a CodeableConcept that defines what free-breathing method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality

ServiceRequest (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | A CodeableConcept that defines the imaging procedure used for intrafraction verification. URL: procedure

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | A CodeableConcept that further describes the imaging used for intrafraction verification. URL: qualifier

ServiceRequest (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | A CodeableConcept that specifies the modality of the radiation used. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, as a Quantity for the beam energy (for external-beam modalities) or CodeableConcept for source isotope (for brachytherapy modalities). URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | A CodeableConcept that defines what type of applicator is used for a brachytherapy source. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | A CodeableConcept that specifies the technique of the radiation used, which is more specific than the modality. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume) (also: extension (radiotherapyDosePlannedToVolume)) | Extension | | Yes | 0..* | Specifies the plan of doses to be delivered to one or more volumes as part of this regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-planned-to-volume

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | A Quantity for the fractional dose prescribed for this volume. URL: fractionDose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | An Integer for how many fractions are prescribed for this volume in the course. URL: fractions

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | A Boolean that indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | A Boolean that indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | A Quantity that provides a more tightly specified metric for the given dose. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | A Quantity that specifies the total dose prescribed for this volume. URL: totalDose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | A Reference to the radiotherapy volume (BodyStructure FHIR resource) whose dose is being described. URL: volume

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume) (also: extension (radiotherapyDosePrescribedToVolume)) | Extension | | Yes | 0..* | Specifies the prescription of doses to be delivered to one or more volumes as part of this regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-to-volume

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | A Quantity that specifies the fractional dose prescribed for this volume. URL: fractionDose

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | An Integer for how many fractions are prescribed for this volume in the course. URL: fractions

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | A Quantity for the total dose prescribed for this volume. URL: totalDose

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | A Reference to the radiotherapy volume (BodyStructure FHIR resource) whose dose is being described. URL: volume

ServiceRequest (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, a CodeableConcept that defines what method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management

ServiceRequest (R4).extension (treatmentIntent) (also: extension (treatmentIntent)) | Extension | | Yes | 0..1 | A CodeableConcept that indicates the clinical intent of this radiotherapy regimen. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent

ServiceRequest (R4).id | id | | | 0..1 | The ServiceRequest FHIR ID.

ServiceRequest (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.

ServiceRequest (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.

ServiceRequest (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).

ServiceRequest (R4).identifier.value | String | | | 1..1 | The value of the identifier.

ServiceRequest (R4).intent | code | | | 1..1 | If the course/phase/plan is prescribed, then this value is original-order. If it is planned, then the value is filler-order.

ServiceRequest (R4).locationReference | Reference | | Yes | 0..* | Reference to the Location FHIR resource where radiotherapy has been requested.

ServiceRequest (R4).meta | Meta

ServiceRequest (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).

ServiceRequest (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this ServiceRequest resource. The following profiles are supported. Profile Name Profile Value Code Intent Radiotherapy Course Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription 1217123003 original-order Radiotherapy Planned Course http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course 1217123003 filler-order Radiotherapy Phase Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription 1222565005 original-order Radiotherapy Planned Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase 1222565005 filler-order Radiotherapy Plan Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription 1255724003 original-order Radiotherapy Treatment Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan 1255724003 filler-order

Profile Name | Profile Value | Code | Intent

Radiotherapy Course Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription | 1217123003 | original-order

Radiotherapy Planned Course | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course | 1217123003 | filler-order

Radiotherapy Phase Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription | 1222565005 | original-order

Radiotherapy Planned Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase | 1222565005 | filler-order

Radiotherapy Plan Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription | 1255724003 | original-order

Radiotherapy Treatment Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan | 1255724003 | filler-order

ServiceRequest (R4).occurrencePeriod | Period | | | 0..1 | The period during which this course/phase/plan is active.

ServiceRequest (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.

ServiceRequest (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.

ServiceRequest (R4).replaces | Reference | | Yes | 0..* | Reference to a ServiceRequest resource that represents the previous version of this course/phase/plan that is replaced by this one.

ServiceRequest (R4).requester | Reference | | | 0..1 | The person or entity responsible for the request.

ServiceRequest (R4).status | code | | | 1..1 | The status of the course/phase/plan. The value should come from the value set http://hl7.org/fhir/R4/ValueSet/request-status

ServiceRequest (R4).subject | Reference | | | 1..1 | Reference to the Patient FHIR resource that this radiotherapy course/plan/phase is planned or prescribed to.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/ServiceRequest/fXG74SEeaQkfhgC534YkPkQ4
```

Sample response


```
{
  "resourceType": "ServiceRequest",
  "id": "fXG74SEeaQkfhgC534YkPkQ4",
  "meta": {
    "lastUpdated": "2026-02-01T09:09:51-06:00",
    "profile": [
```

```
    "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription"
  ]
},
"category": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "1287742003"
      }
    ]
  }
],
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "1217123003"
    }
  ]
},
"intent": "original-order",
"identifier": [
  {
    "use": "usual",
    "system": "urn:hand:display",
    "value": "Bilateral Breast"
  },
  {
    "use": "official",
    "system": "urn:dicom:uid",
    "value": "38dc290f-f773-4489-9a46-b0c2075e89d2"
  }
],
"status": "active",
"requester": {
  "reference": "PractitionerRole/fIidEgs9aUIFFBCsj1TnOXmmbOuEnhxY8XHSG-tyNFpo4"
},
"subject": {
  "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
  "display": "Test,Patient"
},
"occurrencePeriod": {
  "start": "2026-02-01T09:09:51-06:00",
  "end": ""
},
"authoredOn": "2026-02-01T09:09:51-06:00",
"reasonCode": [
  {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "C50",
        "display": "Malignant neoplasm of breast"
      }
    ],
    "text": "Cancer Problem"
  }
],
"reasonReference": [
  {
    "reference": "Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3"
  }
],
"bodySite": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "Breast"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangent"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
```

```
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "7771000",
              "display": "Left"
            }
          ]
        }
      }
    ]
  },
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "76752008",
        "display": "Breast"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangent"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "24028007",
              "display": "Right"
            }
          ]
        }
      }
    ]
  },
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "68171009",
        "display": "Axillary lymph node"
      }
    ],
    "extension": [
      {
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "7771000",
              "display": "Left"
            }
          ]
        }
      }
    ]
  }
],
"locationReference": [
  {
    "reference": "Location/e-iz4EXUrNFfCKJwdsZPd9Q3"
  }
],
"extension": [
  {
    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": 129428001,
          "display": "Preventive"
        }
      ]
    }
  },
  {
```

```
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification",
  "extension": [
    {
      "url": "procedure",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "168537006",
            "display": "Plain X-ray"
          }
        ]
      }
    },
    {
      "url": "qualifier",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "255204007",
            "display": "Multiple"
          }
        ]
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification",
  "extension": [
    {
      "url": "procedure",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/v3/NullFlavor",
            "code": "UNC",
            "display": "un-encoded"
          }
        ],
        "text": "IFV procedure"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification",
  "extension": [
    {
      "url": "procedure",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "168537006",
            "display": "Plain X-ray"
          }
        ]
      }
    },
    {
      "url": "qualifier",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/v3/NullFlavor",
            "code": "UNC",
            "display": "un-encoded"
          }
        ],
        "text": "IFV X-ray qualifier"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique",
  "extension": [
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "1156506007",
            "display": "Photons"
          }
        ]
```

```
      }
    },
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": 1162782007,
            "display": "3D"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-
isotope",
      "valueQuantity": {
        "value": "18",
        "code": "MeV",
        "system": "http://unitsofmeasure.org"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique",
  "extension": [
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "45643008",
            "display": "Electrons"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": 1162782007,
            "display": "3D"
          }
        ]
      }
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-
isotope",
      "valueQuantity": {
        "value": "18",
        "code": "MeV",
        "system": "http://unitsofmeasure.org"
      }
    }
  ]
},
{
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-
to-volume",
  "extension": [
    {
      "url": "volume",
      "valueReference": {
        "reference": "BodyStructure/fVOwxz38-u4Es-qKSYFEHIQ4",
        "display": "Left Breast"
      }
    },
    {
      "url": "totalDose",
      "valueQuantity": {
        "value": "4256",
        "code": "cGy",
        "unit": "cGy",
        "system": "http://unitsofmeasure.org"
      }
    },
    {
      "url": "fractions",
      "valuePositiveInt": "16"
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric",
```

```
          "valueQuantity": {
            "value": 64,
            "unit": "EQD2Gy(a/b=2.5)",
            "system": "http://unitsofmeasure.org",
            "code": "Gy"
          }
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-
    to-volume",
      "extension": [
        {
          "url": "volume",
          "valueReference": {
            "reference": "BodyStructure/f4AD3BX346M0Nhm2jQpdEOg4",
            "display": "Right Breast"
          }
        },
        {
          "url": "totalDose",
          "valueQuantity": {
            "value": "4256",
            "code": "cGy",
            "unit": "cGy",
            "system": "http://unitsofmeasure.org"
          }
        },
        {
          "url": "fractions",
          "valuePositiveInt": "16"
        },
        {
          "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric",
          "valueQuantity": {
            "value": 64,
            "unit": "EQD2Gy(a/b=2.5)",
            "system": "http://unitsofmeasure.org",
            "code": "Gy"
          }
        }
      ]
    },
    {
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-
    to-volume",
      "extension": [
        {
          "url": "volume",
          "valueReference": {
            "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4",
            "display": "Left Axillary Node"
          }
        },
        {
          "url": "totalDose",
          "valueQuantity": {
            "value": "4256",
            "code": "cGy",
            "unit": "cGy",
            "system": "http://unitsofmeasure.org"
          }
        },
        {
          "url": "fractions",
          "valuePositiveInt": "16"
        },
        {
          "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric",
          "valueQuantity": {
            "value": 64,
            "unit": "EQD2Gy(a/b=2.5)",
            "system": "http://unitsofmeasure.org",
            "code": "Gy"
          }
        }
      ]
    }
  ]
}
```


13.1.34 ServiceRequest.Read (Pregnancy Plans) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10276-ServiceRequest-Read-Pregnancy-Plans-R4.html
Epic API ID | 10276
FHIR version | R4

Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/ServiceRequest/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10276

Response elements

Name | Type | Optional | Array | Cardinality | Description
ServiceRequest | Resource | False | | | An instance of the R4 ServiceRequest resource.
ServiceRequest.category | CodeableConcept | False | Yes | | Always returns “pregnancy-plans.”
ServiceRequest.identifier | Identifier | False | Yes | | The identifier.
ServiceRequest.intent | code | False | | | The FHIR intent code. Always returns “plan."
ServiceRequest.locationCode | CodeableConcept | True | Yes | | Planned delivery location.
ServiceRequest.occurrenceDateTime | String | True | | | The working EDD for the pregnancy.
ServiceRequest.status | code | False | | | The status of the request. Supported values include: * Active * Completed * Entered-in-error
ServiceRequest.subject | Reference | False | | | Reference to a patient resource.
ServiceRequest.supportingInfo | Reference | False | Yes | | Reference to the pregnancy episode associated with this plan.

Sample request

```
https://hostname/instance/api/FHIR/R4/ServiceRequest/ePl1Idc84Thyx.sIOzxRrjMChfyQ3Eepkl3SDMNxRKkg3
```

Sample response


```
{
  "resourceType": "ServiceRequest",
  "id": "ePl1Idc84Thyx.sIOzxRrjMChfyQ3Eepkl3SDMNxRKkg3",
  "identifier": [
    {
      "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.728366",
      "value": "853441"
    }
  ],
  "status": "completed",
  "intent": "plan",
  "category": [
    {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/service-request-category",
          "code": "pregnancy-plans",
          "display": "Pregnancy plans"
        }
      ],
      "text": "Pregnancy plans"
    }
  ],
  "subject": {
    "reference": "Patient/eV2PtTDuGgd2W9DJZh3Hv2Gli1hT8E5xgBCXqnAVzSuI3",
    "display": "Smith, Agnes"
  },
  "locationCode": [
    {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.10.728366.35542",
          "code": "1",
          "display": "Hospital"
        }
      ],
      "text": "Hospital"
    }
  ],
  "supportingInfo": [
    {
      "reference": "EpisodeOfCare/eLZ6JoZ4gvHsTAi9c.rgOPQ3",
      "display": "Pregnancy 2"
    }
  ]
}
```


13.1.35 ServiceRequest.Search (External Radiotherapy Summary) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11046-ServiceRequest-Search-External-Radiotherapy-Summary-R4.html
Epic API ID | 11046
FHIR version | R4

Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/ServiceRequest?authored={authored}&category={category}&code={code}&encounter={encounter}&identifier={identifier}&onlyscannable={onlyscannable}&patient={patient}&requester={requester}&status={status}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11046

Description

ServiceRequest resources represent planning summaries at the course, phase, and plan levels. These resources can be sent either prescribed (oncologist has defined what treatment the patient needs) or planned (intended treatment has been planned through dosimetry).

There are six types of ServiceRequests: Course Prescription, Planned Course, Phase Prescription, Planned Phase, Plan Prescription, and Treatment Plan. The table below describes the category, code, and intent that indicates the differences between the types.

Type | Profile | Category | SNOMED Code | Intent
Course Prescription | Radiotherapy Course Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1217123003 | original-order
Planned Course | Radiotherapy Planned Course - CodeX Radiation Therapy v2.0.0 | 108290001 | 1217123003 | filler-order
Phase Prescription | Radiotherapy Phase Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1222565005 | original-order
Planned Phase | Radiotherapy Planned Phase - CodeX Radiation Therapy v2.0.0 | 108290001 | 1222565005 | filler-order
Plan Prescription | Radiotherapy Plan Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1255724003 | original-order
Treatment Plan | Radiotherapy Treatment Plan - CodeX Radiation Therapy v2.0.0 | 108290001 | 1255724003 | filler-order

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | String | Optional | Yes | | Number of results per page.
_id | String | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others are ignored.
_include | String | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_revinclude | String | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | String | Optional | Yes | | Procedure category, this is always a fixed code for all CodeX RT profiles.
code | String | Optional | Yes | | Procedure code, these are fixed values for the CodeX RT profiles to distinguish between courses, phases, and plans.
identifier | String | Optional | Yes | | Identifiers associated with the ServiceRequest. They should be in the format of system|value.
patient | String | Conditional | Yes | | The FHIR ID for the patient associated with the ServiceRequest. Either patient or subject must be provided.
requester | String | Optional | Yes | | Who requested the service.
status | String | Optional | Yes | | Status of the service request: in progress, completed, etc.
subject | String | Conditional | Yes | | The FHIR ID for the patient associated with the ServiceRequest. Either patient or subject must be provided.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
authored | String | Optional | Yes | | When this ServiceRequest was signed.
based-on | String | Optional | Yes | | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile, see the basedOn response element for more details.
body-site | String | Optional | Yes | | A representation of each anatomical site to be treated with radiotherapy.
intent | String | Optional | Yes | | If the course/phase/plan is prescribed, then provide original-order. If it is planned, then provide filler-order.
occurrence | String | Optional | Yes | | The period during which this course/phase/plan is active.
replaces | String | Optional | Yes | | Reference to a ServiceRequest resource that represents the previous version of this course/phase/plan that is replaced by this one.

Response elements

Name | Type | Optional | Array | Cardinality | Description
ServiceRequest (R4) | Resource | | | | An instance of the R4 ServiceRequest resource.
ServiceRequest (R4).authoredOn | dateTime | | | 0..1 | When this ServiceRequest was signed.
ServiceRequest (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Radiotherapy Planned Course: References a ServiceRequest course prescription that this planned course fulfills. Radiotherapy Phase Prescription: References a ServiceRequest course prescription of which this phase prescription is a component. Radiotherapy Planned Phase: Can reference either a ServiceRequest planned course of which this planned phase is a component, or a ServiceRequest phase prescription that this planned phase fulfills. Radiotherapy Plan Prescription: Can reference either a ServiceRequest course prescription of which this plan prescription is a component, or a ServiceRequest phase prescription of which this plan prescription is a component. Radiotherapy Treatment Plan: Can reference either a ServiceRequest planned phase of which this treatment plan is a component, or a ServiceRequest plan prescription that this treatment plan fulfills.
ServiceRequest (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.
ServiceRequest (R4).bodySite.coding | Coding | | Yes | 1..* | The SNOMED code for this anatomical site.

ServiceRequest (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, a CodeableConcept for the laterality of the anatomical site (left, right, bilateral). URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier

ServiceRequest (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, a CodeableConcept for additional qualifiers specifying the anatomical site being treated. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer

ServiceRequest (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.

ServiceRequest (R4).category | CodeableConcept | | Yes | 1..* | Always the SNOMED code for Radiotherapy (procedure): 1287742003.

ServiceRequest (R4).code | CodeableConcept | | | 1..1 | The code of the service being requested. Possible values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)

ServiceRequest (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | A Boolean that indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation

ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, a CodeableConcept that defines what free-breathing method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique

ServiceRequest (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality

ServiceRequest (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification

ServiceRequest (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | A CodeableConcept that defines the imaging procedure used for intrafraction verification. URL: procedure

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | A CodeableConcept that further describes the imaging used for intrafraction verification. URL: qualifier

ServiceRequest (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | A CodeableConcept that specifies the modality of the radiation used. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, as a Quantity for the beam energy (for external-beam modalities) or CodeableConcept for source isotope (for brachytherapy modalities). URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | A CodeableConcept that defines what type of applicator is used for a brachytherapy source. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | A CodeableConcept that specifies the technique of the radiation used, which is more specific than the modality. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume) (also: extension (radiotherapyDosePlannedToVolume)) | Extension | | Yes | 0..* | Specifies the plan of doses to be delivered to one or more volumes as part of this regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-planned-to-volume

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | A Quantity for the fractional dose prescribed for this volume. URL: fractionDose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | An Integer for how many fractions are prescribed for this volume in the course. URL: fractions

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | A Boolean that indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | A Boolean that indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | A Quantity that provides a more tightly specified metric for the given dose. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | A Quantity that specifies the total dose prescribed for this volume. URL: totalDose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | A Reference to the radiotherapy volume (BodyStructure FHIR resource) whose dose is being described. URL: volume

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume) (also: extension (radiotherapyDosePrescribedToVolume)) | Extension | | Yes | 0..* | Specifies the prescription of doses to be delivered to one or more volumes as part of this regimen. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-to-volume

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | A Quantity that specifies the fractional dose prescribed for this volume. URL: fractionDose

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | An Integer for how many fractions are prescribed for this volume in the course. URL: fractions

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | A Quantity for the total dose prescribed for this volume. URL: totalDose

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | A Reference to the radiotherapy volume (BodyStructure FHIR resource) whose dose is being described. URL: volume

ServiceRequest (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, a CodeableConcept that defines what method is used. URL: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management

ServiceRequest (R4).extension (treatmentIntent) (also: extension (treatmentIntent)) | Extension | | Yes | 0..1 | A CodeableConcept that indicates the clinical intent of this radiotherapy regimen. URL: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent

ServiceRequest (R4).id | id | | | 0..1 | The ServiceRequest FHIR ID.

ServiceRequest (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.

ServiceRequest (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.

ServiceRequest (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).

ServiceRequest (R4).identifier.value | String | | | 1..1 | The value of the identifier.

ServiceRequest (R4).intent | code | | | 1..1 | If the course/phase/plan is prescribed, then this value is original-order. If it is planned, then the value is filler-order.

ServiceRequest (R4).locationReference | Reference | | Yes | 0..* | Reference to the Location FHIR resource where radiotherapy has been requested.

ServiceRequest (R4).meta | Meta

ServiceRequest (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).

ServiceRequest (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this ServiceRequest resource. The following profiles are supported. Profile Name Profile Value Code Intent Radiotherapy Course Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription 1217123003 original-order Radiotherapy Planned Course http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course 1217123003 filler-order Radiotherapy Phase Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription 1222565005 original-order Radiotherapy Planned Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase 1222565005 filler-order Radiotherapy Plan Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription 1255724003 original-order Radiotherapy Treatment Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan 1255724003 filler-order

Profile Name | Profile Value | Code | Intent

Radiotherapy Course Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription | 1217123003 | original-order

Radiotherapy Planned Course | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course | 1217123003 | filler-order

Radiotherapy Phase Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription | 1222565005 | original-order

Radiotherapy Planned Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase | 1222565005 | filler-order

Radiotherapy Plan Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription | 1255724003 | original-order

Radiotherapy Treatment Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan | 1255724003 | filler-order

ServiceRequest (R4).occurrencePeriod | Period | | | 0..1 | The period during which this course/phase/plan is active.

ServiceRequest (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.

ServiceRequest (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.

ServiceRequest (R4).replaces | Reference | | Yes | 0..* | Reference to a ServiceRequest resource that represents the previous version of this course/phase/plan that is replaced by this one.

ServiceRequest (R4).requester | Reference | | | 0..1 | The person or entity responsible for the request.

ServiceRequest (R4).status | code | | | 1..1 | The status of the course/phase/plan. The value should come from the value set http://hl7.org/fhir/R4/ValueSet/request-status

ServiceRequest (R4).subject | Reference | | | 1..1 | Reference to the Patient FHIR resource that this radiotherapy course/plan/phase is planned or prescribed to.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/ServiceRequest?patient=eP-
    BmFG0190obAH1wt4wjUw3&category=http://snomed.info/sct|1287742003&code=http://snomed.info/sct|1217123003
```

Sample response


```
{
    "resourceType": "Bundle",
    "type": "searchset",
    "total": 1,
    "link": [
        {
            "relation": "self",
            "url": "https://hostname/instance/api/FHIR/R4/ServiceRequest?patient=eP-BmFG0190obAH1wt4wjUw3"
        }
    ],
 "entry": [
        {
            "link": [
                {
                    "relation": "self",
                    "url": "https://hostname/instance/api/FHIR/R4/ServiceRequest/e214EEAa7RY8csEhIEJW-
    qOpDYFRKCusulwMwwzo333g3"
                }
            ],
            "fullUrl": "https://hostname/instance/api/FHIR/R4/ServiceRequest/e214EEAa7RY8csEhIEJW-
    qOpDYFRKCusulwMwwzo333g3",
            "resource": {
                "resourceType": "ServiceRequest",
                "id": "e214EEAa7RY8csEhIEJW-qOpDYFRKCusulwMwwzo333g3",
```

```
            "meta": {
                "versionId": "1",
                "lastUpdated": "2025-12-17T01:37:37Z",
                "profile": [
                    "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-
planned-course"
                ]
            },
            "extension": [
                {
                    "extension": [
                        {
                            "valueReference": {
                                "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
                            },
                            "url": "volume"
                        },
                        {
                            "valuePositiveInt": 30,
                            "url": "fractions"
                        },
                        {
                            "valueQuantity": {
                                "value": 3000,
                                "unit": "cGy",
                                "system": "http://unitsofmeasure.org",
                                "code": "cGy"
                            },
                            "url": "totalDose"
                        },
                        {
                            "valueBoolean": true,
                            "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-
radiotherapy-primary-plan-dose"
                        }
                    ],
                    "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-
dose-planned-to-volume"
                },
                {
                    "extension": [
                        {
                            "valueReference": {
                                "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
                            },
                            "url": "volume"
                        },
                        {
                            "valuePositiveInt": 30,
                            "url": "fractions"
                        },
                        {
                            "valueQuantity": {
                                "value": 3000,
                                "unit": "cGy",
                                "system": "http://unitsofmeasure.org",
                                "code": "cGy"
                            },
                            "url": "totalDose"
                        },
                        {
                            "valueBoolean": true,
                            "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-
radiotherapy-primary-plan-dose"
                        }
                    ],
                    "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-
dose-planned-to-volume"
                },
                {
                    "extension": [
                        {
                            "valueReference": {
                                "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
                            },
                            "url": "volume"
                        },
                        {
                            "valuePositiveInt": 5,
                            "url": "fractions"
                        },
                        {
                            "valueQuantity": {
                                "value": 1000,
                                "unit": "cGy",
                                "system": "http://unitsofmeasure.org",
                                "code": "cGy"
                            },
                            "url": "totalDose"
```

```
                        },
                        {
                            "valueBoolean": true,
                            "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-
radiotherapy-primary-plan-dose"
                        }
                    ],
                    "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-
dose-planned-to-volume"
                },
                {
                    "valueCodeableConcept": {
                        "coding": [
                            {
                                "system": "http://snomed.info/sct",
                                "code": "129428001",
                                "display": "Preventive - procedure intent"
                            }
                        ]
                    },
                    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent"
                },
                {
                    "extension": [
                        {
                            "valueCodeableConcept": {
                                "coding": [
                                    {
                                        "system": "http://snomed.info/sct",
                                        "code": "1156506007",
                                        "display": "External beam radiation therapy using photons"
                                    }
                                ]
                            },
                            "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality"
                        },
                        {
                            "valueCodeableConcept": {
                                "coding": [
                                    {
                                        "system": "http://snomed.info/sct",
                                        "code": "1162782007",
                                        "display": "Three dimensional external beam radiation therapy"
                                    }
                                ]
                            },
                            "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique"
                        },
                        {
                            "valueQuantity": {
                                "value": 18,
                                "unit": "MeV",
                                "system": "http://unitsofmeasure.org",
                                "code": "MeV"
                            },
                            "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-
radiotherapy-energy-or-isotope"
                        }
                    ],
                    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-
technique"
                },
                {
                    "extension": [
                        {
                            "valueCodeableConcept": {
                                "coding": [
                                    {
                                        "system": "http://snomed.info/sct",
                                        "code": "45643008",
                                        "display": "External beam radiation therapy using electrons"
                                    }
                                ]
                            },
                            "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality"
                        },
                        {
                            "valueCodeableConcept": {
                                "coding": [
                                    {
                                        "system": "http://snomed.info/sct",
                                        "code": "1162782007",
                                        "display": "Three dimensional external beam radiation therapy"
                                    }
                                ]
                            },
                            "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique"
                        },
                        {
```

```
                                "valueQuantity": {
                                    "value": 18,
                                    "unit": "MeV",
                                    "system": "http://unitsofmeasure.org",
                                    "code": "MeV"
                                },
                                "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-
    radiotherapy-energy-or-isotope"
                            }
                        ],
                        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-
    technique"
                    }
                ],
                "identifier": [
                    {
                        "use": "usual",
                        "system": "urn:hand:display",
                        "value": "Bilateral Breast"
                    },
                    {
                        "use": "official",
                        "system": "urn:dicom:uid",
                        "value": "1cd3f693-7eae-4ea5-af0b-02ba378547a7"
                    }
                ],
                "basedOn": [
                    {
                        "reference": "ServiceRequest/e214EEAa7RY8csEhIEJW-qOpDYFRKCusulwMwwzo333g3",
                        "type": "ServiceRequest",
                        "display": "Radiotherapy Course Summary - Prescribed"
                    }
                ],
                "status": "active",
                "intent": "filler-order",
                "category": [
                    {
                        "coding": [
                            {
                                "system": "http://snomed.info/sct",
                                "code": "108290001",
                                "display": "Radiation Oncology And/Or Radiotherapy"
                            },
                            {
                                "system": "http://snomed.info/sct",
                                "code": "1287742003",
                                "display": "Radiotherapy"
                            }
                        ]
                    }
                ],
                "code": {
                    "coding": [
                        {
                            "system": "http://snomed.info/sct",
                            "code": "1217123003",
                            "display": "Radiotherapy course of treatment"
                        }
                    ]
                },
                "subject": {
                    "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
                    "display": "Test, Patient""
                },
                "requester": {
                    "reference": "PractitionerRole/fIidEgs9aUIFFBCsj1TnOXmmbOuEnhxY8XHSG-tyNFpo4"
                },
                "locationReference": [
                    {
                        "reference": "Location/e-iz4EXUrNFfCKJwdsZPd9Q3
    "
                    }
                ],
                "reasonCode": [
                    {
                        "coding": [
                            {
                                "system": "http://hl7.org/fhir/sid/icd-10-cm",
                                "code": "C50",
                                "display": "Malignant neoplasm of breast"
                            }
                        ],
                        "text": "Cancer Problem"
                    }
                ],
                "reasonReference": [
                    {
                        "reference": "Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3"
                    }
```

```
],
"bodySite": [
    {
        "extension": [
            {
                "valueCodeableConcept": {
                    "coding": [
                        {
                            "system": "http://snomed.info/sct",
                            "code": "7771000",
                            "display": "Left"
                        }
                    ]
                },
                "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier"
            },
            {
                "valueCodeableConcept": {
                    "coding": [
                        {
                            "system": "http://snomed.info/sct",
                            "code": "399110001",
                            "display": "Tangential projection"
                        }
                    ]
                },
                "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier"
            }
        ],
        "coding": [
            {
                "system": "http://snomed.info/sct",
                "code": "76752008",
                "display": "Breast structure"
            }
        ]
    },
    {
        "extension": [
            {
                "valueCodeableConcept": {
                    "coding": [
                        {
                            "system": "http://snomed.info/sct",
                            "code": "24028007",
                            "display": "Right"
                        }
                    ]
                },
                "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier"
            },
            {
                "valueCodeableConcept": {
                    "coding": [
                        {
                            "system": "http://snomed.info/sct",
                            "code": "399110001",
                            "display": "Tangential projection"
                        }
                    ]
                },
                "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier"
            }
        ],
        "coding": [
            {
                "system": "http://snomed.info/sct",
                "code": "76752008",
                "display": "Breast structure"

            }
        ]
    },
    {
        "extension": [
            {
                "valueCodeableConcept": {
                    "coding": [
                        {
                            "system": "http://snomed.info/sct",
                            "code": "7771000",
                            "display": "Left"
                        }
                    ]
                },
                "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier"
            }
        ],
        "coding": [
            {
```

```
                            "system": "http://snomed.info/sct",
                            "code": "68171009",
                            "display": "Axillary lymph node structure"
                        }
                    ]
                }
            ]
        },
        "search": {
            "mode": "match"
        }
    }
}
```


13.1.36 ServiceRequest.Search (Pregnancy Plans) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/10277-ServiceRequest-Search-Pregnancy-Plans-R4.html
Epic API ID | 10277
FHIR version | R4
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | api/FHIR/R4/ServiceRequest?patient={patient}&subject={subject}&category={category}&status={status}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10277

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Use “pregnancy-plans” to search for pregnancy plans. This parameter is formatted with a system specified (category=[system]|[code]).
identifier | String | True | | | The pregnancy plan Epic identifier
patient | String | Either the patient or subject must be provided. | | | Reference to a patient resource for which the plan was created. Either the patient or subject must be provided.
status | String | True | | | The status of the request. Possible values include: active completed By default, this service returns all statuses if this element is not specified. Active requests are those with an active associated pregnancy episode. Completed requests are those with a resolved pregnancy episode.
subject | String | Either the patient or subject must be provided. | | | Reference to a patient resource for which the plan was created. Either the patient or subject must be provided.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Unsupported
body-site | String | True | | | Unsupported
instantiates-canonical | String | True | | | Unsupported
intent | String | True | | | The FHIR intent code. Always returns “plan.”
occurrence | String | True | | | The working EDD for the pregnancy.
performer | String | True | | | Unsupported
performer-type | String | True | | | Unsupported
priority | String | True | | | Unsupported
replaces | String | True | | | Unsupported
requisition | String | True | | | Unsupported
specimen | String | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.ServiceRequest | Resource | False | | | An instance of the R4 ServiceRequest resource.
Bundle.entry.resource.ServiceRequest.category | CodeableConcept | False | Yes | | Always returns “pregnancy-plans.”
Bundle.entry.resource.ServiceRequest.identifier | Identifier | False | Yes | | The identifier.
Bundle.entry.resource.ServiceRequest.intent | code | False | | | The FHIR intent code. Always returns “plan.”
Bundle.entry.resource.ServiceRequest.locationCode | CodeableConcept | True | Yes | | Planned delivery location.
Bundle.entry.resource.ServiceRequest.occurrenceDateTime | String | True | | | The working EDD for the pregnancy.
Bundle.entry.resource.ServiceRequest.status | code | False | | | The status of the request. Supported values include: * Active * Completed
Bundle.entry.resource.ServiceRequest.subject | Reference | False | | | Reference to a patient resource.
Bundle.entry.resource.ServiceRequest.supportingInfo | Reference | False | Yes | | Reference to the pregnancy episode associated with this plan.

Sample request

```
https://hostname/instance/api/FHIR/R4/ServiceRequest?patient=eV2PtTDuGgd2W9DJZh3Hv2Gli1hT8E5xgBCXqnAVzSuI3
```

Sample response


```
{
    "resourceType": "Bundle",
    "type": "searchset",
    "total": 1,
    "link": [
        {
            "relation": "self",
            "url":
    "https://hostname/instance/api/FHIR/R4/ServiceRequest?patient=eV2PtTDuGgd2W9DJZh3Hv2Gli1hT8E5xgBCXqnAVzSuI3"
        }
    ],
    "entry": [
        {
            "link": [
                {
                    "relation": "self",
                    "url":
    "https://hostname/instance/api/FHIR/R4/ServiceRequest/ePl1Idc84Thyx.sIOzxRrjMChfyQ3Eepkl3SDMNxRKkg3"
                }
            ],
            "fullUrl":
    "https://hostname/instance/api/FHIR/R4/ServiceRequest/ePl1Idc84Thyx.sIOzxRrjMChfyQ3Eepkl3SDMNxRKkg3",
            "resource": {
                "resourceType": "ServiceRequest",
                "id": "ePl1Idc84Thyx.sIOzxRrjMChfyQ3Eepkl3SDMNxRKkg3",
                "identifier": [
                    {
                        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.728366",
                        "value": "853441"
                    }
                ],
                "status": "completed",
                "intent": "plan",
                "category": [
                    {
                        "coding": [
                            {
                                "system": "http://open.epic.com/FHIR/StructureDefinition/service-request-category",
                                "code": "pregnancy-plans",
                                "display": "Pregnancy plans"
                            }
                        ],
                        "text": "Pregnancy plans"
                    }
                ],
                "subject": {
                    "reference": "Patient/eV2PtTDuGgd2W9DJZh3Hv2Gli1hT8E5xgBCXqnAVzSuI3",
                    "display": "Smith, Agnes"
                },
                "locationCode": [
                    {
                        "coding": [
                            {
                                "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.10.728366.35542",
                                "code": "1",
                                "display": "Hospital"
                            }
                        ],
                        "text": "Hospital"
                    }
                ],
                "supportingInfo": [
                    {
                        "reference": "EpisodeOfCare/eLZ6JoZ4gvHsTAi9c.rgOPQ3",
                        "display": "Pregnancy 2"
                    }
                ]
            },
            "search": {
                "mode": "match"
            }
        }
```


13.1.37 ServiceRequest.Update (External Radiotherapy Summary) (R4)

File: site/apis/13-Clinical-Specialty/1-R4/11047-ServiceRequest-Update-External-Radiotherapy-Summary-R4.html
Epic API ID | 11047
FHIR version | R4
Categories | Clinical Specialty, Radiation Therapy
API type | Industry-Standard
API group | Radiation Therapy API Group

USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | PUT
URL template | /api/FHIR/R4/ServiceRequest/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11047

Description

ServiceRequest resources represent planning summaries at the course, phase, and plan levels. These resources can be sent either prescribed (oncologist has defined what treatment the patient needs) or planned (intended treatment has been planned through dosimetry).

There are six types of ServiceRequests: Course Prescription, Planned Course, Phase Prescription, Planned Phase, Plan Prescription, and Treatment Plan. The table below describes the category, code, and intent that indicates the differences between the types.

Type | Profile | Category | SNOMED Code | Intent
Course Prescription | Radiotherapy Course Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1217123003 | original-order
Planned Course | Radiotherapy Planned Course - CodeX Radiation Therapy v2.0.0 | 108290001 | 1217123003 | filler-order
Phase Prescription | Radiotherapy Phase Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1222565005 | original-order
Planned Phase | Radiotherapy Planned Phase - CodeX Radiation Therapy v2.0.0 | 108290001 | 1222565005 | filler-order
Plan Prescription | Radiotherapy Plan Prescription - CodeX Radiation Therapy v2.0.0 | 108290001 | 1255724003 | original-order
Treatment Plan | Radiotherapy Treatment Plan - CodeX Radiation Therapy v2.0.0 | 108290001 | 1255724003 | filler-order

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ServiceRequest (R4) | Resource | | | | An instance of the R4 ServiceRequest resource.
ServiceRequest (R4).authoredOn | dateTime | | | 0..1 | When this ServiceRequest was signed.
ServiceRequest (R4).basedOn | Reference | | Yes | 0..1 | Reference to a FHIR resource for a prior request or plan that this request fulfills. Varies based on the profile: Radiotherapy Planned Course: References a ServiceRequest course prescription that this planned course fulfills. Radiotherapy Phase Prescription: References a ServiceRequest course prescription of which this phase prescription is a component. Radiotherapy Planned Phase: Can reference either a ServiceRequest planned course of which this planned phase is a component, or a ServiceRequest phase prescription that this planned phase fulfills. Radiotherapy Plan Prescription: Can reference either a ServiceRequest course prescription of which this plan prescription is a component, or a ServiceRequest phase prescription of which this plan prescription is a component. Radiotherapy Treatment Plan: Can reference either a ServiceRequest planned phase of which this treatment plan is a component, or a ServiceRequest plan prescription that this treatment plan fulfills.
ServiceRequest (R4).bodySite | CodeableConcept | | Yes | 0..* | A representation of each anatomical site to be treated with radiotherapy.
ServiceRequest (R4).bodySite.coding | Coding | | Yes | 1..* | The coded representation of the anatomical site.
ServiceRequest (R4).bodySite.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-treatment-location-vs.
ServiceRequest (R4).bodySite.coding.display | String | | | 0..1 | The textual representation of the coded value in the given system.
ServiceRequest (R4).bodySite.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).bodySite.extension (lateralityQualifier) (also: extension (lateralityQualifier)) | Extension | | Yes | 0..1 | If relevant, the laterality of the body site (left, right, bilateral).
ServiceRequest (R4).bodySite.extension (lateralityQualifier).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept | CodeableConcept | | | 0..1 | CodeableConcept container for the locationQualifier.
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the lateralityQualifier.
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the lateralityQualifier. The value must be a member of https://hl7.org/fhir/us/mcode/STU4/ValueSet-mcode-laterality-qualifier-vs.html.
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The display name of the SNOMED code.
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).bodySite.extension (lateralityQualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).bodySite.extension (locationQualifier) (also: extension (locationQualifier)) | Extension | | Yes | 0..* | If relevant, additional qualifiers specifying the body site being treated.
ServiceRequest (R4).bodySite.extension (locationQualifier).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifer
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept | CodeableConcept | | | 0..* | CodeableConcept container for the locationQualifier.
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the locationQualifier.
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the locationQualifier. The value must be a member of https://hl7.org/fhir/us/mcode/STU4/ValueSet-mcode-body-location-qualifier-vs.html.
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The display name of the SNOMED code.
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).bodySite.extension (locationQualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).bodySite.text | String | | | 0..1 | A human-readable description of the anatomical site.
ServiceRequest (R4).category | CodeableConcept | | Yes | 1..* | Categories that apply to this ServiceRequest.
ServiceRequest (R4).category.coding | Coding | | Yes | 0..1 | The coded representation of the category.
ServiceRequest (R4).category.coding.code | String | | | 1..1 | Always the SNOMED code for Radiotherapy (procedure): 1287742003.
ServiceRequest (R4).category.coding.display | String | | | 0..1 | The textual representation of the coded value. Only allowed value: "Radiotherapy (procedure)".
ServiceRequest (R4).category.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).category.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).code | CodeableConcept | | | 1..1 | The coded representation of the ServiceRequest.
ServiceRequest (R4).code.coding | Coding | | Yes | 1..1 | The coded representation of the concept.
ServiceRequest (R4).code.coding.code | String | | | 1..1 | Possible code values: 1217123003 - Radiotherapy course of treatment (regime/therapy) 1222565005 - Radiotherapy treatment phase (regime/therapy) 1255724003 - Radiotherapy treatment plan (regime/therapy)
ServiceRequest (R4).code.coding.display | String | | | 1..1 | The textual representation of the coded value. For course types, the value must be "Radiotherapy course of treatment (regime/therapy)". For phase types, the value must be "Radiotherapy treatment phase (regime/therapy)". For plan types, the value must be "Radiotherapy treatment plan (regime/therapy)".
ServiceRequest (R4).code.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).code.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).extension (courseInvolvesReirradiation) (also: extension (courseInvolvesReirradiation)) | Extension | | Yes | 0..1 | Indicates whether or not this course of radiotherapy is delivering radiation to a target in this patient that was irradiated before.
ServiceRequest (R4).extension (courseInvolvesReirradiation).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-involves-reirradiation
ServiceRequest (R4).extension (courseInvolvesReirradiation).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).
ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique) (also: extension (freeBreathingMotionManagementTechnique)) | Extension | | Yes | | For radiation delivered with free-breathing motion management, defines what free-breathing method is used.
ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-free-breathing-motion-mgmt-technique
ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept | CodeableConcept | | | 1..1 | The CodeableConcept container for the free-breathing method.
ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the free-breathing method.
ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-free-breathing-motion-mgmt-technique-vs
ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.display | String | | | 0..1 | The textual representation of the coded value in the given system.
ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).extension (freeBreathingMotionManagementTechnique).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).extension (imageGuidedRadiotherapyModality) (also: extension (imageGuidedRadiotherapyModality)) | Extension | | Yes | 0..* | For radiation delivered with image guidance, defines what mode of image guidance is used.
ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-image-guided-radiotherapy-modality
ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the image guidance.
ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the imaging guidance.
ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/STU2/ValueSet-codexrt-image-guided-radiotherapy-modality-vs.
ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.
ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).extension (imageGuidedRadiotherapyModality).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).extension (intrafractionVerification) (also: extension (intrafractionVerification)) | Extension | | Yes | 0..* | For radiation regimens that include verification imaging during treatment (intrafraction), defines the method of verification.
ServiceRequest (R4).extension (intrafractionVerification).extension (procedure) (also: extension (procedure)) | Extension | | Yes | 1..1 | Defines the imaging procedure used for intrafraction verification.
ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).url | uri | | | 1..1 | Must be: procedure
ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the procedure.
ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the procedure.
ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-intrafraction-verification-procedure-vs
ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.
ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct
ServiceRequest (R4).extension (intrafractionVerification).extension (procedure).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier) (also: extension (qualifier)) | Extension | | Yes | 0..1 | Further describes the imaging used for intrafraction verification.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).url | uri | | | 1..1 | Must be: qualifier

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the qualifier.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding | Coding | | Yes | 1..* | The coded representation of the qualifier.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-intrafraction-verification-qualifier-vs.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (intrafractionVerification).extension (qualifier).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (intrafractionVerification).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification

ServiceRequest (R4).extension (modalityAndTechnique) (also: extension (modalityAndTechnique)) | Extension | | Yes | 0..* | Describes the source and method of irradiation.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality) (also: extension (modality)) | Extension | | Yes | 1..1 | Specifies the modality of the radiation used.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the modality.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the modality.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-brachytherapy-applicator-type-vs

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (modalityAndTechnique).extension (modality).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope) (also: extension (radiotherapyEnergyOrIsotope)) | Extension | | Yes | 0..* | Further specifies the radiation, giving either the beam energy (for external-beam modalities) or source isotope (for brachytherapy modalities).

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-isotope

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept | CodeableConcept | | | 0..1 | If specifying an isotope, a CodeableConcept representation of the isotope as a SNOMED code.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the isotope.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-isotope-vs.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity | Quantity | | | 0..1 | If specifying a field energy, a Quantity representing the energy in kV, MV, or MeV.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.code | String | | | 1..1 | Must be from the value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-energy-unit-vs.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyEnergyOrIsotope).valueQuantity.value | Decimal | | | 1..1 | Value of the energy.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType) (also: extension (radiotherapyTreatmentApplicatorType)) | Extension | | Yes | 0..1 | Defines what type of applicator is used for a brachytherapy source.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-applicator-type

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the type of applicator.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the type of applicator.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-brachytherapy-applicator-type-vs.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.coding.system | String | | | 1..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (modalityAndTechnique).extension (radiotherapyTreatmentApplicatorType).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique) (also: extension (technique)) | Extension | | Yes | 0..1 | Specifies the technique of the radiation used, which is more specific than the modality.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the technique.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding | Coding | | Yes | 1..1 | Coded representation of the technique.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-radiotherapy-technique-vs.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.display | String | | | 1..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.coding.system | String | | | 1..1 | Must be http://snomed.info/sct

ServiceRequest (R4).extension (modalityAndTechnique).extension (technique).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (modalityAndTechnique).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume) (also: extension (radiotherapyDosePlannedToVolume)) | Extension | | Yes | 0..* | Specifies the plan of doses to be delivered to one or more volumes as part of this regimen.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | Specifies the fractional dose prescribed for this volume.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).url | uri | | | 1..1 | Must be: fractionDose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity | Quantity | | | 1..1 | Quantity of the dose.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity.code | String | | | 1..1 | Must be: cGy

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractionDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | Specifies how many fractions are prescribed for this volume in the course.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractions).url | uri | | | 1..1 | Must be: fractions

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (fractions).valuePositiveInt | positiveInt | | | 1..1 | The number of fractions prescribed.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (pointDose) (also: extension (pointDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is a point dose, meaning that it is used for dose calculation and might not need to be displayed in summaries of the regimen.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (pointDose).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-point-dose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (pointDose).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (primaryPlanDose) (also: extension (primaryPlanDose)) | Extension | | Yes | 0..1 | Indicates whether or not this dose-to-volume is the "primary" dose for a given plan or phase, meaning that its dose can be treated loosely as "the" dose for that phase or plan.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (primaryPlanDose).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-primary-plan-dose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (primaryPlanDose).valueBoolean | boolean | | | 1..1 | Boolean value (true/false).

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric) (also: extension (radiobiologicMetric)) | Extension | | Yes | 0..* | Provides a more tightly specified metric for the given dose.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric).valueQuantity | Quantity | | | 1..1 | The quantity of the radiobiologic metric.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric).valueQuantity.unit | String | | | 1..1 | Must be a value as described in http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (radiobiologicMetric).valueQuantity.value | Decimal | | | 1..1 | The number representing the dose according to the given radiobiologic metric.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | Specifies the total dose prescribed for this volume.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).url | uri | | | 1..1 | Must be: totalDose

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity | Quantity | | | 1..1 | Quantity of the total dose.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity.code | String | | | 1..1 | Must be: cGy

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (totalDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | Identifies a volume to be treated with radiation.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (volume).url | uri | | | 1..1 | Must be: volume

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).extension (volume).valueReference | Reference | | | 1..1 | A reference to the radiotherapy volume (BodyStructure) whose dose is being described.

ServiceRequest (R4).extension (radiotherapyDosePlannedToVolume).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-planned-to-volume

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume) (also: extension (radiotherapyDosePrescribedToVolume)) | Extension | | Yes | 0..* | Specifies the prescription of doses to be delivered to one or more volumes as part of this regimen.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose) (also: extension (fractionDose)) | Extension | | Yes | 0..1 | Specifies the fractional dose prescribed for this volume.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).url | uri | | | 1..1 | Must be: fractionDose

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity | Quantity | | | 1..1 | Quantity of the dose.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity.code | String | | | 1..1 | Must be: cGy

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractionDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractions) (also: extension (fractions)) | Extension | | Yes | 0..1 | Specifies how many fractions are prescribed for this volume in the course.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractions).url | uri | | | 1..1 | Must be: fractions

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (fractions).valuePositiveInt | positiveInt | | | 1..1 | The number of fractions prescribed.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose) (also: extension (totalDose)) | Extension | | Yes | 0..1 | Specifies the total dose prescribed for this volume.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).url | uri | | | 1..1 | Must be: totalDose

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity | Quantity | | | 1..1 | Quantity of the total dose.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity.code | String | | | 1..1 | Must be: cGy

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity.system | String | | | 1..1 | Must be: http://unitsofmeasure.org

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity.unit | String | | | 0..1 | Must be: cGy

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (totalDose).valueQuantity.value | Decimal | | | 1..1 | The number for the dose.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (volume) (also: extension (volume)) | Extension | | Yes | 1..1 | Identifies a volume to be treated with radiation.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (volume).url | uri | | | 1..1 | Must be: volume

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).extension (volume).valueReference | Reference | | | 1..1 | A reference to the radiotherapy volume (BodyStructure) whose dose is being described.

ServiceRequest (R4).extension (radiotherapyDosePrescribedToVolume).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-to-volume

ServiceRequest (R4).extension (respiratoryMotionManagement) (also: extension (respiratoryMotionManagement)) | Extension | | Yes | | For radiation that requires motion management for respiration, defines what method is used.

ServiceRequest (R4).extension (respiratoryMotionManagement).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-respiratory-motion-management

ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept | CodeableConcept | | | 1..1 | The CodeableConcept container for the motion management.

ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding | Coding | | Yes | 1..* | Coded representation of the motion management.

ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/codex-radiation-therapy/ValueSet/codexrt-radiotherapy-respiratory-motion-management-vs

ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.coding.system | String | | | 0..1 | Must be: http://snomed.info/sct

ServiceRequest (R4).extension (respiratoryMotionManagement).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).extension (treatmentIntent) (also: extension (treatmentIntent)) | Extension | | Yes | 0..1 | Indicates the clinical intent of this radiotherapy regimen.

ServiceRequest (R4).extension (treatmentIntent).url | uri | | | 1..1 | Must be: http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent

ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept | CodeableConcept | | | 1..1 | CodeableConcept container for the treatment intent.

ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.coding | Coding | | Yes | 0..1 | Coded representation of the treatment intent.

ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.coding.code | String | | | 1..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-procedure-intent-vs.

ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.coding.display | String | | | 0..1 | The texual repsentation of the coded value in the given system.

ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.coding.system | String | | | 1..1 | Must be http://snomed.info/sct

ServiceRequest (R4).extension (treatmentIntent).valueCodeableConcept.text | String | | | 0..1 | An optional human-readable description of the coded value.

ServiceRequest (R4).id | id | | | 1..1 | The FHIR ID of the ServiceRequest being updated.

ServiceRequest (R4).identifier | Identifier | | Yes | 2..* | Identifiers for this radiotherapy course, phase, or plan.

ServiceRequest (R4).identifier.system | String | | | 1..1 | The identification system or namespace from which the identifier is assigned.
ServiceRequest (R4).identifier.use | String | | | 1..1 | Either usual or official. Identifiers where use = usual are a human-readable display name for the course/phase/plan and may not be unique. Identifiers where use = official are a unique business identifier controlled by the source system (e.g., a DICOM UID or GUID).
ServiceRequest (R4).identifier.value | String | | | 1..1 | The value of the identifier.
ServiceRequest (R4).intent | code | | | 1..1 | If the course/phase/plan is prescribed, then this value is original-order. If it is planned, then the value is filler-order.
ServiceRequest (R4).locationReference | Reference | | Yes | 0..* | Reference to the Location FHIR resource where radiotherapy has been requested.
ServiceRequest (R4).meta | Meta
ServiceRequest (R4).meta.lastUpdated | String | | | 1..1 | The date and time at which this version of the resource took effect (e.g., the time when the most recent updates were made by a user).
ServiceRequest (R4).meta.profile | String | | Yes | 1..* | Describes the profile for this ServiceRequest resource. The following profiles are supported. Profile Name Profile Value Code Intent Radiotherapy Course Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription 1217123003 original-order Radiotherapy Planned Course http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course 1217123003 filler-order Radiotherapy Phase Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription 1222565005 original-order Radiotherapy Planned Phase http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase 1222565005 filler-order Radiotherapy Plan Prescription http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription 1255724003 original-order Radiotherapy Treatment Plan http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan 1255724003 filler-order
Profile Name | Profile Value | Code | Intent
Radiotherapy Course Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription | 1217123003 | original-order
Radiotherapy Planned Course | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-course | 1217123003 | filler-order
Radiotherapy Phase Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-phase-prescription | 1222565005 | original-order
Radiotherapy Planned Phase | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-planned-phase | 1222565005 | filler-order
Radiotherapy Plan Prescription | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-plan-prescription | 1255724003 | original-order
Radiotherapy Treatment Plan | http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-treatment-plan | 1255724003 | filler-order
ServiceRequest (R4).occurrencePeriod | Period | | | 0..1 | The period during which this course/phase/plan is active.
ServiceRequest (R4).occurrencePeriod.end | String | | | 0..1 | The end of the period.
ServiceRequest (R4).occurrencePeriod.start | String | | | 0..1 | The start of the period.
ServiceRequest (R4).reasonCode | CodeableConcept | | Yes | 0..* | The cancer problem to be treated by radiotherapy.
ServiceRequest (R4).reasonCode.coding | Coding | | Yes | 0..1 | Coded representation of the problem.
ServiceRequest (R4).reasonCode.coding.code | String | | | 0..1 | The coded value for the concept in the given system. The value should be from the extensible value set http://hl7.org/fhir/us/mcode/ValueSet/mcode-cancer-disorder-vs.
ServiceRequest (R4).reasonCode.coding.display | String | | | 0..1 | The textual representation of the coded value.
ServiceRequest (R4).reasonCode.coding.system | String | | | 0..1 | The URI for the system in which this coded value is assigned. Preferred values: http://snomed.info/sct http://hl7.org/fhir/sid/icd-10-cm
ServiceRequest (R4).reasonCode.text | String | | | 0..1 | An optional human-readable description of the coded value.
ServiceRequest (R4).reasonReference | Reference | | Yes | 0..* | A reference to a cancer problem Condition resource.
ServiceRequest (R4).replaces | Reference | | Yes | 0..* | Reference to a ServiceRequest resource that represents the previous version of this course/phase/plan that is replaced by this one.
ServiceRequest (R4).requester | Reference | | | 0..1 | The person or entity responsible for the request.
ServiceRequest (R4).status | code | | | 1..1 | The status of the course/phase/plan. The value should come form the value set http://hl7.org/fhir/R4/ValueSet/request-status
ServiceRequest (R4).subject | Reference | | | 1..1 | Reference to the Patient FHIR resource that this radiotherapy course/plan/phase is planned or prescribed to.

Sample request

```
PUT https://hostname/instance/api/FHIR/R4/ServiceRequest/fXG74SEeaQkfhgC534YkPkQ4
{
  "resourceType": "ServiceRequest",
  "id": "fXG74SEeaQkfhgC534YkPkQ4",
  "meta": {
    "versionId": "1",
    "lastUpdated": "2026-02-03T21:15:01Z",
    "profile": [
      "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-course-prescription"
    ]
  },
  "extension": [
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "168537006",
                "display": "Plain X-ray"
              }
            ]
```

```
      },
      "url": "procedure"
    },
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "255204007",
            "display": "Multiple"
          }
        ]
      },
      "url": "qualifier"
    }
  ],
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification"
},
{
  "extension": [
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/v3/NullFlavor",
            "code": "UNC"
          }
        ],
        "text": "IFV procedure"
      },
      "url": "procedure"
    }
  ],
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification"
},
{
  "extension": [
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "168537006",
            "display": "Plain X-ray"
          }
        ]
      },
      "url": "procedure"
    },
    {
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/v3/NullFlavor",
            "code": "UNC"
          }
        ],
        "text": "IFV X-ray qualifier"
      },
      "url": "qualifier"
    }
  ],
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-intrafraction-verification"
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
      },
      "url": "volume"
    },
    {
      "valuePositiveInt": 16,
      "url": "fractions"
    },
    {
      "valueQuantity": {
        "value": 4256,
        "unit": "cGy",
        "system": "http://unitsofmeasure.org",
        "code": "cGy"
      },
      "url": "totalDose"
    },
    {
      "valueQuantity": {
        "value": 64,
        "unit": "EQD2Gy(a/b=2.5)",
```

```
        "system": "http://unitsofmeasure.org"
      },
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric"
    }
  ],
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-
to-volume"
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
      },
      "url": "volume"
    },
    {
      "valuePositiveInt": 16,
      "url": "fractions"
    },
    {
      "valueQuantity": {
        "value": 4256,
        "unit": "cGy",
        "system": "http://unitsofmeasure.org",
        "code": "cGy"
      },
      "url": "totalDose"
    },
    {
      "valueQuantity": {
        "value": 64,
        "unit": "EQD2Gy(a/b=2.5)",
        "system": "http://unitsofmeasure.org"
      },
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric"
    }
  ],
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-
to-volume"
},
{
  "extension": [
    {
      "valueReference": {
        "reference": "BodyStructure/fqYNbuWQOsf.f9Duiycny2g4"
      },
      "url": "volume"
    },
    {
      "valuePositiveInt": 16,
      "url": "fractions"
    },
    {
      "valueQuantity": {
        "value": 4256,
        "unit": "cGy",
        "system": "http://unitsofmeasure.org",
        "code": "cGy"
      },
      "url": "totalDose"
    },
    {
      "valueQuantity": {
        "value": 64,
        "unit": "EQD2Gy(a/b=2.5)",
        "system": "http://unitsofmeasure.org"
      },
      "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiobiologic-metric"
    }
  ],
  "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-dose-prescribed-
to-volume"
},
{
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "129428001",
        "display": "Preventive - procedure intent"
      }
    ]
  },
  "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-procedure-intent"
},
{
  "extension": [
    {
```

```
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "1156506007",
              "display": "External beam radiation therapy using photons"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "1162782007",
              "display": "Three dimensional external beam radiation therapy"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique"
      },
      {
        "valueQuantity": {
          "value": 18,
          "unit": "MeV",
          "system": "http://unitsofmeasure.org",
          "code": "MeV"
        },
        "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-
  isotope"
      }
    ],
    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique"
  },
  {
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "45643008",
              "display": "External beam radiation therapy using electrons"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "1162782007",
              "display": "Three dimensional external beam radiation therapy"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-technique"
      },
      {
        "valueQuantity": {
          "value": 18,
          "unit": "MeV",
          "system": "http://unitsofmeasure.org",
          "code": "MeV"
        },
        "url": "http://hl7.org/fhir/us/codex-radiation-therapy/StructureDefinition/codexrt-radiotherapy-energy-or-
  isotope"
      }
    ],
    "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-radiotherapy-modality-and-technique"
  }
],
"identifier": [
  {
    "use": "usual",
    "system": "urn:hand:display",
    "value": "Bilateral Breast"
  },
  {
    "use": "official",
    "system": "urn:dicom:uid",
    "value": "38dc290f-f773-4489-9a46-b0c2075e89d2"
  }
],
"status": "active",
```

```
"intent": "original-order",
"category": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "108290001",
        "display": "Radiation Oncology And/Or Radiotherapy"
      },
      {
        "system": "http://snomed.info/sct",
        "code": "1287742003",
        "display": "Radiotherapy"
      }
    ]
  }
],
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "1217123003",
      "display": "Radiotherapy course of treatment"
    }
  ]
},
"subject": {
  "reference": "Patient/eP-BmFG0190obAH1wt4wjUw3",
  "display": "Test, Patient"
},
"occurrencePeriod": {
  "start": "2026-02-01T15:09:51Z"
},
"authoredOn": "2026-02-01T15:09:51Z",
"requester": {
  "reference": "PractitionerRole/fIidEgs9aUIFFBCsj1TnOXmmbOuEnhxY8XHSG-tyNFpo4"
},
"locationReference": [
  {
    "reference": "Location/e-iz4EXUrNFfCKJwdsZPd9Q3"
  }
],
"reasonCode": [
  {
    "coding": [
      {
        "system": "http://hl7.org/fhir/sid/icd-10-cm",
        "code": "C50",
        "display": "Malignant neoplasm of breast"
      }
    ],
    "text": "Cancer Problem"
  }
],
"reasonReference": [
  {
    "reference": "Condition/evA9Cs4FuqWRemAD9B321RvjhDZCOh4cL0mszsMdP89s3"
  }
],
"bodySite": [
  {
    "extension": [
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "7771000",
              "display": "Left"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier"
      },
      {
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "399110001",
              "display": "Tangential projection"
            }
          ]
        },
        "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier"
      }
    ],
    "coding": [
      {
```

```
          "system": "http://snomed.info/sct",
          "code": "76752008",
          "display": "Breast structure"
        }
      ]
    },
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "24028007",
                "display": "Right"
              }
            ]
          },
          "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier"
        },
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "399110001",
                "display": "Tangential projection"
              }
            ]
          },
          "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-body-location-qualifier"
        }
      ],
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "76752008",
          "display": "Breast structure"
        }
      ]
    },
    {
      "extension": [
        {
          "valueCodeableConcept": {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "7771000",
                "display": "Left"
              }
            ]
          },
          "url": "http://hl7.org/fhir/us/mcode/StructureDefinition/mcode-laterality-qualifier"
        }
      ],
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "68171009",
          "display": "Axillary lymph node structure"
        }
      ]
    }
  ]
}
```


Sample response

```
Status: OK (200)
Content-Type: application/json; charset=utf-8
```

13.2 Clinical Specialty: STU3

File: site/apis/13-Clinical-Specialty/2-STU3/index.html

* NutritionOrder.Read (STU3)
* NutritionOrder.Search (STU3)
* Observation.Read (Family Situation) (STU3)
* Observation.Search (Family Situation) (STU3)

13.2.1 NutritionOrder.Read (STU3)

File: site/apis/13-Clinical-Specialty/2-STU3/10228-NutritionOrder-Read-STU3.html
Epic API ID | 10228

FHIR version | STU3
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/NutritionOrder/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10228

Description

This resource can read various types of nutrition orders, including oral diets, oral nutrition supplements, enteral nutrition (tube feedings), and infant formula. This resource can also include information from diet orders such as a patient's food allergies, intolerances, and personal/cultural requirements or preferences (such as being kosher or vegan), and includes data on texture and quantity when applicable.

Read interactions typically begin with a client having previously established a relationship, often through querying for NutritionOrder through the search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for a NutritionOrder resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
NutritionOrder | NutritionOrder | False | | | An instance of the STU3 NutritionOrder resource.
NutritionOrder.extension (also: comment) | Extension | True | Yes | | An explanation of dietary recommendations. This element is returned only for organizations in the Netherlands.
NutritionOrder.dateTime | String | False | | | The date and time of an order.
NutritionOrder.id | String | False | | | Identifier.
NutritionOrder.meta | Meta | True | | | The document metadata.
NutritionOrder.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-NutritionAdvice
NutritionOrder.oralDiet | OralDiet | True | | | The details of a diet order.
NutritionOrder.oralDiet.instruction | String | True | | | The name of a diet.
NutritionOrder.oralDiet.texture | Texture | True | Yes | | Texture information.
NutritionOrder.oralDiet.texture.modifier | CodeableConcept | True | | | Describes the texture of food as part of a diet.
NutritionOrder.oralDiet.type | CodeableConcept | True | Yes | | The diet type.
NutritionOrder.patient | Reference | False | | | The patient whom the nutrition orders are related to.
NutritionOrder.status | String | True | | | The status of a nutrition order. Possible statuses include: draft active completed cancelled unknown entered-in-error

Sample request

```
https://hostname/instance/api/FHIR/STU3/NutritionOrder/emV2M.3TKuBneTxDOPf2VYUgryLaXH6Qus6BUOZ8Tg9k3
```

Sample response

```
<NutritionOrder xmlns="http://hl7.org/fhir">
  <id value="emV2M.3TKuBneTxDOPf2VYb5x3vlG-nAgKYCt9fijpOk3"/>
  <status value="active"/>
  <patient>
    <reference value="eFLQ4DCYc6n424WRext9mTQ3"/>
    <display value="Patient, Epic"/>
  </patient>
  <dateTime value="2021-07-01T19:42:38Z"/>
  <oralDiet>
    <type>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.34365"/>
        <code value="3"/>
        <display value="Vegetarian"/>
      </coding>
      <text value="Vegetarian"/>
    </type>
    <texture>
      <modifier>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.34442"/>
          <code value="15"/>
          <display value="Solid"/>
        </coding>
        <text value="Solid"/>
      </modifier>
    </texture>
```

```
    <instruction value="GENERAL DIET PLAN"/>
  </oralDiet>
</NutritionOrder>
```

13.2.2 NutritionOrder.Search (STU3)

File: site/apis/13-Clinical-Specialty/2-STU3/10229-NutritionOrder-Search-STU3.html
Epic API ID | 10229
FHIR version | STU3
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/NutritionOrder?patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10229

Description

This resource can read various types of nutrition orders, including oral diets, oral nutrition supplements, enteral nutrition (tube feedings), and infant formula. This resource can also include information from a patient's diet order such as a patient's food allergies, intolerances, and personal/cultural requirements or preferences (such as being kosher or vegan), and includes data on texture and quantity when applicable.

You can use the search interaction to query for nutrition orders based on a patient.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
patient | String | False | | | FHIR identifier for the patient whose nutrition order details are being requested.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.NutritionOrder | NutritionOrder | False | | | An instance of the STU3 NutritionOrder resource.
Bundle.entry.resource.NutritionOrder.extension (also: comment) | Extension | True | Yes | | An explanation of dietary recommendations. This element is returned only for organizations in the Netherlands.
Bundle.entry.resource.NutritionOrder.dateTime | String | False | | | The date and time of an order.
Bundle.entry.resource.NutritionOrder.id | String | False | | | Identifier.
Bundle.entry.resource.NutritionOrder.meta | Meta | True | | | The document metadata.
Bundle.entry.resource.NutritionOrder.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-NutritionAdvice
Bundle.entry.resource.NutritionOrder.oralDiet | OralDiet | True | | | The details of a diet order.
Bundle.entry.resource.NutritionOrder.oralDiet.instruction | String | True | | | The name of a diet.
Bundle.entry.resource.NutritionOrder.oralDiet.texture | Texture | True | Yes | | Texture information.
Bundle.entry.resource.NutritionOrder.oralDiet.texture.modifier | CodeableConcept | True | | | Describes the texture of food as part of a diet.
Bundle.entry.resource.NutritionOrder.oralDiet.type | CodeableConcept | True | Yes | | The diet type.
Bundle.entry.resource.NutritionOrder.patient | Reference | False | | | The patient whom the nutrition orders are related to.
Bundle.entry.resource.NutritionOrder.status | String | False | | | The status of a nutrition order. Possible statuses include: draft active completed cancelled unknown entered-in-error

Sample request

```
https://hostname/instance/api/FHIR/STU3/NutritionOrder?patient=eFLQ4DCYc6n424WRext9mTQ3
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="1"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/STU3/NutritionOrder?patient=eFLQ4DCYc6n424WRext9mTQ3"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url
    value="https://hostname/instance/api/FHIR/STU3/NutritionOrder/emV2M.3TKuBneTxDOPf2VYUgryLaXH6Qus6BUOZ8Tg9k3"/>
    </link>
    <fullUrl
    value="https://hostname/instance/api/FHIR/STU3/NutritionOrder/emV2M.3TKuBneTxDOPf2VYUgryLaXH6Qus6BUOZ8Tg9k3"/>
```

```
    <resource>
      <NutritionOrder>
        <id value="emV2M.3TKuBneTxDOPf2VYUgryLaXH6Qus6BUOZ8Tg9k3"/>
        <status value="active"/>
        <patient>
          <reference value="eFLQ4DCYc6n424WRext9mTQ3"/>
          <display value="Patient, Epic"/>
        </patient>
        <dateTime value="2021-07-01T19:42:38Z"/>
        <oralDiet>
          <type>
            <coding>
              <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.34365"/>
              <code value="3"/>
              <display value="Vegetarian"/>
            </coding>
            <text value="Vegetarian"/>
          </type>
          <texture>
            <modifier>
              <coding>
                <system value="urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.34442"/>
                <code value="8"/>
                <display value="Pureed"/>
              </coding>
              <text value="Pureed"/>
            </modifier>
          </texture>
          <instruction value="GENERAL DIET PLAN"/>
        </oralDiet>
      </NutritionOrder>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

13.2.3 Observation.Read (Family Situation) (STU3)

File: site/apis/13-Clinical-Specialty/2-STU3/10444-Observation-Read-Family-Situation-STU3.html
Epic API ID | 10444
FHIR version | STU3
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10444

Description

This resource can be used with healthcare organizations in the Netherlands only. This resource cannot be used in any other locales at this time because it has been designed to meet regulatory requirements in the Netherlands.

The FHIR Observation Family Situation API searches for patient observations which include all forms of cohabitation that form a recognizable social unit, of people who are or who are not related and care.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | False | Yes | | This is always "social-history".
Observation.code | CodeableConcept | False | | | Hard coded to SNOMED code 36547003
Observation.comment | string | True | | | Comments on family situation.
Observation.component | BackboneElement | True | Yes | | Can return the following when reported data is available: Care responsibility: The activities the patient carries out to care for a dependent family member, composed of a valueString (SNOMED 406192005). Child: Record of a patient’s child, including their birth date expressed as a valueDateTime. Only appears if a child’s birth was documented with Stork (SNOMED 67822003). Family composition: Make up of a patient’s household, which can include non-family members (SNOMED 224130005). Mapped to the values in GezinssamenstellingCodelijst. Number of children: The number of children a patient has, expressed as a valueQuantity (SNOMED 224118004). Number of children living at home: The number of children a patient has that live with them, expressed as a valueQuantity (SNOMED 55811000146107).

Observation.dataAbsentReason | CodeableConcept | True | | | Always “unknown,” appears only if no data is retrieved.
Observation.status | code | False | | | This value is always "final".
Observation.subject | Reference | False | | | The patient FHIR ID.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/fH6C48xiz47Lxa9k1rwsFPA4
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "fH6C48xiz47Lxa9k1rwsFPA4",
  "meta": {
    "profile": [
      "http://nictiz.nl/fhir/StructureDefinition/zib-FamilySituation"
    ]
  },
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/observation-category",
          "code": "social-history",
          "display": "Social History"
        }
      ],
      "text": "Social History"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "365470003",
        "display": "Finding of family details and household composition (finding)"
      }
    ],
    "text": "Finding of family details and household composition (finding)"
  },
  "subject": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eEvB0tF1AgkK4A2cbxByNpQ3",
    "display": "Mayer, M."
  },
  "comment": "Supportive family environment, no concerns\n",
  "component": [
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "224130005",
            "display": "Household composition (observable entity)"
          }
        ],
        "text": "Household composition (observable entity)"
      },
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "408821002",
            "display": "Lives with partner"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "8871000146101",
            "display": "Lives with underage children"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "224137008",
            "display": "Lives with parents"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "160756002",
            "display": "Lives with relatives"
          }
        ],
        "text": "spouse, children, mother, other family member"
      }
    },
    {
      "code": {
        "coding": [
          {
```

```
            "system": "http://snomed.info/sct",
            "code": "224118004",
            "display": "Number of offspring (observable entity)"
          }
        ],
        "text": "Number of offspring (observable entity)"
      },
      "valueQuantity": {
        "value": 2
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "55811000146107",
            "display": "Number of children living at home (observable entity)"
          }
        ],
        "text": "Number of children living at home (observable entity)"
      },
      "valueQuantity": {
        "value": 2
      }
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "406192005",
            "display": "Caregiver behavior (observable entity)"
          }
        ],
        "text": "Caregiver behavior (observable entity)"
      },
      "valueString": "Ensures mother has transportation to appointments"
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "67822003",
            "display": "Child (person)"
          }
        ],
        "text": "Child (person)"
      },
      "valueDateTime": "2016-06-07"
    },
    {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "67822003",
            "display": "Child (person)"
          }
        ],
        "text": "Child (person)"
      },
      "valueDateTime": "2019-06"
    }
  ]
}
```


13.2.4 Observation.Search (Family Situation) (STU3)

File: site/apis/13-Clinical-Specialty/2-STU3/10457-Observation-Search-Family-Situation-STU3.html
Epic API ID | 10457
FHIR version | STU3
Categories | Clinical Specialty
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Observation?category={category}&code={code}&issued={issued}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10457

Description

This resource can be used with healthcare organizations in the Netherlands only. This resource cannot be used in any other locales at this time because it has been designed to meet regulatory requirements in the Netherlands.

The FHIR Observation Family Situation API searches for patient observations which include all forms of cohabitation that form a recognizable social unit; of people who are or who are not related and who have long-lasting, affective bonds and provide each other with support and care.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | True | | | Use "social-history" to search for family situation observations.
code | token | True | | | SNOMED code 365470003 for family situation observation.
issued | date | True | | | The date ranges the observation is obtained.
patient | reference | Either this or subject must be provided. | | | Reference to a patient resource.
subject | reference | Either this or the patient must be provided. | | | Reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | This is always "social-history".
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | Hard coded to SNOMED code 36547003
Bundle.entry.resource.Observation.comment | string | True | | | Comments on family situation.
Bundle.entry.resource.Observation.component | BackboneElement | True | Yes | | Can return the following when reported data is available: Care responsibility: The activities the patient carries out to care for a dependent family member, composed of a valueString (SNOMED 406192005). Child: Record of a patient's child, including their birth date expressed as a valueDateTime. Only appears if a child's birth was documented with Stork (SNOMED 67822003). Family composition: Make up of a patient's household, which can include non-family members (SNOMED 224130005). Mapped to the values in GezinssamenstellingCodelijst. Number of children: The number of children a patient has, expressed as a valueQuantity (SNOMED 224118004). Number of children living at home: The number of children a patient has that live with them, expressed as a valueQuantity (SNOMED 55811000146107).
Bundle.entry.resource.Observation.dataAbsentReason | CodeableConcept | True | | | Always "unknown," appears only if no data is retrieved.
Bundle.entry.resource.Observation.status | code | False | | | This value is always "final".
Bundle.entry.resource.Observation.subject | Reference | False | | | The patient FHIR ID.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation?code=http://snomed.info/sct|365470003&patient=eEvB0tF1AgkK4A2cbxByNp
    Q3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Observation?code=http://snomed.info/sct%7C365470003&patient=eEvB0t
    F1AgkK4A2cbxByNpQ3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance /api/FHIR/STU3/Observation/fH6C48xiz47Lxa9k1rwsFPA4"
        }
      ],
      "fullUrl": "https://hostname/instance /api/FHIR/STU3/Observation/fH6C48xiz47Lxa9k1rwsFPA4",
      "resource": {
        "resourceType": "Observation",
        "id": "fH6C48xiz47Lxa9k1rwsFPA4",
        "meta": {
          "profile": [
            "http://nictiz.nl/fhir/StructureDefinition/zib-FamilySituation"
          ]
        },
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/observation-category",
                "code": "social-history",
                "display": "Social History"
```

```
      }
    ],
    "text": "Social History"
  }
],
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "365470003",
      "display": "Finding of family details and household composition (finding)"
    }
  ],
  "text": "Finding of family details and household composition (finding)"
},
"subject": {
  "reference": "https://hostname/instance /api/FHIR/STU3/Patient/eEvB0tF1AgkK4A2cbxByNpQ3",
  "display": "Mayer, M."
},
"comment": "Supportive family environment, no concerns\n",
"component": [
  {
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "224130005",
          "display": "Household composition (observable entity)"
        }
      ],
      "text": "Household composition (observable entity)"
    },
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "408821002",
          "display": "Lives with partner"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "8871000146101",
          "display": "Lives with underage children"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "224137008",
          "display": "Lives with parents"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "160756002",
          "display": "Lives with relatives"
        }
      ],
      "text": "spouse, children, mother, other family member"
    }
  },
  {
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "224118004",
          "display": "Number of offspring (observable entity)"
        }
      ],
      "text": "Number of offspring (observable entity)"
    },
    "valueQuantity": {
      "value": 2
    }
  },
  {
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "55811000146107",
          "display": "Number of children living at home (observable entity)"
        }
      ],
      "text": "Number of children living at home (observable entity)"
    },
    "valueQuantity": {
      "value": 2
    }
  },
  {
```

```
            "code": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "406192005",
                  "display": "Caregiver behavior (observable entity)"
                }
              ],
              "text": "Caregiver behavior (observable entity)"
            },
            "valueString": "Ensures mother has transportation to appointments"
          },
          {
            "code": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "67822003",
                  "display": "Child (person)"
                }
              ],
              "text": "Child (person)"
            },
            "valueDateTime": "2016-06-07"
          },
          {
            "code": {
              "coding": [
                {
                  "system": "http://snomed.info/sct",
                  "code": "67822003",
                  "display": "Child (person)"
                }
              ],
              "text": "Child (person)"
            },
            "valueDateTime": "2019-06"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


13.3 Clinical Specialty: Non-FHIR

File: site/apis/13-Clinical-Specialty/3-Non-FHIR/index.html

* GetDepartmentWaitTimes

* GetWaitTimes

13.3.1 GetDepartmentWaitTimes

File: site/apis/13-Clinical-Specialty/3-Non-FHIR/204-GetDepartmentWaitTimes.html
Epic API ID | 204
FHIR version | Non-FHIR
Categories | Clinical Specialty, User Workflow Scheduling
API type | Epic Public
API group | Wait Times Group
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | POST
URL template | api/epic/2014/Scheduling/Utility/GETDEPARTMENTWAITTIMES/GetDepartmentWaitTimes
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
SOAP namespace | urn:Epic-com:Scheduling.2014.Services.Utility
WSDL | Epic.Scheduling.GeneratedServices/Utility.svc?wsdl
Source | https://fhir.epic.com/Specifications?api=204

Remarks

If none of the following optional parameters are set, the extension record set in the Wait Time Extension field in the department record determines the settings for this web service:

* MinutesToLookBack

* StartEvent

* EndEvent

* UseAppointmentTimeAfterStartEvent

If any of these parameters is set, this department-level setting has no effect.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
CalculateForEachDepartment | Boolean | True
CalculateForOverallLevel | Boolean | True
Departments | IDType | False | Yes | | A generic identifier.
EndEvent | IDType | True | | | A generic identifier.
EndEvent.ID | String | False | | | The identifier for this resource.
EndEvent.Type | String | True | | | The type of identifier.
MinutesToLookBack | Int32 | True
StartEvent | IDType | True | | | A generic identifier.
UseAppointmentTimeIfAfterStartEvent | Boolean | True

Response elements

Name | Type | Optional | Array | Cardinality | Description
AverageWaitTime | String | True
DepartmentWaitTimes | DepartmentWaitTimes | True | Yes
DepartmentWaitTimes.AverageWaitTime | String | True
DepartmentWaitTimes.DepartmentIDs | IDType | False | Yes | | A generic identifier.
DepartmentWaitTimes.DepartmentName | String | False
DepartmentWaitTimes.LongestCurrentWaitTime | String | True
DepartmentWaitTimes.PatientsStillWaiting | Int32 | True
DepartmentWaitTimes.TotalPatientsForAverage | String | True
LongestCurrentWaitTime | Int32 | True
PatientsStillWaiting | Int32 | True
TotalPatientsForAverage | Int32 | True

Errors

Code | Severity | Description | Example
NOT-LICENSED | None | | Required license is not activated.
NO-DEPARTMENT-FOUND | None | | Department ID does not resolve.
NO-DEPARTMENTS-FOUND | None | | No department was entered.
NO-START-EVENT-FOUND | None | | Start event ID does not resolve.
NO-END-EVENT-FOUND | None | | End event ID does not resolve.
INVALID-MINUTES-TO-LOOK-BACK | None | | MinutesToLookBack is not in the 1-1440 range.
NO-LEVEL-SPECIFIED | None | | Both CalculateForEachDepartment and CalculateForOverallLevel are false.

Sample request

```
<GetDepartmentWaitTimes xmlns="urn:Epic-com:Scheduling.2014.Services.Utility"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <!--int occurs:0,1-->
  <MinutesToLookBack>35</MinutesToLookBack>
  <!--boolean occurs:0,1-->
  <UseAppointmentTimeIfAfterStartEvent>1</UseAppointmentTimeIfAfterStartEvent>
  <!--boolean occurs:0,1-->
  <CalculateForOverallLevel>1</CalculateForOverallLevel>
  <!--boolean occurs:0,1-->
  <CalculateForEachDepartment>1</CalculateForEachDepartment>
  <!--arrayofidtype occurs:0,1-->
  <Departments>
    <!--idtype occurs:0,-->
    <IDType xmlns="urn:Epic-com:Scheduling.2012.Services.Provider" xsi:nil="false">
      <!--string occurs:0,1-->
      <ID xsi:nil="false">1</ID>
      <!--string occurs:0,1-->
      <Type xsi:nil="false">Internal</Type>
    </IDType>
    <!--idtype occurs:0,-->
    <IDType xsi:nil="true"/>
  </Departments>
  <!--idtype occurs:0,1-->
  <StartEvent xsi:nil="false">
    <!--string occurs:0,1-->
    <ID xmlns="urn:Epic-com:Scheduling.2012.Services.Provider" xsi:nil="false"/>
    <!--string occurs:0,1-->
    <Type xmlns="urn:Epic-com:Scheduling.2012.Services.Provider" xsi:nil="false"/>
  </StartEvent>
  <!--idtype occurs:0,1-->
  <EndEvent xsi:nil="false">
    <!--string occurs:0,1-->
    <ID xmlns="urn:Epic-com:Scheduling.2012.Services.Provider" xsi:nil="false"/>
    <!--string occurs:0,1-->
    <Type xmlns="urn:Epic-com:Scheduling.2012.Services.Provider" xsi:nil="false"/>
  </EndEvent>
```

```
</GetDepartmentWaitTimes>
```

Sample response

```
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">
  <s:Header/>
  <s:Body>
    <GetDepartmentWaitTimesResponse xmlns="urn:Epic-com:Scheduling.2014.Services.Utility">
      <GetDepartmentWaitTimesResult xmlns:i="http://www.w3.org/2001/XMLSchema-instance">
        <AverageWaitTime>14</AverageWaitTime>
        <DepartmentWaitTimes>
          <DepartmentWaitTimes>
            <AverageWaitTime>14</AverageWaitTime>
            <DepartmentIDs xmlns:a="urn:Epic-com:Scheduling.2012.Services.Provider">
              <a:IDType>
                <a:ID>1</a:ID>
                <a:Type>Internal</a:Type>
              </a:IDType>
              <a:IDType>
                <a:ID>1201</a:ID>
                <a:Type>CID</a:Type>
              </a:IDType>
            </DepartmentIDs>
            <DepartmentName>Initial DEP</DepartmentName>
            <LongestCurrentWaitTime>10233</LongestCurrentWaitTime>
            <PatientsStillWaiting>1</PatientsStillWaiting>
            <TotalPatientsForAverage/>
          </DepartmentWaitTimes>
        </DepartmentWaitTimes>
        <LongestCurrentWaitTime>10233</LongestCurrentWaitTime>
        <PatientsStillWaiting>1</PatientsStillWaiting>
        <TotalPatientsForAverage>1</TotalPatientsForAverage>
      </GetDepartmentWaitTimesResult>
    </GetDepartmentWaitTimesResponse>
  </s:Body>
</s:Envelope>
```

13.3.2 GetWaitTimes

File: site/apis/13-Clinical-Specialty/3-Non-FHIR/220-GetWaitTimes.html
Epic API ID | 220
FHIR version | Non-FHIR
Categories | Clinical Specialty
API type | Epic Public
API group | Wait Times Group
USCDI | No
HTTP method | GET
URL template | api/epic/2010/Specialty/EmergencyDepartment/GETWAITTIMES/EmergencyDepartment/WaitTimes?DepartmentID={DepartmentID}&DepartmentIDType={DepartmentIDType}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0
Performs scope validation | False
SOAP namespace | urn:Epic-com:Specialty.2010.Services.EmergencyDepartment
WSDL | Epic.Specialty.GeneratedServices/EmergencyDepartment.svc?wsdl
Source | https://fhir.epic.com/Specifications?api=220

Remarks

Be default, the wait times returned by this web service are the average of wait times completed during specified time frames and patients currently waiting whose wait times exceed the historical average during the specified time frames.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
DepartmentID | String | False
DepartmentIDType | String | False

Response elements

Name | Type | Optional | Array | Cardinality | Description
Events | Event | False | Yes
Events.Description | String | False
Events.EndID | String | False
Events.EndName | String | False
Events.StartID | String | False
Events.StartName | String | False
Events.WaitTimes | WaitTime | False | Yes
Events.WaitTimes.MedianWaitTimeInMinutes | Int64 | False
Events.WaitTimes.PatientCount | Int64 | False
Events.WaitTimes.PercentileWaitTimeInMinutes | Int64 | True | | | The n-th percentile wait time for a time frame.

Events.WaitTimes.TimeFrameInHours | Int64 | False
Events.WaitTimes.WaitTimeInMinutes | Int64 | False
LastUpdatedAt | String | False

Errors

Code | Severity | Description | Example
NO-DEPARTMENT-ID | None | | This fault code will be returned if no department ID was provided
NO-DEPARTMENT-ID-TYPE | None | | This fault code will be returned if no department ID type was provided
NO-DEPARTMENT-FOUND | None | | This fault will be thrown when an error occurs within the web service.

Sample request

```
<GetWaitTimes xmlns="urn:Epic-com:Specialty.2010.Services.EmergencyDepartment">
  <DepartmentID>720733</DepartmentID>
  <DepartmentIDType>Internal</DepartmentIDType>
</GetWaitTimes>
```

Sample response

```
<GetWaitTimesResult>
  <Events>
    <Event>
      <Description>Arrival to Triage</Description>
      <EndID>210</EndID>
      <EndName>TRIAGE COMPLETED</EndName>
      <StartID>50</StartID>
      <StartName>ED ARRIVED</StartName>
      <WaitTimes>
        <WaitTime>
          <PatientCount>44</PatientCount>
          <TimeFrameInHours>3</TimeFrameInHours>
          <WaitTimeInMinutes>40</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>62</PatientCount>
          <TimeFrameInHours>6</TimeFrameInHours>
          <WaitTimeInMinutes>33</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>72</PatientCount>
          <TimeFrameInHours>12</TimeFrameInHours>
          <WaitTimeInMinutes>29</WaitTimeInMinutes>
        </WaitTime>
      </WaitTimes>
    </Event>
    <Event>
      <Description>Triage Time</Description>
      <EndID>210</EndID>
      <EndName>TRIAGE COMPLETED</EndName>
      <StartID>205</StartID>
      <StartName>TRIAGE STARTED</StartName>
      <WaitTimes>
        <WaitTime>
          <PatientCount>39</PatientCount>
          <TimeFrameInHours>3</TimeFrameInHours>
          <WaitTimeInMinutes>7</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>58</PatientCount>
          <TimeFrameInHours>6</TimeFrameInHours>
          <WaitTimeInMinutes>7</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>74</PatientCount>
          <TimeFrameInHours>12</TimeFrameInHours>
          <WaitTimeInMinutes>9</WaitTimeInMinutes>
        </WaitTime>
      </WaitTimes>
    </Event>
    <Event>
      <Description>Triage to Reg</Description>
      <EndID>215</EndID>
      <EndName>REGISTRATION STARTED</EndName>
      <StartID>210</StartID>
      <StartName>TRIAGE COMPLETED</StartName>
      <WaitTimes>
        <WaitTime>
          <PatientCount>27</PatientCount>
          <TimeFrameInHours>3</TimeFrameInHours>
          <WaitTimeInMinutes>29</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>42</PatientCount>
          <TimeFrameInHours>6</TimeFrameInHours>
          <WaitTimeInMinutes>29</WaitTimeInMinutes>
        </WaitTime>
```

```
        <WaitTime>
          <PatientCount>56</PatientCount>
          <TimeFrameInHours>12</TimeFrameInHours>
          <WaitTimeInMinutes>50</WaitTimeInMinutes>
        </WaitTime>
      </WaitTimes>
    </Event>
    <Event>
      <Description>Triage to Room</Description>
      <EndID>55</EndID>
      <EndName>ED ROOMED</EndName>
      <StartID>210</StartID>
      <StartName>TRIAGE COMPLETED</StartName>
      <WaitTimes>
        <WaitTime>
          <PatientCount>17</PatientCount>
          <TimeFrameInHours>3</TimeFrameInHours>
          <WaitTimeInMinutes>8</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>31</PatientCount>
          <TimeFrameInHours>6</TimeFrameInHours>
          <WaitTimeInMinutes>6</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>45</PatientCount>
          <TimeFrameInHours>12</TimeFrameInHours>
          <WaitTimeInMinutes>21</WaitTimeInMinutes>
        </WaitTime>
      </WaitTimes>
    </Event>
    <Event>
      <Description>Room to Dispo</Description>
      <EndID>135</EndID>
      <EndName>KIOSK ARRIVAL TIMED OUT</EndName>
      <StartID>55</StartID>
      <StartName>ED ROOMED</StartName>
      <WaitTimes>
        <WaitTime>
          <PatientCount>11</PatientCount>
          <TimeFrameInHours>3</TimeFrameInHours>
          <WaitTimeInMinutes>75</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>22</PatientCount>
          <TimeFrameInHours>6</TimeFrameInHours>
          <WaitTimeInMinutes>100</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>45</PatientCount>
          <TimeFrameInHours>12</TimeFrameInHours>
          <WaitTimeInMinutes>95</WaitTimeInMinutes>
        </WaitTime>
      </WaitTimes>
    </Event>
    <Event>
      <Description>Dispo to Discharge</Description>
      <EndID>60</EndID>
      <EndName>ED DISCHARGED</EndName>
      <StartID>135</StartID>
      <StartName>KIOSK ARRIVAL TIMED OUT</StartName>
      <WaitTimes>
        <WaitTime>
          <PatientCount>15</PatientCount>
          <TimeFrameInHours>3</TimeFrameInHours>
          <WaitTimeInMinutes>20</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>19</PatientCount>
          <TimeFrameInHours>6</TimeFrameInHours>
          <WaitTimeInMinutes>21</WaitTimeInMinutes>
        </WaitTime>
        <WaitTime>
          <PatientCount>54</PatientCount>
          <TimeFrameInHours>12</TimeFrameInHours>
          <WaitTimeInMinutes>37</WaitTimeInMinutes>
        </WaitTime>
      </WaitTimes>
    </Event>
  </Events>
  <LastUpdatedAt>6/1/2010 1:10 PM</LastUpdatedAt>
</GetWaitTimesResult>
```